PHYSICS GRADE 10: PROPERTIES OF WAVES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 2.0: Waves and the Electromagnetic Spectrum
Sub-Strand	Sub-Strand 2.1: Properties of Waves
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– Nature of waves: transverse and longitudinal waves, mechanical and electromagnetic waves – Wave properties: amplitude, wavelength, frequency, period, and wave speed – The wave equation: $v = f\lambda$ and its application to solve problems – Rectilinear propagation of waves (light rays travelling in straight lines) – Reflection of waves: laws of reflection, plane mirrors, curved reflectors – Refraction of waves: Snell's Law, refractive index, critical angle, and total internal reflection – Diffraction of waves: behaviour of waves passing through gaps and around obstacles – Interference of waves: constructive and destructive interference, superposition principle – Standing (stationary) waves: nodes, antinodes, harmonics on a string – The Doppler Effect: change in observed frequency due to relative motion of source and observer – Wave modulation: amplitude modulation (AM) and frequency modulation (FM) in communication – Applications of wave properties: sonar, ultrasound imaging, radio broadcasting, fibre optics
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) distinguish between transverse and longitudinal waves and give examples of each from everyday Kenyan contexts, b) identify and describe wave properties — amplitude, wavelength, frequency, period, and speed — using diagrams and measurements, c) apply the wave equation $v = f\lambda$ to solve quantitative problems involving wave speed, frequency, and wavelength, d) investigate and explain the phenomena of reflection, refraction, diffraction, and interference of waves with reference to experimental evidence, e) explain the formation of standing waves and identify nodes and antinodes on a vibrating string, f) describe the Doppler Effect and explain its application in medicine (ultrasound), meteorology, and traffic enforcement, g) explain the principles of AM and FM modulation and their role in radio communication in Kenya, h) evaluate real-world applications of wave properties including sonar, fibre optics, and ultrasound imaging, i) appreciate the role of wave physics in modern technology and its contribution to communication and health services in Kenya.
Core Competencies	– Communication and Collaboration: Learners work in pairs and small groups during PhET simulations (Wave Interference, Bending Light) and peer-discussion activities to compare predictions, share observations, and co-construct explanations of

	wave behaviour; teamwork is central to model-building and DQB contributions. – Critical Thinking and Problem Solving: Learners analyse wave diagrams, interpret simulation outputs, and apply the wave equation $v = f\lambda$ to solve multi-step problems; they evaluate competing explanations for phenomena such as why a matatu's horn sounds different as it passes. – Learning to Learn: Learners reflect on their pre-instruction predictions against observed evidence in each lesson cycle, revise their conceptual models iteratively, and self-monitor understanding by updating the Driving Question Board throughout the unit. – Digital Literacy: Learners interact with PhET simulations (Wave Interference, Bending Light, Standing Waves on a String) and curated Khan Academy and YouTube resources to gather evidence; they practise navigating and interpreting digital scientific tools. – Creativity and Imagination: Learners design and present annotated wave-property models using local contexts (e.g., sound waves at a Nairobi matatu stage, light through a glass of water) and propose creative applications of wave modulation in Kenya's communication landscape.
Core Values	– Responsibility: Learners take individual accountability for recording accurate data during simulations and experiments, contributing fairly to group model-building tasks, and maintaining a reliable Driving Question Board that reflects the class's growing understanding. – Respect: Learners demonstrate open-mindedness by listening to peers' alternative explanations during sensemaking discussions and appreciating diverse predictions on the DQB before consensus is reached. – Unity: Learners collaborate across gender and ability lines during group investigation tasks, recognising that understanding wave physics — foundational to Kenya's communication infrastructure — requires collective effort. – Social Justice: Learners explore how equitable access to FM radio and mobile communication (both wave-based technologies) affects rural and urban communities across Kenya, connecting physics to real societal impact.
Science & Engineering Practices	– SEP 1 — Asking Questions and Defining Problems: Learners generate and refine questions for the Driving Question Board in every lesson, starting from the anchoring phenomenon of the matatu horn and radio signal to drive sustained inquiry throughout the unit. – SEP 2 — Developing and Using Models: Learners iteratively build and revise wave-property models (ray diagrams, wavefront sketches, standing-wave string diagrams) across all 12 lessons, culminating in a comprehensive annotated model. – SEP 3 — Planning and Carrying Out Investigations: Learners design observation protocols within PhET simulations (e.g., systematically varying gap width to observe diffraction, changing frequency to observe standing-wave harmonics) and record structured data. – SEP 4 — Analysing and Interpreting Data: Learners extract wave measurements (amplitude, wavelength, frequency) from simulation screenshots and YouTube demonstrations, graph relationships (v vs. f), and interpret patterns. – SEP 6 — Constructing Explanations: Learners write evidence-based explanations for each wave phenomenon (reflection, refraction, diffraction, interference, Doppler) tying back to the anchoring phenomenon and supporting phenomena. – SEP 8 — Obtaining, Evaluating, and Communicating Information: Learners critically evaluate Khan Academy and YouTube resources for accuracy, extract key information on the Doppler Effect and FM modulation, and communicate findings through annotated diagrams and class presentations.
Pertinent & Contemporary Issues (PCIs)	– Life Skills Education (Communication): Learners practise effective verbal and written scientific communication when presenting wave-property explanations and DQB contributions, building skills applicable beyond the classroom. – Environmental Education: The unit's exploration of the greenhouse effect connection (electromagnetic wave absorption by CO_2) and the use of sonar in monitoring Lake Victoria fish populations raises learners' awareness of how wave physics intersects with Kenya's environmental challenges. – Social Cohesion and Values: Group investigation activities are structured to ensure equitable participation across gender,

	linking to Kenya's national goal of closing the gender gap in STEM fields — mirroring the approach in Sub-Strand 1.1. – Citizenship: Learners examine how FM radio broadcasting reaches remote areas of the Rift Valley, Turkana, and the Coast, discussing the role of wave-based communication technology in civic participation and access to information for all Kenyans. – Safety: Learners observe safe laboratory and digital-device practices during simulation sessions, and discuss safety implications of high-frequency wave exposure (e.g., mobile phone radiation awareness).
Career Connections	– Telecommunications Engineer: Understanding wave properties (frequency, wavelength, modulation) is foundational for designing Kenya's mobile and FM radio networks, including Safaricom and KBC infrastructure. – Medical Sonographer / Radiologist: The Doppler Effect and ultrasound applications connect directly to careers in Kenya's growing healthcare sector, including use of ultrasound at hospitals in Nairobi, Mombasa, and Kisumu. – Meteorologist: Kenya Meteorological Department scientists apply the Doppler radar principle to track rainfall patterns over the Rift Valley and Lake Victoria basin. – Marine / Fisheries Scientist: Sonar technology — based on wave reflection — is used in Lake Victoria to locate fish schools, supporting Kenya's fisheries industry. – Broadcast Journalist / Media Technologist: FM and AM modulation principles underpin careers in Kenya's vibrant radio and television broadcast industry (KBC, Radio Citizen, Kiss FM). – Architect / Acoustical Engineer: Knowledge of wave reflection, diffraction, and interference is applied in designing performance halls, mosques, and churches with good acoustics in Kenyan cities. – Optometrist / Optical Engineer: Refraction, total internal reflection, and fibre optics underpin careers in eye care and data communication infrastructure across Kenya.
Focus for Lessons	This unit investigates the fundamental properties of waves — mechanical and electromagnetic — through an inquiry-driven, phenomenon-anchored approach rooted in everyday Kenyan experiences. Beginning with the puzzling observation that a matatu's hooting horn sounds different as it approaches versus recedes, and that Nairobi FM radio signals fade in certain buildings, learners systematically explore wave characteristics, behaviours (reflection, refraction, diffraction, interference), standing waves, the Doppler Effect, and modulation. Throughout the 12 lessons, learners use PhET simulations, curated YouTube demonstrations, and Khan Academy resources as primary evidence-gathering tools, building and refining an annotated wave-property model that grows lesson by lesson. The unit balances quantitative problem-solving ($v = f\lambda$, Snell's Law, refractive index) with qualitative sensemaking, ensuring learners can both calculate and explain wave phenomena in authentic Kenyan contexts.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "Why does the matatu's horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel — and what do these two mysteries reveal about how ALL waves behave?" KICD KEY INQUIRY QUESTIONS: 1. What are the properties of waves? 2. How do waves behave in different media and under different conditions?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Disappearing Matatu Horn and the Fading Radio Signal" A Form 4 student travelling from Nakuru to Nairobi notices two puzzling things on the same journey: (1) As a matatu travelling in the opposite direction passes at high speed on the Nakuru–Nairobi highway, its horn sounds noticeably higher-pitched as it approaches and then suddenly drops to a lower pitch as it moves away — yet the driver of the matatu is not changing the horn at all. (2) When the student's bus enters the long Kabete underpass on the outskirts of Nairobi, the FM radio signal (tuned to Classic 105, 105.2 MHz) cuts out almost completely, but the AM radio signal on the same receiver stays audible — even though both are broadcast from the same city. These two observations — a shifting pitch and a selective signal blackout — seem completely unrelated, yet both can be fully explained by the same underlying set of wave properties. Students find this puzzling because

	sound and radio waves seem like very different things, and it is not obvious why motion should change pitch, or why one type of radio wave passes through a tunnel while another does not. The anchoring phenomenon drives the unit-long investigation: what properties do all waves share, and how do those properties explain everything from the sound of a matatu horn to the reach of a rural FM station in Turkana County?
Storyline Thread	Lesson 1 — PREDICT: Anchoring Phenomenon Launch — "The Matatu Horn and the Fading Radio" Students listen to an audio clip of the Doppler shift (matatu horn) and hear the contrasting AM/FM tunnel scenario. They record written predictions: What is a wave? Why might pitch change with motion? Why might FM fade but AM survive in a tunnel? The Driving Question Board is introduced and populated with initial student questions. Teacher elicits prior knowledge of waves from everyday life (ocean waves at Mombasa, sound at a football match in Kasarani Stadium). Lesson 2 — OBSERVE/EXPLAIN: Nature and Types of Waves Students use PhET: Wave on a String to observe transverse waves, then a slinky demonstration for longitudinal waves. They identify amplitude, wavelength, frequency, period, and wave speed from the simulation. Key vocabulary is established. Students begin populating their first wave-property diagram model. Connection to anchoring phenomenon: Is the matatu horn a transverse or longitudinal wave? What about the FM radio signal? Lesson 3 — OBSERVE/EXPLAIN: The Wave Equation $v = f\lambda$ Students gather quantitative data from PhET: Wave on a String (measuring wavelength and frequency at constant speed). They derive $v = f\lambda$ empirically before the equation is formalised. Worked problems use Kenyan contexts: "Classic 105 FM broadcasts at 105.2 MHz — what is the wavelength of its signal?" and "A fisherman's sonar on Lake Victoria emits a pulse at 40 kHz — what is its wavelength in water?" Students solve problems individually and in pairs. Lesson 4 — OBSERVE/EXPLAIN: Rectilinear Propagation and Reflection of Waves Students use PhET: Geometric Optics (ray model) and a YouTube reflection demonstration to investigate rectilinear propagation and the laws of reflection. They construct ray diagrams for plane mirrors and curved reflectors. Connection to supporting phenomenon 2 (sharp shadow edges). DQB updated: "If light travels in straight lines, how does it bend in a glass of water?" — setting up Lesson 5. Lesson 5 — OBSERVE/EXPLAIN: Refraction and Snell's Law Students use PhET: Bending Light to investigate how light bends at a water–air or glass–air boundary. They measure angles of incidence and refraction, tabulate $\sin\theta_i / \sin\theta_r$ values, and derive the refractive index n empirically. Snell's Law is formalised. Total internal reflection is demonstrated. Connection to supporting phenomenon 1 (bent straw at school canteen). Careers link: fibre-optic cables carrying internet signals to Mombasa via the TEAMS undersea cable. Lesson 6 — OBSERVE/EXPLAIN: Diffraction of Waves Students use PhET: Wave Interference (set to single slit / double slit) to systematically vary gap width and observe the spreading of waves. They record that wider gaps produce less diffraction, and that longer wavelengths diffract more. Connection to anchoring phenomenon: FM waves (shorter wavelength) diffract less around the tunnel walls than AM waves (longer wavelength) — first partial explanation of the radio mystery is constructed. DQB updated with new evidence. Lesson 7 — OBSERVE/EXPLAIN: Interference of Waves Students continue with PhET: Wave Interference (double-source mode) to observe constructive and destructive interference

	patterns. They identify nodal and antinodal lines, relate path difference to superposition. Connection to supporting phenomenon 3 (soap bubble colours). Students sketch their first interference pattern diagram and add it to their growing model. Teacher cold-calls 4 students to explain: "Where is constructive interference happening, and why?" Lesson 8 — OBSERVE/EXPLAIN: Standing Waves Students use PhET: Standing Waves on a String to observe how standing waves form, identify nodes and antinodes, and explore harmonics (1st, 2nd, 3rd). They connect standing wave patterns to the guitar-string supporting phenomenon (Lesson intro). Students calculate the wavelength of each harmonic for a string of fixed length. DQB question addressed: "Why does the guitar string vibrate in sections?" — model updated with standing-wave diagram. Lesson 9 — OBSERVE/EXPLAIN: The Doppler Effect Students watch a YouTube Doppler Effect demonstration (car horn passing a stationary microphone, visualised on an oscilloscope). They use Khan Academy: Doppler Effect reading/video as a follow-up. Students qualitatively explain the pitch shift using wavefront compression/stretching diagrams. They then apply the Doppler formula to a Kenyan traffic scenario ("A traffic officer uses a radar gun on the Thika Superhighway — the car is travelling at 120 km/h…"). This lesson directly resolves the first part of the anchoring phenomenon: the matatu horn mystery is explained. Lesson 10 — OBSERVE/EXPLAIN: Wave Modulation — AM and FM Students use the Khan Academy: FM Modulation resource and a curated YouTube clip on AM vs FM broadcasting. They compare amplitude modulation (AM, longer wavelength, better diffraction around hills and into tunnels) and frequency modulation (FM, higher frequency, higher fidelity, shorter wavelength, less diffraction). Students draw annotated comparison diagrams. This lesson fully resolves the second part of the anchoring phenomenon: why FM fades in the Kabete tunnel but AM does not. DQB is largely answered; students reflect on how far their understanding has grown. Lesson 11 — MODEL BUILDING: Cumulative Model Revision and Application Students compile all diagrams, wave-property tables, ray diagrams, interference patterns, standing-wave sketches, Doppler diagrams, and AM/FM comparison sketches into one comprehensive annotated Unit Model poster (individual or pair). They must explicitly connect each wave property to at least one Kenyan real-world application and label how it relates to either the anchoring or a supporting phenomenon. Teacher circulates and provides structured feedback using a model-quality rubric. Lesson 12 — ASSESS and REFLECT: DQB Resolution, Presentations, and Unit Consolidation Students present their models to the class in a gallery walk format. Peers use sticky notes to give "one star, one question" feedback. The full Driving Question Board is revisited — every original student question is addressed collaboratively. A short formative written assessment (10 questions: 5 calculation, 5 explanation) checks mastery of $v = f\lambda$, Snell's Law, diffraction, Doppler, and modulation. Lesson closes with a careers spotlight: students share which wave-related career (sonographer, telecom engineer, meteorologist, broadcast journalist) they found most relevant to their own interests and why.
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — "The Bent Straw in a Glass of Water": A student at a Nairobi school canteen notices that a straw in a glass of water appears sharply bent at the water surface, even though the straw is perfectly straight. This drives investigation of refraction and Snell's Law, connecting to fibre-optic cables and total internal reflection. – SUPPORTING PHENOMENON 2 — "Shadows and Sharp Edges vs. Fuzzy Edges": When a student holds a hand near a bright torch (mkono wa tochi) in a dark room, the shadow on the wall has a sharp edge — but when sound from outside the classroom goes around the corner of a building, it can still be heard clearly. This contrast drives investigation of diffraction: why do sound waves bend around corners but light waves (seemingly) do not?

	– SUPPORTING PHENOMENON 3 — "Coloured Bands on a Soap Bubble": Children blowing soap bubbles (povu la sabuni) at a Kisumu market notice swirling rainbow colours on the bubble surface — colours that shift and disappear as the bubble pops. This drives investigation of wave interference (constructive and destructive superposition of light waves) and the concept of path difference. – SUPPORTING PHENOMENON 4 — "The Guitar String that Hums at One Pitch but Vibrates in Sections": A student playing a guitar (gitaa) at a school concert in Eldoret notices that plucking the same string at different points produces different, purer tones, and that the string appears to vibrate in distinct sections with still points in between. This drives investigation of standing waves, nodes, antinodes, and harmonics.

LESSON 1 (80 minutes): Anchoring Phenomenon Launch — "The Matatu Horn and the Fading Radio"	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and elicit students' prior knowledge and questions about wave behaviour in everyday Kenyan contexts.
Knowledge	Students will know that waves are disturbances that transfer energy without transferring matter, and that sound and radio signals are both wave phenomena sharing common properties.
Skills	Students will be able to articulate observations about the phenomenon, formulate investigable questions, and record initial predictions using scientific reasoning.
Attitudes	Students will appreciate the relevance of Physics to everyday Kenyan life and demonstrate curiosity and open-mindedness when confronting puzzling natural phenomena.
Key Inquiry Question	1. What are the properties of waves? 2. How do waves behave in different media and under different conditions?
Purpose in Storyline	This is the opening lesson of the unit. It launches the anchoring phenomenon — the matatu horn pitch shift and the Kabete tunnel radio blackout — to surface prior knowledge, generate genuine questions, and open the Driving Question Board (DQB) that will organise the entire unit's investigation.
Safety Notes	No hazardous materials are used in this lesson. If a live audio demonstration is conducted using a speaker or buzzer on a string, ensure the string is securely tied and swung only in a clear open space away from other students' faces. Students with hearing impairments should be seated near the demonstration and provided with a tactile vibration alternative (e.g., hand placed on a resonating surface).

B. LESSON OVERVIEW

Students encounter the anchoring phenomenon through a vivid narrative and short audio-visual stimulus: a matatu horn Doppler shift recorded on the Nakuru–Nairobi highway, and a radio signal dropping out as a vehicle enters the Kabete tunnel. Without any prior instruction on wave properties, students work individually then in pairs to write down what they notice, what they find puzzling, and what they predict might be happening. The class then builds the opening Driving Question Board together, populating it with their questions and initial models. By the end of the lesson every student has a personal written prediction linked to both parts of the phenomenon, and the DQB is displayed prominently for the rest of the unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Anatomy of a skeletal muscle cell] Source: Khan	Students watch/listen to a 90-second teacher-narrated audio-visual stimulus: first, a sound clip (or teacher-produced demonstration) of a buzzer on a string swung overhead and then	SETUP (before class): Tie a buzzer or small Bluetooth speaker playing a continuous tone to a 60 cm string. Prepare a PowerPoint slide showing the Nakuru–Nairobi highway map with the Kabete	STRATEGY: 'Notice-Wonder-Predict' card. This strategy activates prior knowledge and makes student thinking visible before instruction begins. By separating 'noticing' (observable	INDICATOR: Every student produces a completed PREDICT card with at least one written puzzling question and one prediction that references either motion, energy, or signal type.

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Electromagnetic Waves (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	moved rapidly past students, simulating the matatu horn pitch change on the Nakuru–Nairobi highway. Second, a short teacher narration and diagram of the Kabete tunnel scenario — FM 105.2 MHz (Classic 105) cutting out while AM stays audible. Students are given a structured PREDICT card (half an A4 sheet) with three prompts: (1) 'What did you NOTICE?' (2) 'What do you find PUZZLING?' (3) 'Write your BEST GUESS — why does the horn pitch change, and why does FM cut out but not AM?' Students write individually in silence for 5 minutes, then share with a shoulder partner for 2 minutes.	tunnel marked. Print or display PREDICT cards. STEP 1 — Hook narration (3 min): Stand at the front and read aloud dramatically: 'Imagine you are on a bus from Nakuru. A matatu flies past in the opposite lane — WHEEEEE-wooom — the horn pitch drops the instant it passes. Ten minutes later your bus enters the Kabete tunnel and Classic 105 goes dead — silence — but the AM station crackles on. Same city, same receiver, same tunnel. Why?' Show the map slide. Ask: 'Has anyone on this bus journey experienced exactly this?' (Hands — expect 3–6 yes responses. Cold-call 2 students: 'Describe what you heard or noticed.' Wait 10 seconds after each cold-call before accepting the answer.) STEP 2 — Live demonstration (3 min): Swing the buzzer on the string in a horizontal circle above your head. Say: 'Listen carefully — does the pitch change as it comes towards you versus moves away?' Then arc the buzzer in a pendulum swing directly towards and away from the class. Ask: 'What did you notice?' Wait 12 seconds. Cold-call 3 students by name (rotate from the back row first). STEP 3 — Individual PREDICT card (5 min): Distribute PREDICT cards. Say: 'Write your own ideas — there are no wrong answers yet. This is your scientific gut feeling. We will return to this card in every lesson.' Circulate silently; do NOT correct or validate predictions — only encourage	facts) from 'wondering' (questions) and 'predicting' (mechanistic reasoning), students practise the distinction between observation and inference — a foundational science practice. Connecting both halves of the phenomenon (sound + radio) in the same card seeds the cross-cutting concept that different phenomena may share a common underlying mechanism.	Teacher collects cards at end of phase and quickly sorts into three piles during the next phase: (A) predictions that reference wave properties (e.g., 'something about the wave changes'), (B) predictions that reference only surface features (e.g., 'the horn button was pressed'), (C) blank or very sparse. Piles B and C guide targeted questioning in Phase 3.

		specificity: 'Can you add a number or a direction to that idea?' STEP 4 — Pair share (2 min): 'Turn to your shoulder partner, share your puzzling question, then agree on one joint question to bring to the class.' Monitor 3–4 pairs for interesting predictions to highlight anonymously.		
Observe Phase 📹 **VIDEO:** Anatomy of a skeletal muscle cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Electromagnetic Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students rotate through two short evidence stations (10 minutes each, pairs or threes): STATION A — PhET Sound Simulation (on ARES offline tablet or laptop): Students open the 'Sound' PhET simulation and move a listener relative to a stationary source, then move the source relative to a stationary listener. They record on an observation table: frequency setting → perceived pitch (higher/lower/same) for each configuration. STATION B — Diagram Analysis Card: A printed card shows two diagrams: (1) a cross-section of the Kabete tunnel with FM and AM wavelength scales drawn to scale (FM ~2.9 m; AM ~300 m), and the tunnel width (~8 m) labelled. (2) A wave passing through a gap of different sizes. Students annotate: 'Circle the wave that gets through the gap most easily. Why?' After both stations, pairs record three OBSERVATIONS (not explanations) on a class-shared sticky-note board under the heading 'WHAT WE SAW.'	TRANSITION (1 min): 'Before we explain anything, scientists gather evidence. You have 20 minutes at two stations. Your job is only to OBSERVE and RECORD — not to explain yet. Explanations come later.' STATION MANAGEMENT: Assign pairs to stations by row. Set a visible 10-minute timer. Circulate between stations every 2–3 minutes. At STATION A — PhET: If a student says 'the pitch goes up,' press for precision: 'By how much? What did you change in the simulation to make that happen?' Wait 10 seconds. Do NOT introduce the word 'Doppler' yet. At STATION B — Diagram: Point to the tunnel width and FM wavelength drawn to the same scale. Ask: 'What do you notice about the size of the gap compared to the wavelength?' Wait 12 seconds. Cold-call 2 students. If students say 'FM doesn't fit,' accept without affirming: 'Interesting — write that down exactly as you said it.'	STRATEGY: 'Two-Station Evidence Carousel with Observation-Only Protocol.' By explicitly forbidding explanations during observation, this strategy trains students in the NGSS practice of distinguishing data from inference. The side-by-side pairing of a dynamic simulation (sound) with a static diagram (tunnel/wavelength) develops the cross-cutting concept of Scale, Proportion, and Quantity — students begin to notice that wavelength size relative to obstacle size is a measurable, comparable quantity, not just a vague idea.	INDICATOR: Each pair's sticky note contains at least two specific, measurable observations (e.g., 'when source moves toward listener, wave crests are closer together' or 'FM wavelength is about 3 m, tunnel is 8 m wide'). Teacher checks that Station B annotations correctly circle the longer-wavelength wave as passing through more easily. Pairs who circle the shorter wavelength are flagged for targeted questioning in Phase 3.

		WHOLE-CLASS DEBRIEF (5 min): Ask three pairs to read their sticky-note observations aloud. Write key observable facts on the board in a column labelled 'EVIDENCE.' Do not write any explanations yet. Say: 'These are our data. We are not explaining — not yet. But hold these in your mind.'		
Explain Phase 📹 **VIDEO:** [Anatomy of a skeletal muscle cell] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Electromagnetic Waves (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Whole-class structured discussion: Teacher returns sorted PREDICT cards (anonymised) and reads out 2–3 representative predictions from each pile. Students discuss: 'Which prediction best fits the evidence we just gathered, and why?' In groups of four, students draft a 'Best Current Explanation' sentence for each part of the phenomenon — the horn pitch change and the FM blackout — using only the evidence from the stations. Groups record on a mini-whiteboard or large sticky note. A class gallery walk (3 minutes) lets students read all group explanations and place a dot sticker on the one they find most convincing. Teacher facilitates a 5-minute whole-class convergence discussion, drawing out the emerging ideas that (a) motion affects how waves arrive at a listener, and (b) wave size relative to gap size affects whether a wave passes through.	RETURN PREDICT CARDS (2 min): Hand back anonymised predictions (names folded under). Say: 'I am not telling you which ones are correct. Your job is to use the evidence to decide.' GROUP TASK (8 min): 'In your group of four, write ONE sentence that explains the horn pitch change and ONE sentence that explains the FM blackout. Use only what you observed. You have 8 minutes.' Set timer visibly. GALLERY WALK (3 min): Post mini-whiteboards around the room. Students walk silently, read, and place one dot sticker on the explanation they find most evidence-based. No talking during the walk. CONVERGENCE DISCUSSION (5 min): Point to the two or three explanations with the most dots. Ask: 'What do these top explanations have in common?' Wait 15 seconds. Cold-call 3 students from different groups. Write emerging key ideas on the board under the heading 'PATTERNS WE ARE NOTICING.' Look for and name: 'Several groups noticed that the pitch depends on how the matatu is	STRATEGY: 'Claim-Evidence gallery walk with convergence protocol.' Students construct initial explanations (claims) tied explicitly to their observed evidence, then evaluate peer claims using dot voting — a low-stakes peer review. The teacher's role shifts to a facilitator who names patterns without endorsing specific answers, preserving productive uncertainty. This reflects the NGSS practice of Constructing Explanations and Engaging in Argument from Evidence.	INDICATOR: Each group's written explanation contains at least one reference to a quantity or direction (e.g., 'moving toward,' 'wavelength is longer than the gap,' 'crests are closer together'). Teacher checks that no group has written a purely technological explanation (e.g., 'Classic 105 has a weaker transmitter') — if so, teacher prompts: 'Look at the diagram again — what changed in the experiment when you moved the source in the simulation?'

		MOVING relative to the listener — that is a really precise scientific observation.' If the concept of wavelength-to-gap ratio appears, say: 'This group noticed that the SIZE of the wave matters relative to the SIZE of the gap — hold that idea, it becomes very important.' LINK BACK TO PHENOMENON (2 min): Point to the highway map and tunnel diagram. Say: 'Can our current best explanations fully answer the driving question? What is still missing?' Accept 2–3 responses. Write gaps on the board — these become DQB seeds in Phase 4.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Anatomy of a skeletal muscle cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Electromagnetic Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student individually writes 2–3 questions they still cannot answer about the phenomenon on separate sticky notes — one question per note. Questions are posted on the class DQB (a large sheet of paper or whiteboard section divided into two columns: 'The Horn Puzzle' and 'The Tunnel Puzzle'). The teacher reads questions aloud, groups similar questions together by moving sticky notes, and labels emerging clusters with category titles that students suggest. The DQB is photographed and pinned/displayed at the front of the classroom for the remainder of the unit. Students copy the three questions their group finds most important into their notebooks under 'Our Unit Driving Questions.'	SETUP: Pre-draw the DQB template on a large sheet of paper or dedicate a section of the whiteboard. Two columns: 'The Horn Puzzle' and 'The Tunnel Puzzle.' A third column, 'Connections,' is left blank for now. INSTRUCTION (2 min): 'On each sticky note write ONE question you genuinely cannot yet answer about the phenomenon. Start with HOW or WHY. You have 3 minutes — write at least two notes.' Set timer. POSTING (3 min): Students come up row by row and post their notes. While they post, teacher reads a selection aloud: 'I see a question here: WHY does FM cut out but not AM? — that is a precise, testable question. I see another: WHY does the pitch change only when the matatu is MOVING? — excellent.' Avoid answering any question.	STRATEGY: 'Driving Question Board with student-generated clustering.' The DQB externalises and organises the class's collective curiosity, making abstract physics questions feel personally owned. Student-generated clustering (rather than teacher-imposed categories) develops the NGSS practice of Asking Questions and Defining Problems and builds metacognitive awareness — students can see what they know and do not yet know. The physical persistence of the board throughout the unit creates a running narrative of growing understanding.	INDICATOR: The DQB contains at least 15 distinct student questions across both columns. At least 50% of questions are mechanistic (begin with 'How' or 'Why' and reference a physical quantity or process). Teacher notes any column that is sparse (e.g., few questions about the tunnel) as a topic requiring more stimulus in Lesson 2. Students' notebook entries are checked at the start of Lesson 2 to confirm all three questions were recorded.

		CLUSTERING (4 min): Move similar sticky notes together. Ask the class to suggest category names. Likely clusters: 'Wave Size/Wavelength,' 'Wave Speed,' 'Effect of Motion,' 'What is a wave?,' 'Tunnels and Barriers.' Write category labels in a different colour. COMMIT (2 min): 'Copy the three questions your group most wants to answer into your notebook. We will return to this board at the end of every lesson and tick off what we have explained.' Photograph the DQB. LINK FORWARD: 'In our next lesson we will gather real evidence about what makes waves different from each other. Come ready with your PREDICT card — we will look back at your predictions.'		
Model Building Phase 📹 **VIDEO:** Anatomy of a skeletal muscle cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Electromagnetic Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students individually draw an 'Initial Wave Model' in their notebooks in response to the prompt: 'Draw what you think a WAVE looks like as it travels from a matatu horn to your ear. Then draw what you think happens to that wave when it enters a narrow tunnel. Label anything you can.' Students are explicitly told: 'Your model does not need to be correct — it needs to show your current thinking. We will revise it in every lesson.' After 5 minutes of individual drawing, pairs share models and write one similarity and one difference between their two drawings at the bottom of the page.	FRAME THE MODEL (2 min): 'Scientists use models to show how they think something works — even when they are not sure. Your model is a scientific tool, not an art project. Draw arrows, label distances, show direction of travel.' Display a deliberately incomplete teacher example on the board — a simple sine wave arrow pointing from a horn shape toward an ear shape — say: 'Here is my starting model. It is incomplete on purpose. What is missing?' Wait 12 seconds. Cold-call 2 students. Add their suggestions to your model live. INDIVIDUAL DRAWING (5 min): Circulate. Prompt students who draw only a single line: 'Can you	STRATEGY: 'Initial Model — Revise Later protocol.' Asking students to commit an initial model to paper before instruction creates a cognitive anchor — a concrete record of prior understanding that can be explicitly revised as learning progresses. Research shows that students who build and revise models develop deeper conceptual understanding than those who only receive correct models. The deliberate incompleteness of the teacher's model normalises scientific uncertainty and positions revision as progress, not failure. This reflects the NGSS practice of Developing and Using Models.	INDICATOR: Every student's notebook shows a drawn model with at least one labelled feature (e.g., 'horn,' 'wave,' 'ear,' 'tunnel wall') and a written similarity/difference from the pair comparison. Teacher photographs 6–8 models (two from each ability cluster identified from PREDICT cards) to track model revision across the unit. At the start of Lesson 2, teacher projects two anonymised models (one simple, one more detailed) and asks: 'What does each model show well? What is missing?' — this connects directly to new instruction on wave properties.

		show what the wave LOOKS like as it moves — its shape?' Prompt students who draw a detailed sine wave: 'Now draw what happens to that wave at the tunnel entrance — what do you predict?' Do NOT correct wave shape — preserve initial models. PAIR COMPARISON (3 min): 'With your partner, write: (1) one thing your models have in common, and (2) one thing that is different. Write it below your drawing.' CLOSING LINK (2 min): Hold up the PREDICT card from Phase 1. Say: 'You started this lesson with a prediction. You now have a model. In our next lesson, we add evidence about wave properties — amplitude, frequency, wavelength, speed — and we will come back to this model and ask: does our evidence support our drawing, or do we need to revise it? Keep this notebook safe.' Remind students: 'The driving question is still open: why does the horn change pitch, and why does FM fade in the tunnel? We are just beginning.'		

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did every student produce a written prediction and a drawn model — if not, which students need a more scaffolded entry point in Lesson 2 (e.g., a sentence frame: 'I think the pitch changes because ___')? (2) Look at the DQB — are students asking questions about wave properties (frequency, wavelength, speed) or only about technology (transmitters, phones)? If questions are mostly technological, plan a brief 'wave vs. signal' framing at the start of Lesson 2. (3) Did the live buzzer demonstration produce a noticeable pitch shift audible to most students? If the room acoustics were poor, source a short video clip of a matatu horn on the Nakuru–Nairobi highway for Lesson 2. (4) Review the PREDICT card piles: if more than 40% of cards are in Pile C (blank/sparse), the phenomenon may need a more concrete local hook — consider opening Lesson 2 with a 30-second audio clip of an actual Doppler shift before revisiting predictions. (5) Note which students cold-called confidently and which remained silent — plan to cold-call the silent students first in Lesson 2 using low-stakes 'What did you draw in your model?' questions rather than explanatory questions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed (via live buzzer demonstration and PhET Sound simulation) that a moving sound source produces a pitch change — higher as it approaches, lower as it moves away. They also observed from the tunnel diagram that FM waves (~2.9 m wavelength) are comparable in size to the Kabete tunnel width (~8 m), while AM waves (~300 m) are far larger than the tunnel.
What did I learn?	Students learned that sound and radio signals are both wave phenomena, and that waves transfer energy without transferring matter. They learned that a wave can be represented as a drawing showing direction of travel, and that properties such as wavelength may be relevant to how a wave interacts with a physical barrier.
How does this explain the phenomenon?	Students are not yet able to fully explain either part of the phenomenon — this is intentional. Partial emerging explanations include: (a) the pitch change is related to the motion of the matatu relative to the listener, and (b) the FM blackout may be related to the size of the wave relative to the size of the tunnel opening. Both explanations remain incomplete and will be refined across the unit.

LESSON 2 (80 minutes): What Is a Wave? Building Our First Model from Everyday Kenyan Phenomena	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor students' emerging understanding of waves by eliciting prior knowledge, introducing the anchoring phenomenon, and building a first working definition and model of a wave using familiar Kenyan contexts.
Knowledge	Students will be able to define a wave as a disturbance that transfers energy from one point to another without transferring matter, and identify that sound and radio signals are both waves despite appearing very different.
Skills	Students will practise making and recording written predictions, asking investigable questions, and constructing an initial labelled diagram (model) of a wave using observations from the anchoring phenomenon audio clip.
Attitudes	Students will appreciate that curiosity about everyday experiences — such as sounds heard on the Nakuru–Nairobi highway or signals lost in the Kabete tunnel — is a valid starting point for scientific inquiry.
Key Inquiry Question	What are the properties of waves, and how do waves behave in different media and under different conditions?
Purpose in Storyline	This is Lesson 2 of 12 in the evidence-gathering sequence. Students have been introduced to the unit anchoring phenomenon in Lesson 1. In this lesson they listen carefully to an audio clip of the Doppler-shifted matatu horn, engage with the AM/FM tunnel contrast, record written predictions, and populate the Driving Question Board (DQB) with their initial questions. This lesson builds the shared knowledge baseline — a working definition of a wave and a first annotated model — that every subsequent lesson will refine and extend.
Safety Notes	No hazardous materials are used in this lesson. If using a portable speaker at high volume for the audio clip demonstration, ensure volume is kept at a comfortable level (below 85 dB) to protect hearing. Students with hearing impairments should be seated close to the speaker or given visual/tactile wave demonstrations using a slinky on a desk.

B. LESSON OVERVIEW

Students begin by silently listening to a 45-second audio clip (played via the ARES platform or a portable speaker) that simulates the Doppler shift of a matatu horn passing at speed. They then hear a brief narration contrasting the FM signal drop and AM signal survival inside the Kabete tunnel. Before any instruction, every student writes individual predictions: What is a wave? Why might the horn pitch change with motion? Why might FM fade while AM survives in the tunnel? The teacher facilitates a structured Think-Pair-Share to surface prior knowledge — ocean waves at Mombasa beach, the roar of the crowd at Kasarani Stadium, ripples in a cup of chai — and uses these to build a class working definition of a wave. Students construct a first annotated wave diagram in their notebooks. The lesson closes with the creation of the Driving Question Board (DQB): students write questions on sticky notes and the teacher clusters them into three emerging themes (What is a wave? What affects how waves travel? Why do sound and radio behave differently?). These themes will structure the remainder of the 12-lesson unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Wave on a	Students sit quietly and listen to the 45-second ARES audio clip of a matatu horn exhibiting a Doppler	T sets the scene before playing the clip: 'Close your eyes if you are comfortable. Imagine you are	Prediction Writing (Individual Written Prediction): This strategy activates prior knowledge and	Teacher circulates and performs a rapid 'notebook scan' — reading opening sentences of predictions

[String](#) Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Landforms from Wave Erosion and Deposition (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	shift (high pitch approaching, abrupt drop as it passes). Immediately after, they hear a 20-second narration: 'You are on a bus entering the Kabete tunnel. Classic 105 FM — 105.2 MHz — goes silent. The AM station at 1080 kHz stays on.' Every student then individually writes three predictions in their science notebooks under the headings: (1) What do I think a wave is? (2) Why does the horn pitch change as the matatu passes? (3) Why does FM fade in the tunnel but AM does not? No talking is permitted during individual prediction writing (4 minutes). Students are told: 'There are no wrong answers right now — your prediction is evidence of your thinking before new information.'	standing on the verge of the Nakuru–Nairobi highway at Limuru. A matatu is coming fast from Nakuru.' Play the audio clip at moderate volume. After the clip, say: 'Now imagine you are in that same bus entering the Kabete underpass. Classic 105 goes quiet. The AM station stays.' Pause 5 seconds. Then instruct: 'Open your notebooks. At the top of a fresh page, write today's date and the title: MY FIRST WAVE PREDICTIONS. You have 4 minutes — write silently.' Set a visible 4-minute countdown timer. Circulate and scan notebooks without commenting. After 4 minutes, say: 'Pen down. You will keep these predictions. At the end of our 12 lessons together, you will read them again and see how your thinking has changed.'	creates a cognitive anchor. By committing predictions to paper before instruction, students create a personal baseline they can compare against evolving evidence throughout the unit. Writing — rather than just thinking — forces articulation of half-formed ideas and reveals misconceptions the teacher can address in subsequent lessons.	without interrupting. Observable indicators: (1) Student uses the word 'energy' or 'movement' in their wave definition — indicates productive prior knowledge. (2) Student writes 'I don't know' for all three — flags a student needing targeted support in Phase 2. (3) Student writes a scientifically accurate prediction (e.g., 'FM has a shorter wavelength and cannot bend around the tunnel walls') — identifies a student who can be used as a resource during Explain phase. Teacher mentally notes 2–3 students in each category for strategic cold-calling later.
Observe Phase 📹 **VIDEO:** [Video: Wave on a String](#) Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Landforms from Wave Erosion and Deposition (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	The teacher facilitates a structured Think-Pair-Share using three prompt images displayed on the ARES platform screen: (A) an aerial photograph of ocean waves breaking at Mombasa's Nyali Beach, (B) a time-lapse image of a crowd 'wave' rippling through Kasarani Stadium during an AFC Leopards match, (C) a close-up photograph of ripples spreading outward in a cup of chai after a spoon is tapped against the rim. For each image, students first think silently (30 seconds), then discuss with a shoulder partner (1 minute), then the class shares. Students record in their notebooks: 'What is moving? What is NOT moving? What is being carried from place to place?' After the three images, the teacher introduces the PhET Sound Simulation (linked on the ARES	Display Image A (Nyali Beach waves). Say: 'Look carefully. Think silently — what is moving, what is not?' Wait 30 seconds (WAIT TIME). 'Now tell your partner.' After 1 minute, cold-call 3 pairs: 'Pair at the back left — what did you notice?' After each response, press: 'How do you know the water itself is not travelling from Mombasa to the shore?' Repeat for Images B and C. For the Kasarani Stadium image, prompt: 'The people in the stands are not running around the stadium. So what IS travelling?' Accept: 'The movement — the disturbance — the energy.' Write on the board: DISTURBANCE | ENERGY | MATTER. Circle ENERGY and cross out MATTER. Say: 'Keep that in mind.' For PhET: 'Open the Sound simulation on your ARES device. Click the green Play button	Anchored Observation with Contrasting Cases (Think-Pair-Share + Simulation): By presenting three contrasting real-world examples (ocean, stadium, chai cup) before the simulation, students build the concept of 'travelling disturbance' inductively. The PhET simulation then provides a controlled, particle-level view that is impossible to see in real life. The contrast between the particle's back-and-forth motion and the wave's forward travel is the key conceptual move of this phase — directly setting up the formal definition in Phase 3.	Listen for the phrase 'the particle stays near its original position' or equivalent during the PhET debrief cold-calls. If fewer than 3 of 4 cold-called students articulate this idea correctly, the teacher pauses and replays the simulation in slow motion with the whole class before proceeding. Observable misconception to flag: students who say 'the water travels from Mombasa to the beach' — this suggests a conflation of wave speed with medium transport, a misconception to explicitly address in Lesson 4 (wave properties: speed, frequency, wavelength).

	platform). Students open the simulation and set it to 'Air' medium. They observe: (a) the speaker membrane vibrating, (b) the compression and rarefaction patterns travelling outward, (c) what happens to a 'listener' particle — does it travel with the wave or stay near its rest position? Students sketch what they see and annotate: 'The particle moves BACK AND FORTH but does NOT travel forward with the wave.' Duration: 25 minutes total (15 min images + 10 min PhET).	on the speaker. Watch ONE particle — the red dot. Does it travel? Sketch what you see. You have 8 minutes.' Circulate; if students are watching the wave front instead of a single particle, crouch down and point: 'Watch only the red dot. Where does it go? Where does it come back to?' After 8 minutes, cold-call 4 students: 'Amina — describe the red dot's motion in one sentence.' Require full sentences. WAIT TIME: 10 seconds after each cold-call before accepting an answer.		
Explain Phase 📹 **VIDEO:** [Video: Wave on a String] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Landforms from Wave Erosion and Deposition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of four to co-construct a working definition of a wave. Each group receives a half-A3 card and markers. They must write: (a) a one-sentence definition of a wave, (b) two examples from Kenya, (c) one thing a wave carries, and one thing it does NOT carry. Groups display their cards on the wall (gallery walk style). The class reads all cards silently (2 minutes), then the teacher facilitates a whole-class consensus-building discussion to arrive at a shared class definition. The teacher writes the agreed definition on a dedicated 'Class Science Notebook' section of the board: 'A wave is a disturbance that transfers ENERGY from one point to another WITHOUT transferring matter.' Students copy this into their notebooks and immediately annotate it with one arrow pointing to ENERGY (labelled: 'e.g., sound energy from matatu horn → our ears') and one arrow pointing to WITHOUT transferring matter (labelled: 'e.g., air particles near Kabete tunnel do NOT travel into our ears'). Students then draw and label a	Distribute cards and markers. Say: 'In your group of four, you have 6 minutes. Write your group's best definition of a wave, two Kenyan examples, what it carries, and what it does NOT carry. Go.' Set a 6-minute timer. After groups post cards, give 2 minutes of silent gallery walk. Then: 'I am going to cold-call one person from each group to defend one idea on their card. Not the person who wrote it — someone else.' This prevents one student dominating. Cold-call 4 students. After each, ask the class: 'Do you AGREE or DISAGREE? Thumbs up or thumbs down — everyone.' Use thumbs to gauge consensus. When building the class definition, say: 'I hear the word disturbance from Group 2 and movement from Group 3 — are these the same thing? Talk to your partner.' WAIT TIME: 12 seconds. Cold-call 2 students. Arrive at 'disturbance' as the agreed term. For the transverse wave diagram, say: 'I am drawing this as a Lake Victoria wave. When I say CREST, show me with your hand what part of the wave I mean.' Physical gesture	Collaborative Definition Construction + Gallery Walk + Gesture Check: Having groups generate definitions before being given the 'correct' one ensures students are reasoning rather than copying. The gallery walk exposes students to multiple framings of the same idea, building flexible understanding. The physical gesture check for wave anatomy creates a kinesthetic memory anchor — students who can gesture a crest correctly are more likely to label it correctly in an exam or model.	Collect and photograph group definition cards (or teacher circulates and scores them on a 3-point rubric: 3 = includes energy transfer + no matter transfer + correct Kenyan example; 2 = mentions energy but omits matter; 1 = purely descriptive with no energy language). Observable indicator for mastery: Student's individual wave diagram contains all six labels (crest, trough, amplitude, wavelength, rest position, direction of travel) correctly placed. Teacher spot-checks 6 notebooks during diagram drawing by circulating and placing a small tick or question mark next to the diagram — question marks indicate students to revisit in Phase 4.

	transverse wave diagram: crest, trough, amplitude, wavelength, rest position, direction of wave travel, direction of particle vibration. Teacher models this diagram on the board using a 'wave on Lake Victoria' context: 'Imagine a fishing boat at Kisumu port creates a disturbance — the wave travels outward but the water molecules just bob up and down.'	check — all students raise a hand to chest height (crest) or drop it below desk (trough). Correct any students who gesture incorrectly before they draw.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Wave on a String Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives two sticky notes (or cuts two small squares from paper if sticky notes are unavailable). On sticky note 1 (yellow / lighter colour): Write ONE question this lesson has raised for you about waves. On sticky note 2 (pink / darker colour): Write ONE question you still have specifically about the matatu horn OR the Kabete tunnel FM/AM mystery. Students post their sticky notes on the class Driving Question Board — a large sheet of paper or whiteboard section divided into three columns labelled by the teacher: 'Column A: What IS a wave?', 'Column B: What affects how a wave travels?', 'Column C: Why do sound and radio waves behave differently?' The teacher facilitates a 5-minute live clustering exercise, reading selected sticky notes aloud and asking: 'Does this go in Column A, B, or C? Hands up for A… B… C…' The DQB remains displayed for the entire 12-lesson unit. At the start of each subsequent lesson, the teacher will gesture to the DQB and ask: 'Which questions did today's evidence help us answer?'	Distribute sticky notes / paper squares. Say: 'This board is our shared scientific brain for the next 12 lessons. Every question you write is a REAL question — there are no silly ones. Scientists keep question boards like this.' Give 3 minutes for writing. 'Now bring your notes to the board. Stick them anywhere for now.' After all notes are posted, say: 'I am going to read some of these aloud. Help me sort them.' Read 8–10 notes aloud — select strategically to represent all three emerging categories. For each: 'Column A, B, or C? Show me with your fingers — 1, 2, or 3.' Move note to the majority-indicated column. If a note bridges two columns, say: 'Interesting — Kibaki's question seems to belong in BOTH B and C. Let's put it on the border. That might mean there is a connection we haven't discovered yet.' Point to 2–3 questions that directly connect to the matatu horn or Kabete tunnel phenomenon and say: 'These are our BIG questions. We will come back to them in Lessons 5, 8, and 11. Keep them in mind.'	Driving Question Board (DQB) — Question Clustering: The DQB is a visual, public record of the class's collective curiosity. Clustering questions into themes helps students see the structure of the unit's inquiry before content is formally taught. By attributing questions to students by name ('Kibaki's question'), the teacher signals that student thinking drives the inquiry — not just the textbook. This metacognitive move increases student ownership and motivation across the full 12-lesson arc.	Teacher photographs the DQB at the end of the lesson. Analysis indicators: (1) If more than 70% of sticky notes are in Column A ('What IS a wave?'), students are still at surface level — teacher should plan additional anchoring in Lesson 3. (2) If students generate questions about wavelength, frequency, or the speed of light without prompting, this indicates a cohort with strong prior knowledge — teacher can accelerate coverage of basic definitions in Lesson 3. (3) If no student writes a question specifically about the AM/FM tunnel contrast, the teacher should plant this question on the DQB manually and attribute it: 'I noticed no one has asked yet about AM vs FM in the tunnel — I am adding this one myself, because it is going to be very important in Lesson 9.'
Model Building Phase	Students return to their individual prediction page from Phase 1. They draw a box around their	Say: 'Open to your predictions from the start of class. Draw a neat box around them and write	Running Wave Model (Cumulative Model Revision Strategy): A running model that students	At the end of the lesson, teacher collects 6 Running Wave Models (a rotating sample of 6 different

📹 **VIDEO:** Video: Wave on a String Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	original predictions and label it 'BEFORE — Lesson 2.' Beneath the box, they begin their first entry in their unit 'Running Wave Model' — a dedicated 2-page spread they will update after every lesson. For Lesson 2, the Running Wave Model must include: (1) The class definition of a wave (copied verbatim). (2) An annotated transverse wave diagram (from Phase 3). (3) One sentence connecting the wave model to each part of the anchoring phenomenon: 'The matatu horn creates a sound WAVE — a disturbance in air that carries energy to my ears WITHOUT moving the air itself from the matatu to me.' AND 'The FM radio signal from Classic 105 and the AM signal from 1080 kHz are both WAVES — electromagnetic disturbances carrying energy from a Nairobi transmitter to the bus radio.' (4) One 'still wondering' question that the current model cannot yet explain — students write this in a different colour and circle it. These circled questions will be revisited at the model-revision step in the final lesson (Lesson 12).	BEFORE — Lesson 2 in the margin. You will NEVER erase these — they are scientific evidence of where your thinking started.' Pause to allow boxing. 'Now turn to the next clean spread — two full pages. At the top, write: MY RUNNING WAVE MODEL — Lesson 2.' Display the four required model components on the ARES screen as a checklist. Say: 'You have 8 minutes. Work individually. Use diagrams AND words. Your model should be able to explain the matatu horn and the radio signal — use those exact examples.' Set 8-minute timer. At 6 minutes, say: 'Two minutes left. Make sure you have written your STILL WONDERING question in a different colour and circled it.' At the close, cold-call 3 students to read their 'still wondering' questions aloud. After each, say: 'Write that on a sticky note and add it to the DQB under the correct column.' Conclude: 'In Lesson 3, we will investigate wave properties — amplitude, frequency, wavelength, and speed — and those measurements will let us start answering some of the questions on our board, including why FM and AM waves might behave differently in a tunnel.'	update after every lesson makes the growth of understanding visible and personal. By requiring students to explicitly link the model to the anchoring phenomenon at every update, the teacher ensures that abstract wave physics never becomes disconnected from the matatu horn and Kabete tunnel mystery. The 'before box' and 'still wondering' question in a contrasting colour are metacognitive tools that help students track conceptual change — a core NGSS Science and Engineering Practice (SP6: Constructing Explanations; SP8: Obtaining, Evaluating, and Communicating Information).	students each lesson across the 12-lesson unit — ensure every student's model is collected at least twice). Look for: (1) The class wave definition copied accurately — if absent, student was not engaged during Phase 3 consensus-building. (2) The transverse wave diagram includes all six labels — if missing more than two, student needs a labelled diagram scaffold in Lesson 3. (3) The phenomenon connection sentences use 'energy,' 'disturbance,' and 'without transferring matter' — these three terms are the non-negotiable conceptual anchors for this lesson's SLO. (4) The 'still wondering' question is a genuine question (not 'I wonder what waves are' — which has just been answered) — genuine curiosity questions indicate productive epistemic engagement.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. PREDICTION QUALITY: Were students' initial predictions evidence of genuine prior knowledge (ocean waves, stadium sounds) or were they mostly blank? If mostly blank, plan a brief 'wave experiences in Kenya' carousel activity at the start of Lesson 3 to build the experiential base.

2. DQB CLUSTERING: Did the three DQB columns attract roughly equal numbers of questions? A heavily skewed Column A suggests students are not yet connecting wave properties to the phenomenon — consider replaying the audio clip at the start of Lesson 3 before moving into amplitude and frequency.

3. PARTICIPATION EQUITY: Review your cold-call records. Did you reach students from different ends of the classroom, different genders, and different apparent confidence levels? The goal across 12 lessons is for every student to be cold-called at least twice.

4. PHENOMENON CONNECTION: In the model-building phase, did students use the matatu horn and Kabete tunnel as their examples, or did they revert to generic textbook examples (e.g., 'ripples in a pond')? If generic examples dominate, re-anchor at the start of every subsequent lesson with the phrase: 'How does this connect to what the Form 4 student on the Nakuru bus noticed?'

5. PACING: This lesson is dense (prediction + observation + explanation + DQB + model = 5 phases in 80 minutes). If the PhET simulation took more than 12 minutes, trim the gallery walk in Lesson 3 accordingly. If groups finished Phase 3 early, use the surplus time to extend the DQB clustering rather than rushing into model building.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students listened to an audio clip of the Doppler-shifted matatu horn and the AM/FM Kabete tunnel narration. They observed ocean wave images (Nyali Beach, Mombasa), a stadium crowd wave (Kasarani Stadium), and chai cup ripples. They used the PhET Sound Simulation to watch individual air particles move back and forth while the wave disturbance travels forward.
What did I learn?	A wave is a disturbance that transfers energy from one point to another without transferring matter. Transverse waves have crests, troughs, amplitude, wavelength, a rest position, and a direction of travel distinct from the direction of particle vibration. Both the matatu horn sound and the Classic 105 FM / AM radio signals are waves — disturbances that carry energy from a source to a receiver.
How does this explain the phenomenon?	Students began to explain why the horn's pitch change and the FM/AM tunnel contrast are both wave phenomena by connecting the class wave definition to each part of the anchoring phenomenon. Their Running Wave Model (Lesson 2 entry) documents this first explanation attempt, and their 'still wondering' questions on the DQB identify the gaps — particularly around frequency, wavelength, and why different wave types behave differently in confined spaces like the Kabete tunnel — that subsequent lessons (3–12) will resolve.

LESSON 3 (40 minutes): The Wave Equation: Connecting Speed, Frequency, and Wavelength	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will derive and apply the wave equation v = f lambda using empirical data from PhET simulations and solve problems set in Kenyan contexts, advancing their explanation of why FM and AM signals behave differently in the Kabete tunnel.
Knowledge	Students will state the wave equation v = f lambda and explain the mathematical relationship between wave speed, frequency, and wavelength for any wave type.
Skills	Students will gather quantitative data from PhET: Wave on a String, calculate wave speed from measured frequency and wavelength values, and solve multi-step problems involving the wave equation in Kenyan contexts.
Attitudes	Students will demonstrate precision in measurement and recording, and show curiosity by connecting calculated wavelengths to the real-world sizes of the tunnel and radio waves.
Key Inquiry Question	How do waves behave in different media and under different conditions? — specifically, how does the relationship between speed, frequency, and wavelength determine whether a wave can navigate an obstacle like the Kabete tunnel?
Purpose in Storyline	This is the quantitative pivot lesson. Students have qualitatively identified wave properties; now they attach numbers to those properties. The wavelengths they calculate for Classic 105 FM (about 2.85 m) and a typical AM station (about 300 m) will become the critical evidence in Lesson 6 when diffraction is used to fully resolve the radio-tunnel mystery. This lesson also provides the mathematical tool needed for the Doppler calculation in Lesson 9.
Safety Notes	No physical hazards. Ensure screen brightness is comfortable for all students. Students with visual impairments should be seated nearest the projected simulation. All PhET work is browser-based; no downloads required on school devices.

B. LESSON OVERVIEW

Students open the lesson by making a written prediction about what happens to wavelength when frequency is doubled at constant speed. They then use PhET: Wave on a String in a structured data-collection activity, recording at least four frequency-wavelength pairs and calculating wave speed each time to discover the constant relationship v = f lambda. The equation is formalised and applied to two Kenyan problems: the Classic 105 FM wavelength calculation and a Lake Victoria fisherman sonar problem. The lesson closes by updating the Driving Question Board with the new quantitative evidence — students can now state specific wavelengths for FM and AM signals and begin to suspect that size relative to the tunnel opening matters.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Human activities that threaten biodiversity]	Students read the following prompt displayed on the board: 'In Lesson 2 you identified wavelength and frequency on a transverse wave. If the wave travels at a fixed speed	Display the prompt and the two wave diagrams side by side. Circulate quietly during the 2-minute individual writing time — do not offer hints. After partner	Predict-then-test anchoring: written individual prediction followed by paired discussion creates cognitive commitment that makes the subsequent data-collection phase	Teacher scans written predictions during circulation. Look for whether students correctly anticipate an inverse relationship or whether they conflate frequency

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	and you increase the frequency — producing more vibrations per second — what do you think happens to the wavelength? Write your prediction in one sentence and draw a quick sketch comparing a low-frequency wave and a high-frequency wave travelling at the same speed.' Students write individually for 2 minutes, then share with a shoulder partner. The teacher asks three pairs to share aloud. Students also revisit their Lesson 1 predictions about FM and AM signals and note whether they have any new ideas about why the two signals behave differently.	sharing, cold-call three pairs using the class register (not volunteers only). Quote to use: 'You said the wavelength gets shorter — what makes you think that? Does anyone predict the opposite?' WAIT TIME: after asking for the opposite prediction, wait a full 8 seconds in silence before accepting a response. Record each prediction verbatim on the board under the heading 'Our Predictions — Lesson 3'. Do not evaluate correctness yet. Then say: 'We are going to test every one of these predictions with real data today. Keep your prediction written down so you can revisit it at the end.'	more meaningful. Predictions are publicly displayed so students feel accountable to evidence.	with speed. Common misconception to flag: 'higher frequency means faster wave' — note any students who write this so they can be targeted during the Observe phase.
Observe Phase 📹 **VIDEO:** Human activities that threaten biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open PhET: Wave on a String on school computers or tablets (or observe the teacher projection if devices are limited). They set the simulation to 'Oscillate' mode, 'No End' boundary, and turn on the ruler and timer tools. Working in pairs, they complete a structured data table in their exercise books with columns: Frequency (Hz) | Wavelength measured (m, using the on-screen ruler) | Calculated Speed (m/s). They take readings at four different frequency settings — for example 1.0 Hz, 1.5 Hz, 2.0 Hz, and 2.5 Hz — keeping the tension slider constant each time. They measure the wavelength by counting the distance between two consecutive crests using the PhET ruler. They then compute speed as wavelength multiplied by frequency for each row. Teacher pauses the class after the first row is complete and asks: 'What pattern are you seeing in the speed column?' Students complete	Before students begin, demonstrate one complete measurement on the projected simulation: set frequency to 1.0 Hz, pause the wave, use the ruler to measure from one crest to the next, record in the table. Say: 'I am measuring crest to crest — that is the definition of wavelength from Lesson 2. Your job is to repeat this at four different frequencies.' Circulate during data collection. When a pair reports a significantly inconsistent speed value, ask: 'How carefully did you place the ruler? Let us remeasure together.' After all pairs have at least three rows, pause the class and say: 'Look at your speed column — what do you notice? Discuss with your partner for 30 seconds.' WAIT TIME: after posing this to the whole class, wait 30 seconds of partner talk, then cold-call. Quote to use when a student identifies the constant speed: 'Excellent observation. The speed is staying roughly the same even though the	Empirical data collection before equation introduction reverses the typical tell-then-verify sequence. Students experience the invariance of wave speed firsthand, making v = f lambda feel like their own discovery rather than a formula imposed from outside.	Teacher checks data tables during circulation. Correct range for PhET Wave on a String at default tension: wave speed approximately 1.5 to 2.0 m/s (simulation units). Look for: (1) consistent speed values across rows indicating careful measurement; (2) students correctly identifying the inverse relationship between f and lambda; (3) students who have the frequency and wavelength columns swapped — redirect immediately by pointing to the PhET frequency slider label.

	all four rows and write a one-sentence conclusion about what stays constant.	frequency changed. So what must be happening to the wavelength each time frequency goes up?' WAIT TIME: 6 seconds.		
Explain Phase 📹 **VIDEO:** Human activities that threaten biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formalises the wave equation. Students copy into their notebooks: v = f lambda, where v is wave speed in metres per second, f is frequency in hertz, and lambda is wavelength in metres. The teacher works through two Kenyan context problems on the board, and students then solve two further problems independently. Problem 1 (whole class, teacher-led): 'Classic 105 FM broadcasts at 105.2 MHz. Radio waves travel at the speed of light, 3.0 times 10 to the power 8 m/s. Calculate the wavelength of the Classic 105 FM signal.' Solution displayed step by step: lambda = v divided by f = 3.0 times 10 to the power 8 divided by 105.2 times 10 to the power 6 = approximately 2.85 m. Teacher asks: 'Hold up your hands about 2.85 metres apart. That is the length of one complete FM radio wave. Now think about the Kabete tunnel opening — is this wave large or small compared to the tunnel?' Problem 2 (teacher-led): 'A fisherman on Lake Victoria uses a sonar device that emits a pulse at 40 kHz. Sound travels at 1500 m/s in water. Calculate the wavelength of this sonar pulse.' lambda = 1500 divided by 40 000 = 0.0375 m or 3.75 cm. Students then solve two independent problems: (A) 'A typical AM station broadcasts at 567 kHz. Calculate its wavelength in air.' (Answer: approximately 529 m.) (B) 'A student in Kisumu taps a metal school desk. The sound wave travels through the desk at 5100	Write the equation on the board and say: 'This equation is not something I am giving you — your data table just proved it. Wave speed equals frequency multiplied by wavelength.' Underline the equation. Work Problem 1 slowly, narrating each algebraic step: 'I am rearranging v = f lambda to get lambda equals v divided by f.' After the 2.85 m answer, physically gesture with arms spread wide and say: 'This is one FM wavelength. Is that bigger or smaller than the Kabete tunnel entrance? Keep that thought — we will need it in Lesson 6.' For Problem 2, invite a student volunteer to attempt the calculation at the board. Coach with: 'What is your v? What is your f? Now divide.' For independent problems A and B, set a timer for 4 minutes. After time is up, cold-call two different students (one per problem) to read their working aloud before revealing the answer. Quote for wrong answers: 'Thank you for sharing. Let us trace through the steps together and find where the path changed.' WAIT TIME: before revealing answers, wait 5 seconds after asking 'Has everyone got a number?' to allow slower workers to complete their attempt.	Contextualised problem-solving: both teacher-led examples use Kenyan real-world sources (Classic 105, Lake Victoria sonar) so the equation feels functional rather than abstract. The physical gesture for 2.85 m builds spatial intuition that will anchor the diffraction lesson. Peer marking of independent problems creates immediate feedback.	Circulate during the 4-minute independent problem time. Target students who struggled with unit conversion (MHz to Hz, kHz to Hz) — this is the most common error. After marking, ask a show of hands: 'Who got exactly 529 m for problem A?' If fewer than 70 percent of hands are raised, reteach unit conversion immediately using the board before moving on. Exit-check: ask three students at random to state the wave equation without looking at their notes.

	m/s and has a frequency of 1000 Hz. Calculate the wavelength.' (Answer: 5.1 m.) Students mark their own work when answers are revealed.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Human activities that threaten biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board displayed at the front. The teacher reads aloud the original driving question about the matatu horn and the Kabete tunnel radio signal. Students are given two sticky notes each: one green (new evidence) and one yellow (new question). On the green note, each student writes one specific quantitative fact they have just learned — for example 'Classic 105 FM wavelength is about 2.85 m' or 'AM 567 kHz wavelength is about 529 m.' On the yellow note, students write a new question triggered by the lesson — for example 'If the FM wavelength is 2.85 m and the tunnel is wider than that, why does it still fade?' Students place both sticky notes on the DQB. The teacher reads three green notes and three yellow notes aloud to the class and paraphrases: 'We now have numbers attached to our wave properties. The FM wavelength is about 2.85 m, and the AM wavelength is about 529 m. Those two numbers are very different. That difference is going to matter — a lot. Hold that question about the tunnel width, because it is exactly where we are headed in Lesson 6.'	Point to the DQB and say: 'In Lesson 1 you put questions up here based on what puzzled you. In Lesson 2 you added diagrams. Today you are adding numbers — real measurements of real Kenyan radio waves.' After students post their sticky notes, read three green and three yellow notes aloud. Deliberately choose notes that show a range of understanding. For a yellow note that asks a sophisticated question — for example comparing tunnel width to wavelength — say: 'This student is already thinking like a physicist. That question is on our roadmap. We will answer it in Lesson 6.' Do not answer the diffraction question now; preserve the anticipation. End by pointing to the original driving question text and saying: 'Can we fully answer the tunnel mystery yet?' Pause 5 seconds. 'Not yet — but today we have the numbers we need. We are getting closer.'	The DQB update ritualises the evidence-accumulation narrative of the storyline. Green notes externalise new knowledge; yellow notes sustain productive inquiry. Reading notes aloud models scientific communication and validates student thinking publicly.	Collect and read all green sticky notes after the lesson to assess whether students have accurately recorded wavelength values and can distinguish FM from AM numerically. Yellow notes reveal which students are making the conceptual leap toward diffraction — these students can be strategically partnered or cold-called in Lesson 6.
Model Building Phase 📹 **VIDEO:** Human activities	Students open the wave-property diagram they began in Lesson 2. They add a new section labelled 'The Wave Equation: v = f lambda' and draw two annotated side-by-	Circulate and check that students are drawing wave diagrams to a relative scale — the FM wave should have noticeably shorter wavelength spacing than the AM	Cumulative model building makes learning visible over the unit. Adding the equation and the two wavelength values to the existing diagram reinforces the connection	Check that model diagrams show: (1) the wave equation written correctly; (2) FM and AM waves drawn at visibly different wavelength scales; (3) correct

that threaten biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	side wave diagrams: one labelled 'Classic 105 FM: f = 105.2 MHz, lambda approximately 2.85 m, v = 3.0 times 10 to the power 8 m/s' and one labelled 'AM 567 kHz: f = 567 kHz, lambda approximately 529 m, v = 3.0 times 10 to the power 8 m/s'. The FM wave should visibly show many short wavelengths; the AM wave should show very long wavelengths. Students label: frequency increases — wavelength decreases — speed stays the same (for the same medium). They draw a small sketch of the Kabete tunnel and annotate it with a question mark and the note: 'FM lambda = 2.85 m: can this wave get through? AM lambda = 529 m: can this wave get through? — To be answered in Lesson 6.' Students who finish early begin a third diagram showing the Lake Victoria sonar wave with its calculated wavelength.	wave. If a student draws both waves with equal spacing, say: 'Your wavelengths are the same size on the page, but you just calculated them to be 2.85 m and 529 m. Can you show that difference in your drawing?' Praise accurate relative scaling explicitly: 'I can see you have drawn many FM crests in the same space as one AM crest — that is exactly right.' For the Kabete tunnel annotation, prompt: 'You do not need to answer the question yet — just make sure the question is clearly written on your model so you remember to come back to it.' In the final 2 minutes, ask the class to hold up their models and do a quick visual scan. Identify one exemplary model and ask that student to briefly describe one feature they included that others might add.	between qualitative wave properties (Lesson 2) and quantitative relationships (Lesson 3). The tunnel annotation with a question mark sustains productive uncertainty and motivates Lesson 6.	numerical values for frequency and wavelength; (4) the Kabete tunnel sketch with an open question annotated. Students whose models lack the tunnel question mark may not yet be connecting the equation to the anchoring phenomenon — provide a verbal prompt before they leave the lesson.

D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did at least 70 percent of students correctly solve both independent problems, including the unit conversion from MHz and kHz to Hz? If not, plan a 5-minute unit-conversion warm-up at the start of Lesson 4. (2) Did students make the connection between the calculated wavelengths and the tunnel scenario, or did the numbers feel abstract? If the tunnel connection was weak, open Lesson 4 by revisiting the two wavelength values before introducing rectilinear propagation. (3) Were the PhET measurements consistent enough across pairs to confidently derive v = f lambda? If data scatter was large, consider pre-setting four frequency values on the simulation before the lesson to ensure cleaner results. (4) Which students placed sophisticated yellow sticky notes on the DQB — those already asking about diffraction or tunnel width? Note these students as candidates for extended challenge tasks in Lesson 6. (5) Did the lesson stay within 40 minutes? The Explain phase most commonly overruns — if so, move Problem B to the start of Lesson 4 as a warm-up.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Using PhET: Wave on a String, students observed that when frequency was increased while wave speed was held constant, the wavelength decreased in proportion, and that the product of frequency and wavelength remained approximately equal to the wave speed across all four data rows.
What did I learn?	The wave equation v = f lambda shows that wave speed equals frequency multiplied by wavelength. For radio waves travelling at

	the speed of light, Classic 105 FM at 105.2 MHz has a wavelength of approximately 2.85 m, while a typical AM signal at 567 kHz has a wavelength of approximately 529 m — nearly 200 times longer.
How does this explain the phenomenon?	The wave equation gives us the numerical tool to compare FM and AM waves. The very different wavelengths of FM (about 2.85 m) and AM (about 529 m) are our first quantitative clue to explain why the two signals behave differently in the Kabete tunnel. We cannot fully explain the tunnel mystery yet — that requires understanding diffraction in Lesson 6 — but we now have the numbers we need.

LESSON 4 (80 minutes): Wave Types and Properties: Transverse, Longitudinal, and the Language of Waves

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to distinguish between transverse and longitudinal waves, identify and measure wave properties (amplitude, wavelength, frequency, period, wave speed), and connect these properties to the anchoring phenomenon of the matatu horn and the FM radio signal in the Kabete tunnel.
Knowledge	Learners will know: (a) the distinction between transverse waves (oscillation perpendicular to direction of travel) and longitudinal waves (oscillation parallel to direction of travel); (b) the definitions of amplitude, wavelength, frequency, period, and wave speed; (c) the relationship $v = f\lambda$; (d) that sound waves are longitudinal and radio/FM waves are transverse (electromagnetic).
Skills	Learners will be able to: (a) use the PhET Wave on a String simulation to measure wavelength and amplitude from a wave diagram; (b) calculate wave speed using $v = f\lambda$; (c) represent transverse and longitudinal waves with accurate diagrams; (d) classify the matatu horn sound and the FM radio signal by wave type.
Attitudes	Learners will appreciate that precise scientific vocabulary is a tool for unlocking explanations of everyday Kenyan experiences, and will demonstrate curiosity and persistence when navigating the PhET simulation to gather evidence.
Key Inquiry Question	What are the properties of waves, and how do those properties differ between the type of wave that carries a matatu horn's sound and the type that carries an FM radio signal?
Purpose in Storyline	Lesson 4 is the first dedicated evidence-gathering lesson in the wave-properties sequence. Having puzzled over the anchoring phenomenon in Lessons 1–3, students now build the shared scientific vocabulary and wave-type knowledge they will need to explain both mysteries. By the end of this lesson, students can place the matatu horn (longitudinal, mechanical) and the Classic 105 FM signal (transverse, electromagnetic) correctly on their class Driving Question Board, and they begin the first version of their cumulative wave-property diagram model.
Safety Notes	No hazardous materials are used. If a physical slinky is used for the longitudinal wave demonstration, ensure adequate floor space (at least 3 metres) and that students do not overstretch or release the slinky suddenly. Students working on shared devices should handle hardware carefully. Remind learners not to increase simulation speaker volume to uncomfortable levels if the PhET Sound add-on is used.

B. LESSON OVERVIEW

Learners open this lesson by revisiting the anchoring phenomenon — the pitch-shifting matatu horn and the fading FM signal — and making a targeted prediction: are these two waves the same type? They then gather evidence using the PhET Wave on a String simulation (transverse waves) and a teacher-led slinky demonstration (longitudinal waves), identifying amplitude, wavelength, frequency, period, and wave speed. A structured vocabulary-building routine (Frayer models for each term) consolidates language. Learners apply the wave-speed equation $v = f\lambda$ using Kenyan broadcast data (Classic 105 at 105.2 MHz). The lesson closes with students making their first entry on the class Driving Question Board and sketching the initial version of their cumulative wave-property diagram model, explicitly labelling which wave type carries the matatu horn and which carries the FM signal.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Video: Wave on a String] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Transverse Wave (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners read a two-sentence recap of the anchoring phenomenon displayed on the board: 'A Form 4 student hears the matatu horn change pitch on the Nakuru–Nairobi highway. Inside the Kabete tunnel, the FM signal on Classic 105 (105.2 MHz) cuts out but the AM signal survives.' Each learner writes a silent individual response (2 minutes) in their science notebook to the prompt: 'Do you think sound from the matatu horn and the FM radio signal are the same type of wave, or different types? Draw a quick sketch of what you imagine each wave looks like as it travels.' Learners then share with a shoulder partner (Think-Pair) for 90 seconds before three pairs are cold-called to share with the class. Predictions are left unresolved and posted on a 'Our Predictions' sticky-note column on the Driving Question Board.	Display the phenomenon recap on the board. Say exactly: 'Before we gather any new evidence today, I want you to commit to a prediction in writing — no right or wrong answers yet.' Set a visible 2-minute timer. Circulate silently; do NOT correct or affirm any sketch. After 2 minutes, say: 'Turn to your shoulder partner. You have 90 seconds — share your sketch and explain your reasoning.' After 90 seconds, cold-call exactly 3 pairs using the class random-name stick. For each pair, ask: 'What did you predict, and what is your reason?' Allow 10 seconds of WAIT TIME after each response before moving on. Do not evaluate answers — record key words from each response on the board under 'Predictions.' Say: 'We will test every one of these predictions with evidence today.'	Think-Pair-Share with written commitment: Requiring learners to write before discussing prevents anchoring to the loudest voice and surfaces genuine prior conceptions about wave types — which the teacher can address specifically during the Explain phase.	Observable indicator: Every learner has a sketch and at least one written sentence in their notebook before pair discussion begins. Teacher scans for the key misconception — learners who draw both waves identically (e.g., both as identical sine curves) are flagged for targeted questioning during the Observe phase. Cold-call responses reveal whether learners distinguish 'sound' and 'radio' as different phenomena at all.
Observe Phase 📹 **VIDEO:** [Video: Wave on a String] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Transverse Wave (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners gather evidence in two parts. PART A (20 min) — PhET Wave on a String: Working in pairs on devices, learners open the PhET Wave on a String simulation. They follow a structured observation guide (printed or on-screen): (1) Set to 'Oscillate' mode, 'No End.' Observe the wave shape. Sketch it in notebooks and label the direction the string moves versus the direction the wave travels. (2) Increase amplitude using the slider — record: what changes? What stays the same? (3) Increase frequency — record the same. (4) Freeze the wave using the pause button and use the on-screen ruler to measure	For Part A: Distribute or display the structured observation guide. Say: 'Your job is not to play — your job is to be a scientist collecting data. Every number goes in the table; every observation gets a sketch.' Circulate every 3 minutes. At each pair, ask one probing question chosen from: 'What happened to the wavelength when you increased the frequency?' / 'What stayed constant when you changed the amplitude?' / 'Which direction is the string moving right now?' Allow 8 seconds of WAIT TIME for each question. If a pair is stuck on measuring wavelength, say: 'A full wavelength goes from	Structured Data Table + Annotated Sketch: The combination of quantitative measurement (wavelength, frequency, calculated speed) and qualitative labelled sketches forces learners to engage both halves of scientific observation — pattern recognition through numbers and conceptual representation through diagrams. The tape on the slinky coil makes the abstract 'direction of particle oscillation' concrete and visible.	Observable indicators: (1) Every pair's data table has at least three rows of frequency-wavelength-speed data from the PhET simulation. (2) Every learner's notebook has a labelled slinky sketch with 'compression' and 'rarefaction' marked. (3) During cold-call on slinky direction, correct response ('coils move back and forth in the SAME direction as the pulse') is given by at least 3 of 4 students — if not, teacher immediately re-runs the demonstration with a second pulse before proceeding.

	one full wavelength in cm. Record three trials at different frequencies and complete the table: Frequency (Hz) | Wavelength (cm) | Calculated speed $v = f\lambda$ (cm/s). (5) Answer in notebooks: 'In this wave, does the string move in the same direction as the wave travels, or perpendicular to it?' PART B (10 min) — Slinky Longitudinal Wave Demonstration: Teacher and one volunteer student stretch a slinky along the floor at the front of the room. Teacher sends a compression pulse by pushing one end forward quickly. Learners observe from their seats and answer in notebooks: 'Sketch what the slinky looks like. Mark where it is compressed and where it is stretched (rarefied). Which direction do the coils move? Which direction does the pulse travel?' A second volunteer holds a small piece of tape on one coil so the class can track individual coil motion.	one crest to the very next crest — use the ruler and count in cm.' For Part B: Before starting, say: 'Watch the tape on that single coil. I want you to track where it goes.' After the demonstration, ask the class (whole-group, hands down, cold-call 4 students): 'Tell me: which direction did the tape move?' Then ask: 'Which direction did the pulse travel down the slinky?' After responses, say: 'This is your evidence. Write it down — we will use it in Phase 3.'		
Explain Phase 📹 **VIDEO:** Video: Wave on a String Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Transverse Wave (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Vocabulary Frayer Models (12 min): Learners construct Frayer models (Definition / Characteristics / Example / Non-Example) for five terms in their notebooks: amplitude, wavelength, frequency, period, wave speed. They work individually for 6 minutes using their simulation data as evidence, then cross-check with their partner for 2 minutes, then the class builds a shared master Frayer model for 'wavelength' on the board (4 minutes) as a model. PART B — Wave Type Classification (8 min): Teacher displays a T-chart: 'Transverse Waves | Longitudinal Waves.' The class co-constructs definitions using evidence from Parts A and B	For Frayer Models: Circulate and check that 'example' cells contain Kenyan contexts (e.g., wavelength example: 'the distance between two consecutive crests of the sound wave from a matatu horn'). If learners write abstract examples, redirect: 'Can you connect this to something from our journey from Nakuru?' For the T-chart: Build it by asking, not telling. Say: 'From your PhET sketch — in a transverse wave, which direction does the string move relative to the wave's travel?' Cold-call 2 students. WAIT TIME: 12 seconds. For the $v = f\lambda$ calculations: After the volunteer writes the FM solution, say: 'Class, check their working — do you agree? Thumbs	Frayer Model Vocabulary Routine + Cognitive Tension Hold: Frayer models anchor technical vocabulary to learner-generated examples before formal instruction, reducing rote memorisation. Deliberately withholding the tunnel explanation after presenting the FM vs AM wavelength data maintains productive cognitive dissonance — learners carry an unresolved question into Phase 4, which drives deeper engagement with the DQB.	Observable indicators: (1) All five Frayer models are complete with a Kenyan-context example in each. (2) In the T-chart, learners correctly place 'matatu horn' under Longitudinal and 'FM signal' under Transverse without prompting. (3) FM wavelength calculated correctly as ≈ 2.85 m (accept 2.8–2.9 m); AM wavelength calculated correctly as ≈ 333 m (accept 330–340 m). Teacher circulates and ticks off correct answers during individual work time. Any learner with an incorrect $v = f\lambda$ calculation receives immediate peer-correction before the class moves on.

	of the Observe phase. Learners then answer in notebooks: 'Is the matatu horn's sound wave transverse or longitudinal? What is your evidence?' and 'Is the FM signal from Classic 105 transverse or longitudinal? What is your evidence?' PART C — Applying $v = f\lambda$ with Kenyan Data (8 min): Teacher presents: 'Classic 105 broadcasts at 105.2 MHz. The speed of all electromagnetic waves in air is 3×10^8 m/s. Calculate the wavelength of the Classic 105 FM signal.' Learners solve individually, then a volunteer writes working on the board. Teacher then says: 'A standard AM station in Nairobi broadcasts at 900 kHz. Calculate that wavelength too.' Learners compare the two wavelengths and record: 'Which wave has the longer wavelength — FM or AM? How might this connect to what happened in the Kabete tunnel?'	up or thumbs down silently.' Address any thumbs-down. After the AM calculation, say: 'Look at your two wavelengths. The FM wavelength is roughly 2.85 m. The AM wavelength is roughly 333 m. Keep that difference in mind — we will come back to it.' Do NOT yet explain the tunnel — let learners sit with the cognitive tension.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Wave on a String Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Transverse Wave (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the ORANGE sticky note, they write one new piece of evidence they gathered today that helps answer the driving question (e.g., 'The matatu horn is a longitudinal wave — air particles vibrate in the same direction as the sound travels'). On the BLUE sticky note, they write one new question this lesson has raised for them (e.g., 'Why does a shorter wavelength wave get blocked by the tunnel but a longer one doesn't?' or 'Does the speed of the matatu change the wavelength or the frequency of the horn?'). Learners post both sticky notes on the class DQB in the designated 'Lesson 4 Evidence' column and 'New Questions' column	Distribute sticky notes. Say: 'Your orange note must contain specific evidence — not just a feeling or guess. It must include a wave property we named today.' Allow 3 minutes of silent writing. Then say: 'Post your notes now — orange in Evidence, blue in New Questions.' After posting, say: 'You have 3 minutes to walk along the board. Read quietly. Put a dot on any question that is also in your head.' After the gallery walk, read the top 3 dotted questions aloud. For each, say: 'We do not have the answer to this yet — and that is exactly why we are scientists. We will add this to our investigation plan.' Circle the three questions on the DQB with a red marker. Say: 'Notice that two of these questions	Two-Colour Sticky Note DQB Routine: Separating 'evidence gathered' (orange) from 'new questions raised' (blue) trains learners in the disciplinary habit of distinguishing claims from questions — a core science and engineering practice. The dot-vote surfaces collective curiosity and gives the teacher real-time data on which phenomena are most puzzling to the class.	Observable indicators: (1) Every orange sticky note references at least one named wave property from today's lesson (amplitude, wavelength, frequency, period, wave speed, transverse, or longitudinal). (2) Every blue sticky note is phrased as a genuine question (contains a question mark and is not a restatement of a fact). (3) Teacher reads the three top-dotted questions — if none relate to the anchoring phenomenon, teacher prompts: 'Does anyone have a question about the matatu horn or the Kabete tunnel specifically?' and adds a teacher-generated card if needed to ensure the phenomenon remains the anchor.

	respectively. The class conducts a 3-minute 'gallery walk' along the DQB, silently reading three other learners' sticky notes and putting a small dot sticker on any question they also have. The three most-dotted questions are read aloud by the teacher and left explicitly unresolved — flagged as 'Questions we will investigate next.'	are things we can actually test or calculate. That is what we will do in Lessons 5 and 6.'		
Model Building Phase 📹 **VIDEO:** Video: Wave on a String Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Transverse Wave (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open a fresh page in their science notebooks labelled 'Wave Property Diagram — Version 1 (Lesson 4).' They draw and annotate two side-by-side diagrams: (1) A transverse wave (labelled: crest, trough, amplitude, wavelength, direction of wave travel, direction of particle oscillation). They write inside the diagram box: 'Example: FM signal from Classic 105 (105.2 MHz), $\lambda \approx$ 2.85 m.' (2) A longitudinal wave with compression and rarefaction zones marked (labelled: compression, rarefaction, wavelength, direction of wave travel, direction of particle oscillation). They write inside: 'Example: Sound from matatu horn on Nakuru–Nairobi highway.' Below both diagrams, learners write the wave speed equation $v = f\lambda$ and one worked example using Classic 105 data. At the bottom of the page, they write the 'Version 1 Limitations' box: 'What my model does NOT yet explain: (a) Why the pitch of the matatu horn changes as it passes. (b) Why the FM signal disappears in the Kabete tunnel but AM does not.' This limitations box is explicitly left blank-space for future additions — it will be updated in every subsequent lesson.	Say: 'Every great scientist keeps a running model — a diagram they update as they learn more. Today you are starting Version 1. It will be incomplete, and that is fine. In fact, I want you to tell your model what it cannot yet explain.' Display a partially completed teacher model on the board (with one label missing) and say: 'I have deliberately left one label off my model. Can someone tell me which label is missing and where it goes?' Cold-call 1 student. WAIT TIME: 10 seconds. After the student answers, say: 'Correct — add that label to your own diagram now.' Circulate for the final 5 minutes. Check that EVERY diagram has: two wave types, at least 5 labels each, the $v = f\lambda$ equation with correct worked example, AND the Version 1 Limitations box with both bullet points written. For any learner who finishes early, say: 'Add one more thing to your model: mark on the transverse wave diagram where you would measure the amplitude if the wave were the Classic 105 signal at maximum broadcast power.'	Cumulative Annotated Model with Explicit Limitations: Requiring learners to articulate what their model does NOT yet explain is a deliberate epistemic practice drawn from NGSS Science and Engineering Practice 2 (Developing and Using Models). It prevents premature closure, keeps the anchoring phenomenon alive as an unresolved driver, and creates a personal intellectual 'to-do list' that motivates future lessons.	Observable indicators: (1) Both wave-type diagrams are present and contain all required labels (crest/trough or compression/rarefaction, amplitude, wavelength, direction arrows for wave travel and particle motion). (2) The Kenyan example (Classic 105, matatu horn) is written inside each diagram box. (3) The $v = f\lambda$ worked example yields $\approx$ 2.85 m for the FM signal. (4) The Version 1 Limitations box contains BOTH bullet points in the learner's own words. Exit criterion: Teacher will not dismiss the class until a quick show-of-hands confirms that every learner has both diagrams labelled and the Limitations box filled — any learner who has not reached this point is given the last 2 minutes of class time with direct teacher or peer support.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal: (1) During the PhET simulation, did pairs engage with the structured observation guide as scientists (recording data in the table) or as players (exploring randomly)? If the latter, what structured scaffold — such as a pre-filled partial table or a timed checkpoint — would redirect behaviour next time? (2) When cold-called on the slinky direction question, how many of the 4 students gave the correct answer on the first attempt? If fewer than 3 were correct, consider beginning Lesson 5 with a repeat slinky demonstration before any new content. (3) Review the orange sticky notes on the DQB: did learners use precise vocabulary (e.g., 'longitudinal,' 'wavelength,' 'compression') or vague language (e.g., 'the wave goes that way')? The ratio of precise to vague language is your proxy for vocabulary acquisition. (4) Were the Version 1 Limitations boxes completed with both bullet points, or did learners leave them blank? A blank box suggests learners do not yet see the model as a living document — consider a brief norm-setting discussion at the start of Lesson 5. (5) How equitably did you distribute cold-calls across gender, seating zones, and confidence levels? Review your tally and adjust in the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students used the PhET Wave on a String simulation to observe transverse waves and watched a slinky demonstration of longitudinal waves. They measured wavelength using the on-screen ruler at multiple frequencies, recorded data in a structured table, tracked individual coil motion in the slinky using a tape marker, and posted evidence and new questions on the class Driving Question Board.
What did I learn?	Students learned that waves are classified as transverse (particle oscillation perpendicular to wave travel, e.g., the FM radio signal from Classic 105) or longitudinal (particle oscillation parallel to wave travel, e.g., the sound from the matatu horn). They defined and applied amplitude, wavelength, frequency, period, and wave speed, and used $v = f\lambda$ to calculate that the Classic 105 FM signal has a wavelength of approximately 2.85 m while a 900 kHz AM signal has a wavelength of approximately 333 m.
How does this explain the phenomenon?	The matatu horn produces a longitudinal mechanical wave — air particles compress and rarefy in the same direction as the sound travels from the matatu to the student's ear. The Classic 105 FM radio signal is a transverse electromagnetic wave with a wavelength of about 2.85 m. The dramatic difference in wavelength between FM (≈ 2.85 m) and AM (≈ 333 m) is now visible in the data, though the full explanation of why this causes the FM signal to disappear in the Kabete tunnel — involving wave diffraction around obstacles — remains an open question for the next lessons.

LESSON 5 (80 minutes): The Wave Equation: Connecting Frequency, Wavelength, and Speed ($v = f\lambda$)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to derive, apply, and interpret the wave equation $v = f\lambda$ using real-world Kenyan wave phenomena, building toward a full explanation of why AM and FM signals behave differently inside the Kabete tunnel.
Knowledge	Learners will know that: (a) the speed of a wave equals the product of its frequency and wavelength ($v = f\lambda$); (b) for a given medium, wave speed is constant, so frequency and wavelength are inversely proportional; (c) FM radio waves (e.g., Classic 105 at 105.2 MHz) have a much shorter wavelength than AM radio waves, which directly determines their ability to diffract around tunnel walls.
Skills	Learners will be able to: (a) derive the wave equation $v = f\lambda$ from definitions of frequency and wavelength; (b) calculate wave speed, frequency, or wavelength given two of the three quantities; (c) use PhET Sound Simulation and LabXchange Wave Interference tools to gather quantitative evidence linking frequency to wavelength; (d) construct and annotate a scaled wave diagram comparing AM and FM wavelengths.
Attitudes	Learners will appreciate that: (a) a single elegant equation connects measurable wave properties across all wave types; (b) mathematical relationships in Physics explain everyday Kenyan experiences such as radio reception and the sound of a matatu horn; (c) precision and accuracy in measurement and calculation matter in real engineering contexts such as radio tower placement in Turkana County.
Key Inquiry Question	How do the speed, frequency, and wavelength of a wave relate to each other — and how does this relationship explain why Classic 105 FM cuts out inside the Kabete tunnel while AM stays audible?
Purpose in Storyline	In Lessons 1–4, learners established that waves transfer energy, identified wave properties (amplitude, frequency, wavelength, period), and observed reflection and diffraction qualitatively. Lesson 5 is the quantitative pivot: learners derive and wield $v = f\lambda$ as a mathematical tool, calculate the actual wavelengths of the FM and AM signals from the anchoring phenomenon, and use those numbers to begin explaining the tunnel mystery quantitatively. This sets up Lessons 6–7, which will treat diffraction and interference in depth.
Safety Notes	No hazardous materials or high-voltage equipment are used in this lesson. Learners using laptop/tablet screens should maintain appropriate viewing distance. If the PhET simulation is run on shared devices, ensure each device is sanitised before group rotation. Remind learners not to share earphones during the Sound simulation activity.

B. LESSON OVERVIEW

Learners begin by predicting how changing a wave's frequency affects its wavelength and speed, using the matatu horn and the Kabete tunnel as motivating contexts. They then gather quantitative evidence through the PhET Sound Simulation (manipulating frequency and observing wavelength changes at fixed speed) and the LabXchange Wave Interference tool (measuring wavelengths of overlapping waves). In the Explain phase, the class derives $v = f\lambda$ from first principles, applies it to calculate the actual wavelengths of Classic 105 FM (105.2 MHz) and a representative AM signal (567 kHz, KBC Radio), and compares these to the approximate diameter of the Kabete tunnel (~8 m). The Driving Question Board is updated with new quantitative evidence, and learners revise their cumulative wave models to include labelled v, f, and λ annotations. The lesson is grounded throughout in Kenyan radio broadcasts, the matatu horn Doppler context, and rural connectivity challenges in counties such as Turkana and Marsabit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Proof of the hyperbola foci formula Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are presented with the following written prompt displayed on the board: 'Classic 105 FM broadcasts at 105.2 MHz and radio waves travel at 300,000,000 m/s. If you could freeze one moment of that radio wave in space above Nairobi, how long — in metres — do you think one complete wave cycle would be? Write your best estimate and your reasoning in your exercise book. Then predict: if a different station broadcasts at a lower frequency, would its wavelength be longer, shorter, or the same?' Learners work individually for 4 minutes, then share with a shoulder partner for 2 minutes. Each pair records both estimates on a sticky note and posts it on the class Prediction Wall (one column labelled SHORT < 5 m, one labelled 5–50 m, one labelled LONG > 50 m). The class silently observes the distribution of sticky notes on the wall.	Display the prompt and say verbatim: 'Before we calculate anything, I want to know what your gut tells you. How long is one FM radio wave? Write your thinking — not just a number, but WHY you think that.' Allow 4 minutes of individual silent work. Circulate and note 3–4 interesting or contrasting estimates to cold-call later. After pair-share, facilitate sticky-note posting. Then cold-call exactly 3 pairs — one from each column — using: 'Tell us your estimate and the reasoning behind it.' After each response, wait 10 seconds before commenting, then ask the class: 'Does anyone want to challenge or build on that idea?' Do NOT confirm or correct estimates yet. Close with: 'Hold your predictions — we are about to gather evidence that will either surprise you or prove you right.'	Prediction Wall / Claim-Evidence-Reasoning Primer — by making and displaying predictions publicly before instruction, learners activate prior knowledge about frequency and wavelength, create cognitive dissonance that motivates the evidence phase, and have a concrete record to revisit and revise, reinforcing metacognitive awareness of how their thinking changes.	Observable indicator: Every learner has a written prediction with at least one line of reasoning in their exercise book before the pair-share. Teacher scans books during circulation. Red flag: learner writes only a number with no reasoning — prompt with 'What does frequency mean to you? What does wavelength look like in your mind?'
Observe Phase 📹 **VIDEO:** Proof of the hyperbola foci formula Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Activity A — PhET Sound Simulation (12 min, groups of 3): Learners open the PhET Sound simulation. Each group works through a structured evidence-gathering protocol printed on a task card: (1) Set the speaker frequency to 200 Hz. Use the 'measure' tool to record the distance between two adjacent compressions (this is λ). Record speed displayed. (2) Change frequency to 400 Hz. Repeat measurement. (3) Change to 800 Hz. Repeat. Record all data in a three-column table: Frequency (Hz) | Wavelength (m) | Speed (m/s). (4) Write one sentence describing the pattern. Activity B	Before releasing groups, say: 'You are detectives. Your job is not to play with the simulation — your job is to collect precise evidence about how frequency and wavelength are connected. Every number must go in your table.' Circulate and spend no more than 90 seconds per group. Use these targeted questions: 'What happened to the wavelength when you doubled the frequency?' (wait 10 seconds) and 'Did the speed change when you changed the frequency? What does that tell you?' (wait 12 seconds). During the data share, cold-call 4 groups to read their λ values aloud. After the ratios are on the board, pause for 15	Structured Data Table + Ratio Analysis — requiring learners to compute ratios ($\lambda_{200}/\lambda_{800}$ and f_{800}/f_{200}) rather than just record numbers forces them to see the inverse proportionality numerically before it is formalised algebraically. This bridges concrete simulation data to abstract mathematical relationships.	Observable indicator: Each group's data table contains at least three frequency-wavelength pairs with consistent speed values (~340 m/s for sound in air). Teacher checks that the ratio column is completed. Red flag: groups whose speed values vary wildly — this indicates measurement error; prompt with 'How are you measuring the wavelength? Show me where you placed the measurement tool.'

	— LabXchange Wave Interference (8 min, same groups): Learners open the LabXchange Wave Interference simulation. They create two waves of different frequencies and observe how wavelength changes as frequency increases at constant speed. They sketch one screenshot-equivalent diagram in their books, labelling λ_1 and λ_2. Whole-class data share (5 min): One representative from each group writes their 200 Hz and 800 Hz wavelength values on the board in a class data table. Learners calculate the ratio $\lambda_{200}/\lambda_{800}$ and the ratio f_{800}/f_{200} and compare them.	seconds of silent observation, then ask: 'What do you notice about these two ratios?' Cold-call 2 learners who have not yet spoken today.		
Explain Phase 📹 **VIDEO:** Proof of the hyperbola foci formula Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Derivation (8 min): Learners follow a guided derivation in their books. The teacher writes the scaffolded steps on the board and learners fill in the blanks on a printed or copied framework: Step 1 — Definition: In one second, a source vibrating at frequency f produces f complete waves. Step 2 — Each wave has length λ. Step 3 — Therefore, the total length of wave produced per second = f × λ. Step 4 — But 'length of wave produced per second' is the wave speed v. Step 5 — Therefore: v = fλ. Application to the Phenomenon (7 min): Learners individually calculate: (a) Wavelength of Classic 105 FM: $\lambda = v/f = 3\times10^8 \div 105.2\times10^6 \approx 2.85$ m. (b) Wavelength of KBC AM Radio (567 kHz): $\lambda = 3\times10^8 \div 567\times10^3 \approx 529$ m. They then compare both wavelengths to the approximate width of the Kabete tunnel (~8 m) and write a one-sentence inference: 'The FM wavelength (≈2.85 m) is much smaller than the tunnel opening, while the AM	During derivation, say verbatim: 'Do not copy blindly — at each step, ask yourself: does this make physical sense?' After Step 4, pause and cold-call 2 learners: 'In your own words, why does f times λ equal speed?' (wait 12 seconds each). After the FM/AM calculations, display the comparison: 'λ_FM ≈ 2.85 m versus tunnel width ≈ 8 m. λ_AM ≈ 529 m.' Ask the class: 'Which wave — FM or AM — has a wavelength closest in size to the tunnel opening? What might that mean for how the wave interacts with the tunnel?' Wait 15 seconds. Cold-call 3 learners. Do not give the full diffraction answer yet — say: 'You are circling the answer. We will confirm it in Lesson 6 when we study diffraction. For now, record your best inference.' During gallery rotation, circulate and note 2 misconceptions heard to address in the DQB phase.	Guided Algebraic Derivation + Anchored Calculation — deriving v = fλ from definitions (rather than presenting it as a formula to memorise) builds conceptual ownership. Immediately applying it to the specific FM and AM frequencies from the anchoring phenomenon makes the algebra meaningful and creates a quantitative 'bridge' to the upcoming diffraction explanation, maintaining narrative coherence across the lesson sequence.	Observable indicator: Learner exercise books show correct calculations for both FM and AM wavelengths with units, and a written inference sentence connecting wavelength size to the tunnel. Teacher spot-checks 6 books during the gallery rotation. Red flag: learners who write λ = v×f (multiplying instead of dividing) — address immediately with: 'If frequency goes up, what happens to wavelength according to v = fλ? Should wavelength get bigger or smaller?'

	wavelength (≈529 m) is far larger — this difference in wavelength relative to the tunnel size must be key to explaining the signal blackout.' Gallery comparison (5 min): Learners rotate to see three anchor posters on the wall showing: (1) a matatu horn wave diagram with labelled v, f, λ; (2) a scaled comparison of FM vs AM wavelengths; (3) the class Prediction Wall. They add a tick or cross to their original sticky note based on whether their prediction was close.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Proof of the hyperbola foci formula Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two new sticky notes: (1) a CLAIM note (blue): one sentence stating something they now know for certain, backed by today's evidence (e.g., 'The FM signal has a wavelength of ~2.85 m because v = fλ and the speed of radio waves is 3×10^8 m/s'). (2) a NEW QUESTION note (yellow): one question today's lesson raised that they cannot yet answer (e.g., 'Why does wavelength size compared to the obstacle determine whether a wave passes through or not?'). Learners place their notes on the class Driving Question Board under the appropriate columns: WHAT WE NOW KNOW | WHAT WE STILL WONDER | WHAT CONNECTS TO THE MATATU HORN. The teacher reads aloud 3 selected claim notes and 3 new question notes, and the class votes (thumbs up/sideways/down) on whether each claim is well-supported by today's evidence.	Say: 'The Driving Question Board is our collective memory. Every note must be traceable to evidence — not opinion.' Allow 3 minutes of silent individual writing. Then facilitate posting (2 min). Read the selected notes aloud and after each claim, ask: 'What evidence from today's simulation or calculation supports this claim?' (wait 10 seconds, cold-call 1 learner per note). After the new questions are read, say: 'These yellow notes are your research agenda. Keep them — Lesson 6 on diffraction will answer the question about wavelength and obstacles directly.' Point to the two unresolved mysteries on the board (the matatu horn and the tunnel blackout) and ask: 'Has today's lesson brought us closer to answering either mystery? Which one, and how?' (wait 12 seconds, cold-call 2 learners).	Evidence-Backed Claim Wall + Public Peer Vetting — requiring learners to distinguish between a claim and a question, and then submit claims to class scrutiny (thumbs vote), builds scientific argumentation skills and reinforces the norm that knowledge in Physics must be evidence-based. The three-column board structure makes the progression of the storyline visible and cumulative.	Observable indicator: Every learner has posted at least one claim note that references a specific number or observation from today (not a vague generalisation). Teacher checks that claim notes in the WHAT WE NOW KNOW column cite v = fλ or a calculated wavelength value. Red flag: claim notes that say only 'waves have frequency and wavelength' without quantitative evidence — prompt the learner to add: 'How do you know? What number proves it?'
Model Building Phase 📹 **VIDEO:**	Learners retrieve their cumulative wave model diagram begun in Lesson 1 (a labelled drawing of the anchoring phenomenon — the	Say: 'Your model is not a pretty picture — it is a scientific argument. Every label must be a number you calculated or	Cumulative Annotated Model Revision — requiring learners to add quantitative labels (calculated values with units) to a persistent	Observable indicator: Learner models show correct calculated values for all three wavelengths (sound horn, FM, AM) with units

Proof of the hyperbola foci formula Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	matatu on the Nakuru–Nairobi highway and the bus entering the Kabete tunnel). Today, they add the following required annotations to the model: (1) On the matatu horn sound waves: label v (≈340 m/s in air), f (horn frequency, estimated ~400 Hz), and calculate and label λ (≈0.85 m). (2) On the FM radio signal: label v (3×10^8 m/s), f (105.2 MHz), and λ (≈2.85 m). (3) On the AM radio signal: label v (3×10^8 m/s), f (567 kHz), and λ (≈529 m). (4) Add a scale bar beneath the tunnel diagram comparing the three wavelengths visually. Learners also add a 'Still unexplained' annotation arrow pointing to the tunnel entrance with the note: 'WHY does the short λ FM signal get blocked while the long λ AM signal passes through? → Lesson 6: Diffraction.' Pairs do a 90-second model swap: each partner must find one thing to praise and one label that needs correction or more detail.	measured, not a guess.' Circulate during the 7 minutes of individual model work. Target your attention to the scale bar: 'Has anyone drawn the AM wavelength to the same scale as the FM wavelength? What does that reveal visually?' (wait 10 seconds). During the pair swap, listen for the quality of peer feedback. After 90 seconds, cold-call 2 learners: 'What did your partner correct on your model, and do you agree with the correction?' Close the lesson by pointing to the anchoring phenomenon poster at the front and saying: 'When you started this unit, the Kabete tunnel blackout was mysterious. Today you can say exactly how long the FM wave is and exactly how long the AM wave is. In Lesson 6, you will find out why that difference in size is the key to the whole mystery.'	visual model that spans the full lesson sequence creates a tangible record of growing understanding. The 'Still unexplained' arrow deliberately leaves a productive knowledge gap that sustains curiosity and connects forward to Lesson 6, maintaining narrative drive across the storyline.	labelled, and a scale bar that correctly represents the relative sizes. Teacher checks that the 'Still unexplained' annotation is present and pointed at the tunnel. Exit check: teacher asks the whole class to hold up their models — scan for missing labels. Red flag: learners who have drawn FM and AM waves the same size — address with: 'Your calculation says one is 2.85 m and one is 529 m. How can they look the same on your diagram?'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the ratio analysis in the Observe phase (computing $\lambda_{200}/\lambda_{800}$ and f_{800}/f_{200}) successfully help learners see the inverse relationship before the formula was introduced, or did learners still treat v = fλ as a formula to memorise? What adjustment would strengthen the bridge from data to algebra? (2) When learners calculated λ_FM ≈ 2.85 m and λ_AM ≈ 529 m, how many spontaneously connected the size difference to the tunnel observation without teacher prompting? If fewer than half did so, consider adding a more explicit comparison step — e.g., a physical demonstration using a door frame (≈1 m wide) as a proxy for the tunnel to ask: 'Which wave fits through this gap?' (3) Were the cold-call targets (3 in Predict, 4 in Observe, 2+3 in Explain, 2+1 in DQB, 2 in Model) achieved, and did the 10–15 second wait times hold? If learners answered too quickly or too slowly, adjust the complexity of the cold-call question, not the wait time. (4) Review the DQB yellow 'new question' notes: do they cluster around diffraction and wave-obstacle interaction? If not, the Lesson 6 launch may need an additional bridge activity. (5) Did any Kenyan-context connections — Classic 105, KBC, Kabete tunnel, Turkana County radio reach — generate genuine student engagement or questions? Document any student-generated Kenya-context examples for use in future lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Using the PhET Sound Simulation, learners measured the wavelength of sound waves at three different frequencies (200 Hz, 400 Hz, 800 Hz) while speed remained constant (~340 m/s), and recorded the data in a three-column table. Using the LabXchange

	Wave Interference tool, learners observed that higher-frequency waves had shorter wavelengths. The class computed the ratios $\lambda_{200}/\lambda_{800}$ and f_{800}/f_{200} and found them equal, providing numerical evidence of the inverse relationship between frequency and wavelength at constant speed.
What did I learn?	Learners derived the wave equation $v = f\lambda$ from first principles (number of waves per second × length of each wave = distance covered per second = speed). They applied the equation to calculate: (a) the wavelength of Classic 105 FM (105.2 MHz) ≈ 2.85 m; (b) the wavelength of KBC AM Radio (567 kHz) ≈ 529 m. They discovered that for a given wave speed, frequency and wavelength are inversely proportional — doubling the frequency halves the wavelength. They also connected these wavelengths to the size of the Kabete tunnel (~8 m) as a quantitative first step toward explaining the FM signal blackout.
How does this explain the phenomenon?	The wave equation $v = f\lambda$ explains why Classic 105 FM and AM radio signals behave so differently inside the Kabete tunnel: FM waves at 105.2 MHz have a wavelength of only ~2.85 m, which is much smaller than the tunnel opening (~8 m), while AM waves at 567 kHz have a wavelength of ~529 m, far larger than the tunnel. This difference in wavelength relative to obstacle size is the quantitative foundation for the diffraction explanation to be completed in Lesson 6 — longer wavelengths diffract more readily around and through obstacles, while shorter wavelengths are more easily blocked.

LESSON 6 (40 minutes): Diffraction of Waves: Why AM Survives the Kabete Tunnel

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To investigate how waves spread when they pass through gaps or around obstacles, and to use this property to construct the first complete explanation of why FM radio fades inside the Kabete tunnel while AM radio does not.
Knowledge	Learners will be able to: (1) define diffraction as the spreading of waves when they pass through a gap or around an obstacle; (2) state that diffraction is most pronounced when the gap width is comparable to or smaller than the wavelength; (3) state that longer wavelengths diffract more than shorter wavelengths through the same gap; (4) apply these principles to contrast AM (wavelength ~529 m) and FM (wavelength ~2.85 m) signals passing through the Kabete tunnel (~8 m opening).
Skills	Learners will: manipulate variables (gap width, wavelength/frequency) in PhET Wave Interference simulation; record observations systematically in a structured data table; sketch wavefront diagrams showing diffraction patterns for wide and narrow gaps; construct a written qualitative explanation linking diffraction to the AM/FM tunnel phenomenon.
Attitudes	Learners will appreciate that systematic observation and evidence-based reasoning — rather than guessing — allow scientists and engineers to solve real communication problems; they will show curiosity by asking follow-up questions about how diffraction affects mobile phone signals in buildings and valleys across Kenya.
Key Inquiry Question	How do waves behave in different media and under different conditions? (Specifically: how does the relationship between wavelength and gap size determine whether a wave spreads or travels straight through an opening?)
Purpose in Storyline	Lesson 6 provides the first quantitatively grounded resolution of the FM/AM tunnel mystery that has anchored the unit since Lesson 1. Learners already know from Lesson 3 that FM wavelength (~2.85 m) is far smaller than AM wavelength (~529 m) and that the tunnel opening is ~8 m. Today they discover WHY those numbers matter: diffraction depends on the ratio of wavelength to gap size. This resolves the radio half of the anchoring phenomenon and motivates Lesson 7 (interference), where they will explore what happens when diffracted waves from two sources overlap.
Safety Notes	No physical hazards. Learners using school computers or tablets should be reminded not to download or install any software. If using a shared computer lab, ensure screens are visible to the teacher during the simulation. Learners with photosensitive epilepsy should be seated away from rapidly flickering simulation displays.

B. LESSON OVERVIEW

Learners open the PhET Wave Interference simulation (water-wave mode, single-slit setting) and conduct a structured inquiry varying two independent variables — gap width and wave frequency — while observing the degree of spreading (diffraction) of wavefronts beyond the gap. They record qualitative and semi-quantitative observations, sketch wavefront diagrams, and state two evidence-based rules about diffraction. The lesson culminates in a whole-class sense-making discussion in which learners explicitly connect the diffraction rules to the AM/FM wavelengths calculated in Lesson 3 and to the physical dimensions of the Kabete tunnel, thereby constructing a coherent causal explanation for the anchoring phenomenon. The Driving Question Board is updated with this new evidence, and learners add a diffraction diagram to their growing unit model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Rational equation word problem Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Wavelength (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their exercise books to the Driving Question Board notes from Lesson 1. The teacher displays two diagrams side by side on the board: (A) a narrow doorway with straight arrows (wavefronts) approaching it, and (B) a wide doorway with the same arrows. Learners are given 2 minutes to write individual silent predictions in response to two prompts: (1) Sketch what you think the waves look like AFTER they pass through each doorway — do they continue straight, or do they spread sideways? (2) If you change the doorway to be the width of a pencil, what happens? Using the wavelength values from Lesson 3 (FM ~2.85 m, AM ~529 m) and the tunnel opening (~8 m), learners also predict which signal — AM or FM — will spread more into the tunnel and why. Predictions are written in silence so every learner commits to an answer before seeing peers' ideas.	Teacher says: 'Before we touch the simulation, I want YOUR thinking on paper. Look at these two doorways. What do the waves look like on the other side? Write and sketch — no talking for two minutes.' [WAIT TIME: 2 minutes of enforced silence.] Teacher circulates, reading over shoulders without commenting, noting the range of predictions (some learners will draw straight rays through both gaps; others may sketch spreading for the narrow gap). After 2 minutes: 'Turn and share your sketch with your partner — you have 30 seconds each.' Teacher then cold-calls three pairs with contrasting predictions and records key phrases on the board under the heading PREDICTIONS: e.g., 'Waves go straight through,' 'Waves spread out from the gap,' 'Narrow gap spreads more.' Teacher says: 'Let us keep ALL of these on the board. By the end of this lesson you will have evidence to decide which prediction the data supports.'	Prediction-before-observation (anchoring prior knowledge): by committing to a sketch, learners surface naive conceptions (e.g., waves always travel straight) that the simulation will directly confront. Recording predictions publicly on the board creates accountability and sets up a predict-observe-explain cycle.	Teacher scans written predictions during the 2-minute silence. Look for: (a) learners who draw NO spreading for either gap — they hold a ray-optics mental model that will need explicit challenge; (b) learners who correctly predict more spreading for the narrow gap — they can be positioned as resources during the explain phase. Note whether any learner spontaneously references the FM/AM wavelengths from Lesson 3 — this indicates high transfer readiness.
Observe Phase 📹 **VIDEO:** Rational equation word problem Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Wavelength (Flexbook) Source: CK-12 🔍 Search ARES	Learners open PhET Wave Interference at https://phet.colorado.edu/en/simulations/wave-interference on a school computer or tablet. They select INTERFERENCE, then choose WATER mode, then SLITS, and set the number of slits to ONE. Working in pairs, they complete a structured investigation table (drawn in their exercise books before the lesson or provided as a printed sheet) with the following two experiments: EXPERIMENT 1 — Varying gap	Before learners begin: 'You have EXACTLY 12 minutes for both experiments. Experiment 1 first — do not skip to Experiment 2 early. I will call time at 6 minutes.' Teacher circulates continuously, pausing at pairs who are watching the simulation without recording, and prompting: 'What is your sketch for this slit width? Write it now — if you do not record it, it did not happen.' At the 6-minute mark: 'Stop Experiment 1. Read your two observation sentences to your partner. Do they agree? Now	Structured inquiry with a data table forces learners to vary ONE variable at a time — this is explicit practice of controlled investigation. The requirement to write two summary observation statements before discussion ensures every learner constructs their own language for the pattern, rather than borrowing language from a more vocal peer.	Teacher checks data tables for: (a) correct directional trends — spreading INCREASES as gap width DECREASES, and spreading INCREASES as wavelength INCREASES (frequency decreases); (b) whether sketches show curved wavefronts beyond the slit (correct) or straight wavefronts (misconception — treat this immediately by asking the learner to zoom in on the simulation and look again). If more than one third of pairs have the wrong trend for

[for similar readings]	width (frequency held constant at medium): Learners set the slit width to VERY SMALL, then SMALL, then MEDIUM, then LARGE. For each setting they observe the wavefront pattern beyond the slit and record: (i) a quick sketch of the wavefront shape, (ii) whether the spreading is WIDE ANGLE, MODERATE, SLIGHT, or NONE. EXPERIMENT 2 — Varying frequency/wavelength (gap width held constant at SMALL): Learners set frequency to LOW (long wavelength), then MEDIUM, then HIGH (short wavelength). For each they record the same observations. Learners are directed to pay specific attention to the angle through which waves spread beyond the slit. After completing the table, each learner writes two summary observation statements in their own words before discussion begins.	move to Experiment 2.' At the 12-minute mark: 'Simulations closed. Pens down. Read your two summary statements silently once more.' [WAIT TIME: 30 seconds.] Teacher then asks: 'Who is willing to share their summary statement for Experiment 1?' Cold-calls a learner who was observed recording confidently. Writes their statement on the board verbatim, then asks the class: 'Do you agree, partially agree, or disagree? Thumbs up, sideways, or down.' Addresses disagreements before moving to Experiment 2 summary.		either variable, pause the whole class and re-run one demonstration on the projector before proceeding.
Explain Phase 📹 **VIDEO:** [Rational equation word problem] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Wavelength (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Learners use their observation tables to construct TWO evidence-based diffraction rules, written as IF-THEN statements in their books: Rule 1: 'IF the gap width is small compared to the wavelength, THEN the waves spread (diffract) through a wide angle beyond the gap.' Rule 2: 'IF the wavelength is long compared to the gap width, THEN diffraction is greater.' Teacher formalises the definition: 'Diffraction is the spreading of waves when they pass through a gap or around an obstacle — it is greatest when the gap is about the same size as, or smaller than, the wavelength.' The class then applies these rules directly to the anchoring phenomenon using the numbers from Lesson 3. Teacher	Teacher opens the explain phase with a direct challenge to the class predictions on the board: 'Look at your prediction from the start of the lesson. Raise your hand if your prediction was correct.' [WAIT TIME: 5 seconds — allow hands to rise without rushing.] 'Raise your hand if your prediction was partially correct.' 'Raise your hand if the data surprised you.' Teacher normalises surprise: 'Being surprised by evidence is a sign you are doing real science. Adjust your model — that is what scientists do.' Teacher then says: 'I want ONE volunteer to explain in their own words why FM fades in the Kabete tunnel, using the word DIFFRACTION and the numbers we calculated in Lesson 3. You	IF-THEN rule construction is a disciplinary literacy strategy that forces learners to express causal relationships rather than mere descriptions. Applying rules to specific numbers (FM 2.85 m vs AM 529 m vs tunnel 8 m) requires quantitative reasoning and prevents surface-level pattern matching. The mobile-phone extension is a transfer task — it tests whether learners can apply the concept to a new, Kenya-relevant context without teacher scaffolding.	Listen carefully to the paired AM/FM application discussion. Target misconception: some learners may think FM fades because FM is a weaker signal, not because of diffraction — address this explicitly by asking, 'If we put a much more powerful FM transmitter in Nairobi, would the signal survive in the tunnel?' (No — power and diffraction are independent.) Cold-call the extended mobile-phone question to three different learners; their answers reveal whether diffraction as a concept (not just a memorised example) has been understood.

	writes on the board: FM wavelength ~2.85 m, AM wavelength ~529 m, Kabete tunnel opening ~8 m. Learners work in pairs for 3 minutes to answer: 'For FM waves, the gap (~8 m) is MUCH larger than the wavelength (~2.85 m) — predict the diffraction angle. For AM waves, the gap (~8 m) is MUCH smaller than the wavelength (~529 m) — predict the diffraction angle.' Pairs share conclusions. Teacher draws a large board diagram: two wavefront diagrams of the Kabete tunnel — one showing FM waves passing straight through with minimal spreading (most energy continues forward and is not captured inside the tunnel), and one showing AM waves bending widely around and into the tunnel interior. Learners copy both diagrams with labels into their exercise books. The teacher explicitly states: 'This is why your Classic 105 FM signal disappears in the Kabete tunnel — the FM wavelength is tiny compared to the tunnel opening, so the waves do not bend around the tunnel walls and reach your radio antenna inside the bus. The AM signal, with its 529-metre wavelength, diffracts around and into the tunnel easily.'	have 20 seconds to prepare your sentence.' [WAIT TIME: 20 seconds.] Cold-calls a learner. After the learner speaks, teacher asks the class: 'Did they use the word diffraction? Did they use a number? Give them a thumbs up for each one they included.' Teacher then extends: 'Think about this — mobile phone signals in Kenya operate at around 900 MHz, which gives a wavelength of about 33 centimetres. Would a mobile signal diffract easily around a large concrete building in Nairobi CBD? Discuss with your partner for 1 minute.' [WAIT TIME: 1 minute.] Takes two responses and connects to everyday Kenyan experience of dropped calls near tall buildings.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Rational equation word problem] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos]	Learners return to the class Driving Question Board, which has been displayed continuously since Lesson 1. They review the original questions posted there, particularly those relating to: 'Why does FM fade in the tunnel but AM does not?' Each learner writes one sticky note (or one line in their exercise book if sticky notes are unavailable) with the heading NEW EVIDENCE and completes	Teacher says: 'Go back to our Driving Question Board. Find the question about FM and AM in the tunnel. We have been carrying that question since Lesson 1. Do we have enough evidence NOW to answer it? Write your answer on a sticky note — use the word DIFFRACTION, use at least one number, and mention the wavelength of either FM or AM.' [WAIT TIME: 2 minutes for writing.]	Updating the DQB with evidence-anchored sticky notes makes the cumulative nature of the inquiry storyline visible to learners — they can physically see questions being answered over time. Posting a new, unanswered question immediately prevents premature closure and models how scientific knowledge generates new questions rather than terminating inquiry.	Read all sticky notes added to the DQB during the lesson or immediately after. Look for: (a) correct causal language — 'because the FM wavelength is much smaller than the tunnel opening, so it does not diffract significantly'; (b) incorrect causal language — 'because FM is faster' or 'because FM has more energy' — these are misconceptions to address at the start of Lesson 7.

📖 **HTML:** [Wavelength (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	the sentence: 'We now know that FM fades in the Kabete tunnel because...' Learners place their sticky notes on or near the relevant DQB question. The teacher also writes a new question card on the DQB in a different colour: 'If two diffracted waves from the same source meet each other on the other side of a gap, what happens where they overlap?' — this bridges to Lesson 7 on interference. Learners read this new question and in one sentence predict what they think will happen when two spreading waves collide.	Teacher reads three sticky notes aloud (choosing a range of quality) and models adding them to the board. Teacher then points to the new question card: 'Here is a question our evidence today CANNOT yet answer. Two waves are spreading out from the same gap — what happens when they meet? Hold that thought — it is the starting point of Lesson 7.' Teacher does NOT answer the question; the goal is to sustain productive curiosity.		The one-sentence prediction about overlapping waves provides baseline data for planning the Lesson 7 predict phase.
Model Building Phase 📹 **VIDEO:** [Rational equation word problem] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Wavelength (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes, learners open their growing unit model diagram (begun in Lesson 2 and updated in each subsequent lesson). They add a new labelled section titled DIFFRACTION. In this section they draw TWO wavefront diagrams side by side: (A) a narrow gap where wavelength is comparable to gap width — showing curved, widely spread wavefronts; (B) a wide gap where gap width is much larger than wavelength — showing nearly straight wavefronts with only slight edge spreading. Beneath the diagrams they write the two IF-THEN diffraction rules. In the corner of the section they draw a miniature version of the Kabete tunnel diagram with FM and AM labels, and write: 'AM (~529 m wavelength) diffracts into the tunnel. FM (~2.85 m wavelength) does not.' They draw an arrow connecting this section of the model to their Lesson 3 wavelength calculations — explicitly showing how v = fλ from Lesson 3 fed into the diffraction	Teacher says: 'You have 5 minutes to update your unit model. Three things MUST appear: (1) two wavefront diagrams — one for a small gap, one for a large gap; (2) your two IF-THEN rules; (3) a connection arrow to your Lesson 3 wavelength values. If you finish early, draw a third diagram showing a mobile phone signal trying to get around a Nairobi CBD building.' Teacher circulates and checks that the connection arrow to Lesson 3 is present — this is the key cross-lesson link. Teacher uses the final 60 seconds to narrate the storyline so far: 'Lesson 2: we learned what waves are. Lesson 3: we found that FM and AM have VERY different wavelengths. Today, Lesson 6: we discovered WHY those different wavelengths matter — diffraction. The tunnel mystery is cracked. But we have one more wave behaviour to understand before we fully explain everything — and that is what happens when two waves meet. See you in Lesson 7.'	Adding a connection arrow from the Lesson 6 diffraction section to the Lesson 3 wavelength calculations is a deliberate structural move: it externalises the logical chain of reasoning (frequency determines wavelength, wavelength relative to gap size determines diffraction, diffraction determines whether the signal reaches the radio inside the tunnel). This supports learners in seeing science as a coherent explanatory framework rather than a set of disconnected topics.	Before learners close their books, teacher does a quick show-me scan: 'Hold up your model page — I need to see two wavefront diagrams and one connection arrow.' This whole-class visual check takes 20 seconds and immediately identifies learners who need to complete the model at the start of Lesson 7. Teacher also asks one exit-ticket question verbally: 'In one word — what wave property did we investigate today?' Expected answer: diffraction. Any learner who says reflection or refraction needs targeted follow-up.

	explanation of Lesson 6.			

D. TEACHER REFLECTION
After the lesson, consider: (1) Did the majority of learners correctly apply the two diffraction rules to the AM/FM numbers without prompting? If not, the Explain phase needs more guided practice in Lesson 7 before moving to interference. (2) Were learners able to draw wavefront diagrams that showed curved spreading rather than straight rays? If many diagrams still showed straight rays, revisit the PhET simulation at the start of Lesson 7 with a class demonstration on the projector. (3) Did any learner raise the question of what happens when FM signals reach a valley or a hill in Turkana or Marsabit County? This is a rich extension that connects to rural connectivity and the Kenyan government's digital inclusion agenda — consider using it as a warm-up scenario in Lesson 10 when AM vs FM modulation is formally compared. (4) Was the 40-minute timing realistic? The simulation section (Observe Phase) most commonly runs long. If needed, reduce Experiment 2 to two frequency settings instead of three. (5) Note which learners demonstrated high transfer ability in the mobile-phone extension question — these learners can serve as peer tutors during the Lesson 11 model-building session.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	When waves pass through a gap in PhET Wave Interference, the waves spread out on the other side. The spreading (diffraction) is greatest when the gap is narrow, and is also greater when the wavelength is long. A very wide gap produces almost no spreading — the waves continue almost straight through.
What did I learn?	Diffraction is the spreading of waves when they pass through a gap or around an obstacle. Diffraction is most pronounced when the gap width is similar to or smaller than the wavelength. Longer wavelengths (lower frequencies) diffract more through the same gap than shorter wavelengths (higher frequencies).
How does this explain the phenomenon?	The FM radio signal from Classic 105 (wavelength ~2.85 m) barely diffracts when it enters the Kabete tunnel opening (~8 m wide) because the gap is much larger than the FM wavelength — the signal travels nearly straight and most of its energy does not bend around to reach the radio antenna inside the bus. The AM signal (wavelength ~529 m) diffracts very strongly because its wavelength is far larger than the tunnel opening — the AM signal bends widely around and into the tunnel, keeping the signal audible. This is why FM fades but AM survives in the Kabete tunnel.

LESSON 7 (40 minutes): Interference of Waves: When Waves Meet, Add, and Cancel	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will investigate how two wave sources interact to produce regions of reinforcement and cancellation, explaining the coloured bands on soap bubbles and advancing the unit model toward full resolution of the anchoring phenomenon.
Knowledge	Students can state the principle of superposition and distinguish constructive interference (path difference = $n\lambda$) from destructive interference (path difference = $(n + 0.5)\lambda$); identify nodal and antinodal lines in a two-source interference pattern; and connect path difference to the resulting amplitude at any point.
Skills	Students can use PhET Wave Interference in double-source mode to observe and sketch interference patterns; measure and compare path differences to whole and half wavelengths; annotate diagrams with nodal and antinodal lines; and communicate pattern descriptions using precise wave vocabulary.
Attitudes	Students demonstrate curiosity when unexpected dark and bright bands appear; they persist through the abstract concept of path difference by grounding it in the visible simulation; and they appreciate that mathematics (path difference ratios) can predict physical phenomena observed in everyday Kenyan life.
Key Inquiry Question	What are the properties of waves? / How do waves behave in different media and under different conditions?
Purpose in Storyline	Lesson 6 established that wavelength controls diffraction — giving the first partial answer to the FM-versus-AM tunnel mystery. Lesson 7 now reveals that when two wave trains meet they do not simply add their energies uniformly; instead they produce a structured pattern of loud-and-quiet or bright-and-dark zones. This prepares students for standing waves (Lesson 8) and underpins the full Doppler and modulation explanations in Lessons 9 and 10. It also explains why soap-bubble colours (supporting phenomenon 3) form distinct coloured rings rather than a uniform colour — a detail students have been curious about since the DQB was first populated.
Safety Notes	Lesson is simulation-based. Ensure screen brightness is comfortable to avoid eye strain during the 15-minute PhET observation block. If any student has photosensitive epilepsy, set the PhET animation speed to Slow before the session begins and alert the student in advance.

B. LESSON OVERVIEW
Students open PhET Wave Interference in double-source mode and systematically explore what happens when two identical wave sources emit simultaneously. They observe alternating bright (constructive) and dark (destructive) bands radiating outward, measure path differences at selected points, and build the conceptual link between path difference and superposition. The lesson closes with four cold-called students explaining nodal and antinodal lines in their own words. The soap-bubble colour phenomenon is introduced as a visual anchor, and the DQB is updated with the new evidence that interference adds a second layer of explanation to why signals can fade — not only from diffraction but from cancellation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students are shown a photograph	Display the soap-bubble	Anchored prediction with a Kenyan	Teacher reads written predictions

📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	of a soap bubble filmed at a Nairobi street market stall, displaying vivid bands of colour — red, green, and violet — arranged in rings across the bubble surface. The teacher asks: Both sides of the soap film are reflecting light at the same time. The two reflected waves meet each other. Predict in your exercise book — will the two waves always add up to give a brighter result, or could something else happen? Give a reason. Students write a silent individual prediction for 2 minutes, then share with a neighbour for 1 minute. The teacher takes a quick show-of-hands poll: How many of you predict the waves will always add up and get brighter? How many predict sometimes they cancel? The spread of predictions is noted on the board without resolution — it remains a live question for the Observe phase. The teacher then links explicitly to prior learning: In Lesson 6 we saw that longer wavelengths spread more when passing through a gap. Today we investigate what happens after waves have spread — when two sets of waves from different sources actually overlap in the same space.	photograph on the projector (source: Wikimedia Commons, free licence). Read the prediction prompt aloud and then say: I will not tell you the answer yet — that is what the simulation is for. You have 90 seconds of silent thinking time starting now. [Wait 90 seconds — circulate and read over shoulders without commenting.] After sharing with a neighbour say: Put your hand up if your neighbour had a completely different prediction from yours. Cold-call two pairs to share their disagreement. Say: Good — we have two competing ideas. Let the simulation decide. Teacher quote for transition: Last lesson we learned that waves spread around obstacles. Today we ask a harder question — what do waves do TO EACH OTHER when they meet?	visual artefact (soap bubble at a Nairobi market stall). The photograph connects the abstract physics to a familiar childhood experience of blowing soap bubbles — ubiquitous across Kenya. The show-of-hands poll surfaces prior misconceptions (students commonly assume wave energies always add) and creates productive cognitive conflict that the simulation will resolve.	during the 90-second wait; looks for the misconception that two waves always produce a brighter result. This informs emphasis during the Explain phase — the teacher will specifically address why cancellation is not intuitive.
Observe Phase 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos	Students open PhET Wave Interference at https://phet.colorado.edu/en/simulations/wave-interference on the school laptops or tablets (one device per pair). They select the INTERFERENCE tab and switch to two-source mode with water waves displayed. Step-by-step guided observation: Step 1 — Run the simulation with default settings for 1 minute. Sketch the pattern you see in your exercise book.	Distribute or display the observation table. Say: You have 15 minutes with the simulation. Your job is a scientific one — you are measuring, not just watching. Collect numbers. At the 7-minute mark do a mid-observation pause: Everybody pause the simulation. Look at your path-difference column for the strongly rippled points. What do you notice about those numbers? [Wait 20 seconds.] Now look at the calm	Structured data-collection observation using the PhET double-source water-wave simulation. The five-step sequence moves students from qualitative pattern recognition to quantitative path-difference measurement, building the empirical foundation for the superposition principle. Switching to light mode at Step 5 previews the universality of wave interference — all waves interfere — reinforcing the unit theme that	Teacher checks path-difference tables during circulation. Key checkpoint: Are students finding that path differences at rippled points cluster near whole-number multiples of the wavelength, and path differences at calm points cluster near half-integer multiples? If fewer than half the pairs are seeing this pattern, the teacher pauses the class and does one worked example on the board before the Explain phase.

📖 **HTML:** Wave Interference (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Label any regions that appear calm and any regions that appear strongly rippled. Step 2 — Switch to Measure mode and use the on-screen ruler to find three points that are strongly rippled and measure the distance from each point to Source 1 and Source 2. Record both distances in a table. Calculate the difference (path difference) for each point. Step 3 — Find three calm points and repeat the path-difference measurement. Step 4 — Change the frequency slider to a higher frequency and note what happens to the spacing of the calm and rippled regions. Record your observation. Step 5 — Return to default frequency, then change the simulation to LIGHT mode and observe the same double-source pattern using light waves — record whether the pattern looks similar or different in structure. Students record all data in a structured table provided on a half-A4 worksheet (teacher prints or draws on board): Point | Distance from S1 | Distance from S2 | Path Difference | Calm or Rippled?	points. What do you notice? [Wait 20 seconds.] Do not tell me yet — finish your measurements and we will discuss in the Explain phase. Circulate actively. For pairs who finish early, prompt: What would happen if the two sources vibrated COMPLETELY OUT OF STEP with each other from the start — predict, then try it in the simulation. For pairs who are struggling to read the ruler, model one measurement on the projector. Quote: Science is measurement. A beautiful pattern means nothing until we can describe it with numbers.	sound, water, and electromagnetic waves share the same underlying behaviour.	
Explain Phase 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook) Source: CK-12	The teacher leads a whole-class data consolidation. Five pairs read out their path-difference values for strongly rippled points; five pairs read values for calm points. The teacher records all values on the board in two columns. Students are invited to identify the pattern. The teacher then introduces the principle of superposition formally: When two waves meet at a point, the resultant displacement is the algebraic sum of the individual displacements. Two diagrams are drawn on the board side-by-side — Constructive Interference (crest	During data consolidation, say: I am looking for a pattern in these numbers — you tell me what it is. [Wait 25 seconds after writing all values on the board before accepting answers.] After students identify the pattern, affirm and formalise: Exactly. Whole-number multiples of wavelength give reinforcement. Half-integer multiples give cancellation. That is the principle of superposition in action. For the cold-call sequence: after naming each student, say: Take 30 seconds to organise your thoughts. You may look at your	Class data consolidation followed by teacher-drawn canonical diagrams and formal vocabulary introduction. The cold-call structure (with mandatory 30-second wait time) ensures that the conceptual explanation is student-articulated, not only teacher-transmitted. Connecting the soap-bubble phenomenon at the end of the Explain phase closes the loop opened in the Predict phase and rewards the curiosity generated by the photograph.	Quality of cold-call responses reveals whether students can (a) identify nodal lines visually, (b) state the path-difference condition for constructive interference in algebraic form, (c) predict the effect of wavelength change (addressing higher-order thinking), and (d) apply superposition to a new context (soap bubbles). Teacher notes which students struggle with (b) — these students are targeted in the Model Building phase.

🔍 Search ARES for similar readings	meets crest, trough meets trough; resultant amplitude = 2A; path difference = 0, λ, 2λ …) and Destructive Interference (crest meets trough; resultant amplitude = 0; path difference = 0.5λ, 1.5λ, 2.5λ …). Students copy both diagrams into their exercise books and annotate with the path-difference conditions. Vocabulary formalised: nodal lines (lines of destructive interference — permanently calm), antinodal lines (lines of constructive interference — permanently strongly rippled). Students label these on their earlier simulation sketch. Connection to soap bubbles: The teacher explains — the soap film has two surfaces. Light reflects from the outer surface and from the inner surface. These two reflected waves travel slightly different path lengths. Where the path difference equals a whole number of wavelengths of red light, red constructive interference occurs — you see red. Where it equals half a wavelength, red is cancelled — you see the complementary colour. This is why the colours form rings as the film thickness changes across the bubble. Cold-call sequence (4 students, 30-second wait between each): Student 1 — Point to a nodal line on the class diagram. Explain in one sentence why that line is permanently calm. Student 2 — What path difference condition must be satisfied for constructive interference? Give the condition in terms of λ. Student 3 — If the wavelength increases, what happens to the spacing between nodal lines? Use your simulation observation as evidence. Student 4 — How does	diagram. [Time 30 seconds visibly.] After each response, redirect to the class: Does everyone agree? Is there anything to add or correct? For Student 4, if the answer is incomplete, prompt: Think about what we said about path difference — does every part of the bubble have the same thickness? What does that mean for path difference at different points? Quote to close the phase: Interference is not random chaos — it is perfectly predictable. Where the maths says cancellation, the wave cancels. Where the maths says reinforcement, the wave reinforces. Every coloured ring on that soap bubble is obeying an equation.		

	wave interference help explain why the colours on a soap bubble are not all the same colour everywhere?			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher displays the current DQB (built since Lesson 1 and updated in Lesson 6 with the diffraction-based partial answer to the FM/AM tunnel mystery). Students are given two sticky notes each. On the YELLOW sticky note they write: One new thing I now know that I did not know at the start of this lesson — expressed as a complete sentence. On the GREEN sticky note they write: One question that interference raises for me — something I am still not sure about or want to know more about. Students place yellow notes on the EVIDENCE panel of the DQB and green notes on the STILL WONDERING panel. The teacher reads three or four yellow notes aloud and explicitly maps them to the anchoring phenomenon: Our model of the Kabete tunnel situation now has two layers — diffraction (from Lesson 6) AND interference (from today). FM signals that do enter the tunnel via diffraction can also undergo destructive interference if reflected waves from tunnel walls arrive with the right path difference. Our explanation is getting richer. The teacher then highlights one green sticky note that anticipates Lesson 8: Someone is asking — can interference happen within a single wave bouncing back on itself? That question will be answered in Lesson 8 on standing waves.	Hand out sticky notes before the DQB activity. Say: You have 3 minutes — one sentence on each note. I am going to read some of your yellow notes aloud, so make sure your sentence is complete and clear. After placing notes, read 3 yellow notes and explicitly say: This piece of evidence is now part of our shared class model. Point to the DQB and say: Look at where we started in Lesson 1. We had a matatu horn mystery and a radio mystery that seemed completely unrelated. Look at how much of the board is now filled with evidence. You are building a scientific model, piece by piece. Preview Lesson 8: The standing-wave question on that green note — can a single wave interfere with itself? — is one of the most beautiful ideas in physics. We tackle it next lesson.	The two-colour sticky-note protocol distinguishes evidence (new knowledge) from inquiry (new questions), keeping the DQB intellectually alive rather than becoming a static record. The teacher explicitly connects today's evidence to the two-layer explanation of the tunnel phenomenon, reinforcing storyline coherence and showing students that their accumulating model has real explanatory power.	Teacher scans yellow notes for quality — complete sentences with correct vocabulary (constructive, destructive, path difference, nodal, antinodal). Notes that contain misconceptions (for example, claiming that destructive interference destroys energy permanently) are photographed and addressed anonymously at the start of Lesson 8.
Model Building	Students spend the final 6 minutes adding an interference diagram to	Say: You have 6 minutes to update your unit model. Your	Cumulative model building ensures that each lesson leaves a	Teacher physically checks each model diagram during the 6-minute

Phase 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	their growing unit model (begun in Lesson 2 and updated each lesson). The diagram must include: (1) Two point sources S1 and S2 drawn and labelled; (2) At least two antinodal lines (labelled AN, drawn as solid arcs or lines radiating outward) and at least two nodal lines (labelled N, drawn as dashed lines); (3) A path-difference annotation on one antinodal line (e.g., path difference = λ) and one nodal line (e.g., path difference = 0.5λ); (4) A brief note in the margin connecting interference to the soap-bubble photograph and one sentence connecting it to the anchoring phenomenon (the Kabete tunnel radio signal). Students who finish early add a second connection: how could destructive interference theoretically make an FM signal disappear at a specific point inside the tunnel even if the wave had successfully diffracted in? The teacher circulates and checks that all four required elements are present, giving verbal feedback using the prompt: What is the path difference here, and is that constructive or destructive?	diagram must have all four elements I have listed on the board — write them down now. [Display the four bullet points.] I will come around and check. If I find a diagram missing one element I will ask you to add it before the lesson ends. For the anchoring-phenomenon sentence, prompt any student who writes only about soap bubbles: How does this also connect to the tunnel? Think about reflected FM waves inside the tunnel walls meeting the incoming diffracted FM waves. Quote when circulating: Your model is your thinking made visible. A future scientist reading your model should be able to understand interference without asking you a single question. Close the lesson by saying: Today we discovered that waves do not simply pass through each other politely — they add and subtract in a perfectly predictable way. That predictability is what makes wave physics so powerful. In Lesson 8 we will see what happens when a wave interferes with its own reflection — and the result will explain why a guitar string vibrates in sections.	permanent, visual artefact in the student record. The four required elements are scaffolded so that every student can meet the minimum, while the extension task (FM destructive interference inside the tunnel) challenges high-attaining students. The closing teacher quote reinforces scientific habits of mind — precision, predictability, and communication.	circulation for the four required elements. Students who are missing the path-difference annotation (the most abstract requirement) are identified for targeted re-teaching at the start of Lesson 8. The exit evidence is the completed interference diagram itself — collected in the exercise book rather than handed in, but visible to the teacher during circulation.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did the cold-call wait times (30 seconds each) feel uncomfortable — and did students use them productively? If students remain silent even after 30 seconds, the path-difference concept may need a visual re-explanation using a ruler and chalk on the floor at the start of Lesson 8. (2) Were path-difference measurements in the Observe phase accurate enough to reveal the whole-number versus half-integer pattern? If not, consider preparing a pre-measured screenshot of the PhET simulation with path distances already marked, to be used as a fallback in future lessons. (3) Did the soap-bubble connection land successfully — did students articulate the colour-ring explanation in their own words during cold-call 4? If fewer than half the class could do so, add a 3-minute revisit at the start of Lesson 8 before moving to standing waves. (4) Check sticky-note green questions for any that anticipate electromagnetic wave interference (e.g., Wi-Fi dead spots, mobile network black spots in Nairobi high-rises) — these can be woven into Lesson 10 on modulation as an enrichment link.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Two wave sources in PhET Wave Interference produced a pattern of alternating strongly rippled regions and calm regions radiating outward from the sources. Strongly rippled points had path differences close to whole-number multiples of the wavelength. Calm points had path differences close to half-integer multiples of the wavelength. Increasing frequency made the pattern finer with more closely spaced nodal lines.
What did I learn?	When two waves meet, the resultant displacement at any point is the algebraic sum of the individual displacements (principle of superposition). Constructive interference occurs where path difference = $n\lambda$ ($n = 0, 1, 2, \ldots$), producing antinodal lines of maximum amplitude. Destructive interference occurs where path difference = $(n + 0.5)\lambda$, producing nodal lines of zero amplitude. These conditions are fixed in space for two coherent sources, creating a stable interference pattern.
How does this explain the phenomenon?	The coloured rings on a soap bubble form because light reflects from both surfaces of the thin soap film. The two reflected waves have a path difference that depends on the local film thickness. Where the path difference matches a whole number of wavelengths of a particular colour, that colour undergoes constructive interference and is seen brightly. Where the path difference is a half-integer multiple of that wavelength, the colour is cancelled by destructive interference. Different colours have different wavelengths, so different colours are constructively reinforced at different thicknesses, producing the ring pattern. For the Kabete tunnel, FM waves that do diffract into the tunnel can also undergo destructive interference when reflected waves from the tunnel walls meet the incoming waves with a path difference of half a wavelength — adding a second mechanism to the FM signal loss first explained by diffraction in Lesson 6.

LESSON 8 (40 minutes): Standing Waves: When a Wave Meets Itself

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of this lesson, students will be able to explain how standing waves form on a string under fixed boundary conditions, identify nodes and antinodes, calculate the wavelengths of the first three harmonics for a string of known length, and connect standing wave patterns to real-world vibrating systems such as guitar strings and radio antenna resonance.
Knowledge	Students know that a standing wave forms when two waves of equal frequency and amplitude travel in opposite directions along the same medium and superpose. Nodes are points of permanent zero displacement; antinodes are points of maximum displacement. The fundamental frequency (1st harmonic) has one antinode and two nodes. The nth harmonic has n antinodes. For a string of fixed length L, the wavelength of the nth harmonic is given by lambda-n equals 2L divided by n.
Skills	Students can use the PhET Waves on a String simulation to systematically vary frequency and observe the formation of standing wave patterns. They can measure and record node positions, antinode positions, and wavelength for each harmonic. They can calculate wavelength using the formula lambda-n = 2L/n and cross-check using v = f multiplied by lambda.
Attitudes	Students appreciate that patient, systematic investigation of a simulation can reveal patterns that lead to mathematical generalisations. They show curiosity about why only certain frequencies cause large vibrations in a medium, and they connect this curiosity to phenomena they have personally observed, such as a guitar string or a singing bowl at a Maasai cultural festival.
Key Inquiry Question	How do waves behave in different media and under different conditions? This lesson addresses the special condition of two identical waves travelling in opposite directions in a bounded medium, showing that the result is not a travelling wave at all but a stationary pattern of nodes and antinodes.
Purpose in Storyline	Lesson 8 extends the superposition principle established in Lesson 7 to the special case where reflected waves interfere with incident waves in a bounded medium. This creates standing waves — a concept that will resurface in Lesson 9 (Doppler Effect) when discussing resonance in instrument chambers, and in Lesson 10 (AM and FM modulation) when discussing antenna length tuned to a specific wavelength. It also deepens the model students are building toward fully explaining the anchoring phenomenon: the matatu horn and the radio tunnel mystery both involve wave behaviour shaped by boundaries and wavelength.
Safety Notes	No significant physical hazards. Ensure screen brightness is set comfortably to protect eyesight during extended simulation use. If a physical string-and-vibrator demonstration is available as an extension, ensure the vibrator is clamped securely to the bench and students do not touch the string while it is vibrating.

B. LESSON OVERVIEW

Students open the lesson by recalling the principle of superposition and being shown a short photo of a guitar player at the Koroga Festival in Nairobi, prompting the question: why does a plucked guitar string vibrate in distinct sections rather than a single continuous wave? In the Predict phase, students write predictions about what happens when a wave on a string reflects off a fixed wall and meets the incoming wave. In the Observe phase, students use PhET: Waves on a String (set to Oscillate mode with one end fixed) to discover that only certain frequencies produce stable patterns. In the Explain phase, students formalise the concepts of nodes, antinodes, and harmonics, derive the wavelength formula lambda-n = 2L/n, and solve calculation problems set in Kenyan contexts. The DQB is updated to record that standing waves explain why the guitar string vibrates in sections and to flag that antenna resonance (connected to FM and AM broadcasting) also depends on standing wave principles. Students update their cumulative model diagrams.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Wave on a String Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Standing Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects a photograph of a guitarist performing at the Koroga Festival at Two Rivers Mall in Nairobi. Students are asked to look at the blurred motion of a plucked guitar string and recall what they already know about waves on strings from Lessons 2 and 3. Each student writes a 3-sentence prediction in their exercise books in response to the prompt on the board: A wave on a guitar string travels to the end of the string and reflects back. What do you predict happens when the reflected wave meets the original wave? Will the result be the same as the double-source interference you observed in Lesson 7? Students then share predictions with a neighbour in a Think-Pair exchange lasting 90 seconds before one pair is cold-called to share with the class. The teacher records two or three contrasting predictions on the board without evaluating them, explicitly telling students: we are going to use evidence from the simulation to decide which prediction is closest to what actually happens.	Teacher opens with the Koroga Festival guitar photograph on the projector and says: Last lesson we saw what happens when two separate sources send waves toward each other. Today I want you to think about a different situation — one wave source, one fixed end. The wave bounces back. What then? Give students 2 minutes of silent writing time for their predictions. After 90 seconds of pair discussion, cold-call one pair using the class register. Avoid affirming or denying predictions. Say: That is an interesting prediction — keep it in mind as you observe the simulation. WAIT TIME: After cold-calling, pause for 5 full seconds before accepting any further comment, allowing all students to process the shared prediction. Write the two most divergent student predictions on the board under the heading: Our Predictions — We Will Test These.	Prediction journaling activates prior knowledge of superposition from Lesson 7 and creates cognitive dissonance: students who predict a pattern identical to travelling-wave interference will be surprised by the stability and stationarity of standing waves. Writing before discussing prevents anchoring on the first voice heard.	Teacher scans written predictions during the pair-share. Look for whether students mention superposition, cancellation, or reinforcement from Lesson 7. Students who write only that the string will vibrate more have not yet connected to superposition and will need targeted questioning during the Observe phase. No formal score — diagnostic only.
Observe Phase 📹 **VIDEO:** Video: Wave on a String Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:**	Students open PhET: Waves on a String on laptops or tablets (or observe the teacher projection if devices are 1-per-3 students). Settings: select Oscillate, set one end to Fixed End, damping to zero, tension to high. Step 1 (5 minutes): Students freely explore by slowly increasing frequency from 0 Hz upward and recording in a table what they see at each approximate frequency. They are	Circulate continuously. At approximately the 4-minute mark, if groups have not yet found a stable pattern, prompt: Try frequencies between 1.5 Hz and 4 Hz and watch for a pattern where some parts of the string stop moving altogether. After most groups have found at least two harmonics, pause the class with a hand signal and ask one student to project their screen: Can you	The structured data table scaffolds systematic observation and prevents random clicking. The pattern-statement task (lambda = _____ times L) bridges raw observation to algebraic generalisation, preparing students for the formal derivation in the Explain phase. Sketching each harmonic pattern creates visual memory anchors that will appear again in the model-building phase.	Teacher checks tables during circulation. Target check: do students correctly record that the 1st harmonic has 1 antinode and 2 nodes, the 2nd harmonic has 2 antinodes and 3 nodes, the 3rd harmonic has 3 antinodes and 4 nodes? Students who record equal numbers of nodes and antinodes have a misconception to address during Explain. Teacher makes a mental note of two students to

[Standing Waves (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	told: note frequencies at which a clear, stable, repeating pattern appears — count and sketch the number of regions of large motion (antinodes) and the points that do not move (nodes). Step 2 (5 minutes): Students systematically record data for the 1st, 2nd, and 3rd clear stable patterns they find, filling in a structured table with columns: Harmonic number, Approx. frequency (Hz), Number of antinodes, Number of nodes, Sketch of pattern. Step 3 (2 minutes): Students measure (using the simulation ruler tool) the wavelength in each harmonic by noting the distance between successive nodes and doubling it, and record this against string length L shown in the simulation. They compare wavelength to string length for each harmonic and write a pattern statement: In harmonic 1, lambda equals _____ times L; in harmonic 2, lambda equals _____ times L; etc.	describe what you see at this frequency? Count the still points out loud with me. WAIT TIME: After asking the class to count nodes together, wait 4 seconds before confirming. If a student misidentifies a node as an antinode, say: Look at that point again — does it ever move, or is it always still? Let the simulation answer, not me. Prompt the wavelength measurement step by saying: Use the ruler tool — measure from one still point to the next still point. What is that distance compared to the full length of the string? Write a sentence.		cold-call during the Explain phase.
Explain Phase 📹 **VIDEO:** [Video: Wave on a String](#) Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Standing Waves (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	The teacher projects a clean standing wave diagram showing a string of length L with the 1st, 2nd, and 3rd harmonics drawn side by side. Students label their own diagrams in their books as the teacher explains. Key explanation points covered: (1) Definition of a standing wave — formed by superposition of two waves of equal frequency and amplitude moving in opposite directions; unlike a travelling wave, a standing wave does not transfer energy along the string. (2) Nodes — points of permanent destructive interference, always zero displacement. (3) Antinodes — points of permanent constructive interference, maximum	Begin the Explain phase by saying: Let us look at what your data is telling us. In every case where you saw a stable pattern, how many half-waves fitted into the string? Count with me on the diagram. Formalise the derivation on the board step by step, narrating: If n half-wavelengths fit into length L, then L equals n times lambda over 2. Rearranging — who can tell me what lambda equals? WAIT TIME: After posing the rearrangement question, wait 6 seconds. Accept a student answer. If incorrect, say: Let us rearrange together — multiply both sides by 2, then divide both sides by n. For Problem A, cold-call the student noted earlier for targeted support	The side-by-side harmonic diagram and the step-by-step derivation connect the visual pattern from the simulation to the algebraic formula. Problem B deliberately links standing waves to the FM antenna, advancing the narrative toward the full resolution of the anchoring phenomenon in Lesson 10. This cross-lesson coherence reinforces that the unit is building one unified explanation, not isolated topics.	Board solutions for Problems A and B are checked publicly. Teacher asks the class: Thumbs up if the working on the board matches yours; thumbs sideways if you got a different number. If more than 4 thumbs are sideways, pause and rework the calculation with explicit unit-checking before moving on. Exit criterion for this phase: every student should have correct answers and derivation written in their exercise book before the DQB update.

	displacement. (4) Deriving the wavelength formula: for the nth harmonic, n half-wavelengths fit into the string length L, therefore L = n multiplied by lambda-n divided by 2, rearranging to give lambda-n = 2L divided by n. Students copy this derivation step by step. (5) Two worked problems in Kenyan contexts: Problem A — A guitar player at the Koroga Festival Festival tunes a string of length 0.65 m. What is the wavelength of the 1st, 2nd, and 3rd harmonics? (Answers: 1.30 m, 0.65 m, 0.43 m.) Problem B — A radio antenna designed to resonate at the FM frequency of Classic 105 (105.2 MHz) acts as a half-wave dipole. If the antenna is 1.4 m long, what frequency does it resonate at? (Students use v = f multiplied by lambda with lambda = 2 multiplied by 1.4 m = 2.8 m and v = 3 multiplied by 10 to the power 8 m/s; f = approximately 107 MHz — close to Classic 105, reinforcing the anchoring phenomenon connection.) Students solve Problem A individually (3 minutes), then Problem B in pairs (3 minutes). Two students write solutions on the board.	and say: Read out your working — tell us what L is and which harmonic you are calculating. For Problem B, make the explicit link to the anchoring phenomenon: Notice — an FM antenna is designed to be exactly half a wavelength long. That is a standing wave condition built into the hardware. This is one reason FM signals behave differently from AM signals in tunnels — their hardware and wavelengths are tuned to very different scales.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Wave on a String] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:**	Students return to the class Driving Question Board displayed at the front. The teacher reads aloud the question placed on the DQB in Lesson 2: Why does the guitar string vibrate in sections rather than as one smooth wave? Students are asked to write a three-sentence answer on a sticky note using the sentence starters provided: A standing wave forms because... Nodes are the places where... This explains the guitar string because... Students post	Guide the DQB update by saying: We now have enough evidence to close one question and open a new one. Point to the guitar-string question and say: I want three sentences — not one word answers. What causes a standing wave? What is a node? And how does this explain sections of vibration? After students post sticky notes, read one aloud and model how to evaluate it: This note says nodes form because of destructive interference — is that	The green-dot closure ritual gives students a visible sense of epistemic progress — a question that was genuinely open in Lesson 2 is now answered by their own evidence. The new prompt card maintains productive tension toward the anchoring phenomenon resolution, ensuring the DQB remains a living document rather than a static display.	Teacher reads at least 6 sticky notes during or after posting. Look for: correct use of the word superposition, correct identification of destructive interference at nodes, and a clear link to the guitar string or another vibrating object. Sticky notes that mention only the word standing wave without explaining the cause indicate surface-level understanding and flag those students for additional questioning during Model Building.

Standing Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	their sticky notes on the DQB next to the original question, physically marking it as answered with a green dot. The teacher then adds a new prompt card to the DQB in the In Progress column: If FM antenna length is tuned to a specific wavelength using standing wave principles, how does this connect to why FM fades in the Kabete tunnel but AM does not? Students briefly discuss in pairs whether they can yet fully answer this question, then agree as a class that it remains partially open — they have a new piece of the puzzle but the full answer will come in Lesson 10.	consistent with what we derived? Yes — let us give it a green dot. Then introduce the new prompt card and say: We now know FM antennas are tuned using half-wave standing wave conditions. Does that fully explain the tunnel mystery? Thumbs up if yes, thumbs sideways if no or partly. Accept both and say: Hold that question — Lessons 9 and 10 will close it. WAIT TIME: Before introducing the new DQB card, pause for 5 seconds after reading the existing guitar-string question to allow students to mentally rehearse their sticky note sentence.		
Model Building Phase 📹 **VIDEO:** Video: Wave on a String Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Standing Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open their cumulative Unit Model diagram (started in Lesson 2 and updated in every subsequent lesson). They add a new section labelled Standing Waves. In this section they must draw and label: (a) A diagram of the 1st, 2nd, and 3rd harmonics on a string of fixed length L, with nodes and antinodes clearly labelled and the wavelength of each harmonic expressed in terms of L. (b) An arrow connecting this diagram to the superposition principle established in Lesson 7, with the annotation: Standing wave = superposition of two coherent waves moving in opposite directions. (c) A small sketch of a guitar string and a radio dipole antenna annotated with the phrase: Resonance — the system vibrates strongly only at frequencies whose wavelength fits the standing wave condition. (d) One sentence connecting standing waves to the anchoring phenomenon: FM antennas are tuned by standing wave conditions	Circulate and provide structured feedback using the unit model rubric (accuracy of diagram labels, correct formula, presence of connections to prior lessons, presence of a link to the anchoring phenomenon). Say to any student whose diagram lacks a connection arrow: Your standing wave diagram is accurate — now add an arrow back to your superposition diagram from Lesson 7. They are the same idea — one wave is just reflected. To students who finish the extension calculation, say: What does that number — 264 metres — tell you about AM broadcasting infrastructure in Kenya? Why would KBC need such a tall tower? This prompts a qualitative discussion that anticipates Lesson 10. Spend the final 2 minutes asking one student to hold up their model and describe the new section to the class in two sentences. WAIT TIME: After asking for the two-sentence description, wait 5 seconds before prompting further,	Cumulative model building makes the knowledge structure visible and forces students to represent connections across lessons, not just within a single lesson. The extension calculation creates a memorable quantitative anchor — a 264-metre AM antenna versus a 1.4-metre FM dipole — that will make the AM/FM comparison in Lesson 10 feel concrete and surprising. Labelling the diagram section explicitly as connected to Lessons 7 and 10 rehearses the integrative thinking required for the Lesson 11 poster.	Model diagrams are the primary artefact of this phase. Teacher checks for: correct node/antinode labelling, correct formula lambda-n = 2L/n written on the diagram, at least one cross-lesson connection arrow, and the anchoring phenomenon sentence. Students who are missing the formula on their diagram are prompted before leaving: Before you close your book, write the formula lambda-n equals 2L divided by n next to your diagram and circle it. This is the exit condition for the lesson.

	to a specific wavelength, which is one reason FM and AM signals behave differently near boundaries like tunnel walls. Students who finish early are challenged to calculate the length of an AM 567 kHz KBC Radio Kenya antenna if it is also a half-wave dipole (lambda = v/f = 3 x 10^8 / 567000 approximately 529 m; half-wave = 264 m — illustrating vividly why AM towers are enormous compared to FM dipoles).	allowing the student to organise their response.		

D. TEACHER REFLECTION
After the lesson, consider: Did most students successfully identify all three harmonics in the simulation, or did groups get stuck at the 2nd harmonic? Were the two worked problems at the right challenge level — if Problem B (the antenna calculation) felt too abstract, next time introduce it as a teacher-led example rather than a pair task. Check the sticky notes on the DQB — if fewer than half the class correctly mentioned destructive interference when describing nodes, plan a 5-minute re-explanation at the start of Lesson 9 using the superposition diagram. Reflect on whether the extension calculation (AM antenna length) sparked genuine curiosity — if it did, open Lesson 10 with the question: If an AM antenna is 264 metres tall, how does KBC afford to broadcast nationally? This keeps the Kenyan context vivid and locally meaningful.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the PhET simulation, when the frequency was set to specific values, the string stopped looking like a travelling wave and instead formed a pattern of sections — some parts moved up and down with large amplitude while other points never moved at all. When the frequency was increased to the next stable pattern, one more section appeared.
What did I learn?	These stable patterns are called standing waves. They form when a wave reflects off the fixed end and superposes with the incoming wave. Points of permanent zero displacement are called nodes (permanent destructive interference) and points of maximum displacement are called antinodes (permanent constructive interference). For a string of length L, the wavelength of the nth harmonic is given by lambda-n = 2L divided by n.
How does this explain the phenomenon?	A guitar string vibrates in sections because only certain frequencies — those whose wavelength satisfies the condition that n half-wavelengths fit exactly into the string length — produce stable standing waves. At other frequencies, destructive interference prevents a stable pattern from building up. This same standing wave principle determines the length of FM and AM radio antennas: an FM dipole for Classic 105 at 105.2 MHz is about 1.4 m long, while an AM dipole for KBC at 567 kHz would be about 264 m long. This dramatic difference in scale is one more reason FM and AM waves behave so differently near tunnel boundaries — a thread we will complete in Lesson 10.

LESSON 9 (80 minutes): Diffraction and Interference: Why Waves Bend Around Obstacles and Cancel or Reinforce Each Other	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how waves diffract around obstacles and through gaps, and how two overlapping waves produce interference patterns — linking these behaviours to why AM radio survives the Kabete tunnel while FM does not.
Knowledge	By the end of the lesson, the learner should be able to: (a) define diffraction as the bending of waves around obstacles or through gaps; (b) state the conditions that maximise diffraction (gap ≈ wavelength); (c) distinguish between constructive and destructive interference; (d) apply the principle of superposition to predict the resultant displacement of two overlapping waves.
Skills	By the end of the lesson, the learner should be able to: (a) use the LabXchange Wave Interference simulation to gather quantitative evidence of constructive and destructive interference; (b) draw superposition diagrams to predict resultant wave patterns; (c) analyse PhET Sound simulation data to relate interference to real audible dead-spots; (d) construct an annotated model diagram linking diffraction and interference to the Kabete tunnel phenomenon.
Attitudes	The learner appreciates that careful evidence gathering — using both digital simulations and analytical reasoning — is necessary before drawing conclusions about puzzling real-world observations such as the FM signal blackout.
Key Inquiry Question	How do waves behave in different media and under different conditions?
Purpose in Storyline	Lessons 1–8 established wave properties (speed, frequency, wavelength, reflection, refraction, the Doppler effect). Lesson 9 now addresses the remaining half of the Kabete tunnel mystery: not only does FM bend less (shorter wavelength, covered in refraction/diffraction intro), but inside the tunnel the signal also breaks up due to interference between reflected waves and the incoming signal. By the end of this lesson students have enough evidence to attempt a full explanation of BOTH phenomena in the driving question.
Safety Notes	All activities use digital simulations (LabXchange and PhET) and paper-based superposition drawing. No hazardous materials or electrical apparatus are used. Ensure screen brightness is comfortable to avoid eye strain during extended simulation work. If the school uses a shared computer lab, students must not alter system settings or download unauthorised software.

B. LESSON OVERVIEW
Students move through a PREDICT → OBSERVE → EXPLAIN → DQB → MODEL sequence focused on diffraction and interference. They begin by predicting what happens to a water wave passing through a narrow gap (connecting to why sound enters a room through a door even when you cannot see the speaker). They then gather evidence using the LabXchange Wave Interference simulation (constructive/destructive interference) and the PhET Sound simulation (audible dead-spots caused by destructive interference). In the Explain phase, students apply the Principle of Superposition to drawn wave diagrams and map their findings onto the Kabete tunnel context — FM waves ($\lambda \approx 2.85$ m) barely diffract around the tunnel mouth compared with AM waves ($\lambda \approx 300$ m), AND reflected FM waves inside the tunnel create destructive interference dead-zones, together explaining the signal blackout. The lesson closes with DQB updates and a cumulative model revision that now incorporates diffraction and interference arrows alongside the Doppler, reflection, and refraction annotations from earlier lessons.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are presented with the 'Door-Gap Challenge': the teacher plays a short audio clip of a Nairobi street vendor shouting 'Nakuru! Nakuru! Nakuru!' from outside a partially open classroom door while students sit inside with their eyes closed. Students record in their science notebooks: (1) Can you hear the vendor even though the sound source is not in line-of-sight? (2) Draw a top-down diagram of the door gap and show with arrows how you think the sound travels. (3) Predict: if the door gap were made very narrow — say, 2 cm — would you hear more or less? Explain your reasoning. Students then make a second prediction specifically for the driving question: 'In your notebook, sketch the Kabete tunnel mouth. Draw FM waves (short) and AM waves (long) approaching it. Predict which type diffracts more into the tunnel and why.' Pairs share their sketches with the pair behind them (Think-Pair-Share) before whole-class cold-call.	Teacher opens with the audio clip (or mimics the vendor call if no device is available). SAY: 'Eyes closed. Listen carefully. Where is the sound coming from relative to the door?' After 20 seconds, SAY: 'Open your eyes. In your notebooks, answer the three prediction questions — you have 4 minutes. Go.' Circulate and note 2–3 interesting or divergent sketches to reference later. After 4 minutes, SAY: 'Share your door-gap diagram with the pair behind you — 90 seconds.' After sharing, cold-call THREE students (choose one who drew straight arrows, one who drew curved arrows, one uncertain): 'Amina, describe your diagram.' [WAIT 10 s] 'Brian, yours was different — explain.' [WAIT 12 s] 'Zawadi, which diagram do you think is more accurate and why?' [WAIT 15 s]. Do NOT confirm correct answers yet. Transition: 'You've made your predictions. Now let's gather evidence to test them.'	Predict-Observe-Explain (POE) Launch — activating prior knowledge about sound propagation and creating cognitive dissonance. By asking students to commit a prediction to paper before seeing evidence, the teacher ensures that new observations will be processed as confirming or contradicting a held idea, deepening encoding. The door-gap scenario anchors diffraction in a familiar Kenyan daily-life context (street vendor outside a school) before formalising it.	Observable indicator: Each student's notebook contains a labelled top-down diagram with arrows attempting to show the sound path through the door gap, AND a second sketch showing FM vs AM waves at the tunnel mouth. Teacher scans 6–8 notebooks during circulation and flags any student who draws sound travelling only in straight lines (no bending) — these students need targeted questioning during the Observe phase. Key diagnostic question: 'Does the student's diagram show any bending at the gap edges?'
Observe Phase 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook) Source: CK-12	Students work in pairs at devices (computers, tablets, or teacher-projected screen) across two sequential simulation activities. ACTIVITY A — LabXchange Wave Interference (12 min): Students open the Wave Interference simulation and complete a structured evidence-collection table with columns: [Gap Width Setting | Wave Pattern Observed | Spreading: Wide/Narrow | Sketch of wavefront shape]. They test at least THREE gap widths: (i) gap much larger than wavelength, (ii) gap approximately equal to wavelength, (iii) gap much smaller	Before releasing to simulations, SAY: 'You are scientists collecting evidence to answer our driving question. Your job is to fill in your evidence table — do not skip the sketches, those are your data.' Circulate during Activity A; at the 6-minute mark, prompt any pair still on the first gap width: 'You need at least three gap settings before we debrief. Move to the next setting now.' During Activity B, SAY to the whole class at the 5-minute mark: 'Raise your hand if you have found a quiet spot — a dead-zone.' [Count hands aloud: 'Seven pairs — excellent. Three	Structured Evidence Collection with Two-Source Triangulation — students use two independent simulations (wave tank analogy via LabXchange; acoustic analogy via PhET) to gather converging evidence for the same phenomenon. Recording data in a structured table prevents superficial clicking and ensures students externalise observations. The class-shared data table on the board introduces a community-of-science norm: individual data becomes class evidence.	Observable indicator 1: Evidence table has entries for all three gap-width conditions with sketches. Observable indicator 2: Interference diagram has at least two 'C' and two 'D' labels in correct positions. Observable indicator 3: PhET dead-spot positions recorded with at least three entries. Teacher checks connecting sentence in evidence log for the phrase 'destructive interference' or equivalent conceptual language. Flag any student who describes the dead-spots as 'the sound disappears' without referencing overlapping

🔍 Search ARES for similar readings	than wavelength. Students record which condition produces the most spreading (diffraction). They then switch to the two-source interference mode, identify at least TWO nodal lines (destructive interference) and TWO antinodal lines (constructive interference), and annotate a printed or drawn wavefront diagram with 'C' (constructive) and 'D' (destructive) labels. ACTIVITY B — PhET Sound Simulation (10 min): Students set up two in-phase speakers separated by 1 m in the simulation and use the 'listen' tool to move across the wavefront. They record the positions of at least THREE loud regions and THREE quiet regions (dead-spots). They note: 'The dead-spots are caused by ____.' A class data table is filled in on the board (or shared doc) as pairs report their quiet-spot positions. Final 3 min: Students write one sentence in their evidence log connecting the PhET dead-spots to the Kabete tunnel FM blackout.	pairs still looking — try moving the listener tool slowly across the wavefront.'] At the 20-minute mark of this phase, pause the class: cold-call TWO pairs to share their most surprising observation. SAY: 'Fatuma, what surprised you most in the interference simulation?' [WAIT 12 s]. 'Korir, same question.' [WAIT 12 s]. Then direct: 'Write your connecting sentence about the tunnel NOW — one sentence, evidence log, 2 minutes.'		waves — prompt: 'Two waves are arriving at that point — what are they doing to each other?'
Explain Phase 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook) Source: CK-12 🔍 Search ARES for similar	Students transition from evidence to explanation in three steps. STEP 1 — Superposition Drawing Exercise (8 min): Each student receives (or draws) a worksheet showing two sinusoidal waves of equal amplitude and frequency: Wave A and Wave B. In Scenario 1, the waves are in phase (crests align). In Scenario 2, the waves are exactly out of phase (crest aligns with trough). Students add the displacements point-by-point at 8 marked positions to draw the resultant wave for each scenario, then label: 'Constructive interference — amplitude doubles' and 'Destructive interference —	Distribute or project the superposition worksheet. SAY: 'You have 8 minutes to add the waves together point by point. This is not guessing — you are doing the mathematics of waves.' Circulate; if students are unsure about point-by-point addition, model ONE point on the board: 'At this position Wave A has displacement +2 cm, Wave B has displacement +2 cm. What is the resultant? [WAIT 8 s] Yes — +4 cm. Now you do the rest.' For the concept map, use cold-call to populate each node: 'Njenga, what condition maximises diffraction?' [WAIT 12 s]. 'Akinyi, what does the	Bridging Representations — students move from simulation visuals (Phase 2) to mathematical superposition drawing (step 1) to verbal/symbolic concept mapping (step 2) to written explanation (step 3). Each representation reinforces the same underlying idea at increasing levels of abstraction, ensuring the explanation is not surface-level recall but multi-representational understanding. The concept map explicitly chains physical quantities (wavelength values, tunnel dimensions) to the observed phenomenon, making the reasoning falsifiable and specific.	Observable indicator 1: Superposition worksheet — resultant wave in Scenario 1 has doubled amplitude; resultant in Scenario 2 is a flat line. Teacher collects 5 random worksheets for rapid check. Observable indicator 2: Exit sentence contains TWO distinct reasons (diffraction AND interference) and correctly identifies AM as the survivor. Teacher reads 4–5 exit sentences aloud (anonymised) to the class as a quick norm-setting check before moving to Phase 4. Flag students who conflate diffraction and interference (treat them as the same thing) — they need the

readings	amplitude = zero.' STEP 2 — Class Concept Map (7 min): Working as a whole class with teacher facilitation, students co-construct a concept map on the board connecting: Diffraction → [gap ≈ wavelength] → maximum spreading; Two coherent wave sources → Principle of Superposition → Constructive interference / Destructive interference; FM wavelength (2.85 m) vs Tunnel mouth width (~4 m) → poor diffraction; Reflected FM waves inside tunnel + incoming FM waves → destructive interference → dead-zones; AM wavelength (~300 m) vs Tunnel mouth (~4 m) → excellent diffraction → AM survives. STEP 3 — Exit sentence completion (5 min): Students complete in notebooks: 'The FM radio signal cuts out in the Kabete tunnel because (a) __________ [diffraction reason] and (b) __________ [interference reason], whereas AM survives because __________.'	Principle of Superposition say?' [WAIT 15 s]. 'Oduya, given FM wavelength is about 2.85 metres and the tunnel mouth is about 4 metres wide, will FM diffract a lot or a little?' [WAIT 15 s]. After the concept map is complete, SAY: 'Copy this into your notes — it is your explanation scaffold.' For exit sentences, SAY: 'Complete both blanks. You have 4 minutes. Use the concept map if you need it, but try from memory first.'		concept map re-explained with explicit contrast.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook)	Students return to the class Driving Question Board, which has been maintained since Lesson 1. Each student receives two sticky notes (or writes in their DQB section of their notebook). On the FIRST sticky note (green = evidence gained): students write one piece of evidence from today's lesson that helps answer either DQB question ('What properties do all waves share?' or 'Why does the FM signal disappear in the tunnel?'). On the SECOND sticky note (orange = remaining questions): students write one question that today's lesson has NOT yet answered or a new	Hand out sticky notes (or direct students to the DQB page in their notebooks). SAY: 'Green note: one piece of evidence from today that moves us closer to answering our driving question. Orange note: one thing you still don't understand or a new question today raised. You have 3 minutes.' After posting, read THREE orange notes: 'This student asks: why don't FM waves just go around the whole tunnel wall? Interesting — class, 30 seconds, who has a partial answer?' [WAIT 20 s, cold-call one student]. Update the progress tracker visibly on the board. SAY: 'We have now gathered evidence	Driving Question Board Curation — a metacognitive public-record strategy. By physically posting evidence and questions, students externalise their epistemic state, making both growth and remaining gaps visible to themselves and peers. Reading orange (unresolved) notes aloud normalises productive confusion and models scientific practice: good investigations generate new questions. Mapping completed evidence nodes on the progress tracker builds narrative coherence across the 12-lesson sequence.	Observable indicator: Every student has a green sticky note that references a specific concept from today (diffraction, interference, superposition, wavelength comparison) — not a vague statement like 'I learned about waves.' Teacher scans all green notes; any that are vague get a brief individual prompt: 'Can you make this more specific — which property, which evidence?' Orange notes are monitored for misconceptions that need to be addressed in Lessons 10–11 planning.

Source: CK-12 🔍 Search ARES for similar readings	question it has raised. Students post their notes in the correct columns on the board. The teacher then reads three orange sticky notes aloud and invites brief responses from the class (30 seconds each). The teacher explicitly maps today's lesson onto the DQB progress tracker: 'We now have evidence for: rectilinear propagation ✓, reflection ✓, refraction ✓, Doppler effect ✓, diffraction ✓, interference ✓. What is still missing from a complete explanation?' Students name: resonance, stationary waves (upcoming Lessons 10–11).	on six wave properties. Lessons 10 and 11 will give us the last two pieces — resonance and stationary waves — and then in Lesson 12 we will put the entire explanation together. You are almost there.'		
Model Building Phase 📹 **VIDEO:** Reflection and Interference of Mechanical Waves in One-Dimension - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Wave Interference (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative wave model diagram, which they have been revising since Lesson 1. This diagram shows the Nakuru–Nairobi highway scene (Doppler effect annotations) and the Kabete tunnel scene (reflection and refraction annotations added in earlier lessons). Today, students add TWO new annotated layers to the tunnel scene: (1) DIFFRACTION LAYER — at the tunnel mouth, students draw wavefronts for FM (tightly spaced, λ ≈ 2.85 m) and AM (widely spaced, λ ≈ 300 m) waves arriving at the tunnel opening (~4 m wide). They annotate: 'FM: gap ≈ 1.4λ → limited diffraction, weak signal inside' and 'AM: gap ≈ 0.013λ → strong diffraction, signal bends into tunnel.' (2) INTERFERENCE LAYER — inside the tunnel, students draw incoming FM wavefronts and reflected FM wavefronts from the tunnel walls converging. They mark two or three 'X' points labelled 'Destructive interference → FM dead-zone' and one or two '★'	SAY: 'Get out your cumulative model diagrams. You are adding two new layers today — diffraction at the tunnel mouth and interference inside the tunnel. Use the wavelength values we calculated: FM ≈ 2.85 m, AM ≈ 300 m, tunnel mouth ≈ 4 m. You have 7 minutes.' Circulate and prompt: 'Show me where the dead-zones would be — mark them with an X.' If a student draws FM and AM identically, ask: 'The tunnel mouth is 4 metres. The FM wavelength is 2.85 metres. The AM wavelength is 300 metres. Which one fits through the gap more easily in terms of diffraction?' [WAIT 12 s]. At 7 minutes, SAY: 'Pair-share — 90 seconds. Use the feedback prompt on the board: one thing clear, one thing needing a label.' Close with: 'Your model now explains five of the six wave behaviours we have studied. One more lesson on resonance and stationary waves, and your model will be complete.'	Cumulative Model Revision (CMR) — students maintain a single evolving diagram across all 12 lessons rather than drawing a new one each time. This strategy mirrors authentic scientific modelling practice: models are refined as evidence accumulates, not discarded and restarted. The explicit layering (diffraction layer, interference layer) ensures students see how today's concepts integrate with — rather than replace — prior knowledge. The two-sentence caption forces precision: students must distinguish the two mechanisms rather than conflating them.	Observable indicator 1: Cumulative model now contains BOTH a diffraction annotation at the tunnel mouth (with FM and AM wavelength values cited) AND an interference annotation inside the tunnel (with X and ★ markers). Observable indicator 2: Two-sentence model caption correctly separates the diffraction explanation from the interference explanation. Teacher collects 6 model diagrams (random sample) to review before Lesson 10 and identify any students who still conflate the two mechanisms or who have incorrect wavelength comparisons. These students receive a targeted feedback slip at the start of Lesson 10.

	points labelled 'Constructive interference → FM hot-spot.' Students write a two-sentence model caption: 'This model shows that the FM blackout is caused by two compounding effects: [sentence 1 — diffraction]. [sentence 2 — interference].' Final 2 minutes: pair-share model diagrams for peer feedback using the prompt 'One thing I see clearly / One thing that needs a label.'			

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) EVIDENCE QUALITY — Did students' superposition worksheets show genuine point-by-point addition, or were they guessing the shape? If more than 30% of collected worksheets showed incorrect resultant waves, plan a 5-minute re-teach at the start of Lesson 10 using a physical rope demonstration before moving to resonance. (2) DQB ORANGE NOTES — What were the most common unresolved questions? If many students asked 'why does diffraction depend on gap size relative to wavelength?' this gap needs to be addressed in the Lesson 10 warm-up, as it underpins resonance cavity understanding. (3) SIMULATION ACCESS — If device access was limited and only the teacher could project, did students still complete their own evidence tables, or did they passively watch? If passive watching occurred, redesign the activity as a whole-class 'stop-and-sketch' protocol where the teacher pauses the simulation every 3 minutes for students to draw what they see. (4) TUNNEL MODEL ACCURACY — Did students' model diagrams correctly distinguish FM diffraction (limited) from AM diffraction (extensive) using specific wavelength values? If students wrote only qualitative statements ('FM is shorter so it diffracts less') without citing the 2.85 m and 300 m values, the next lesson should begin with a brief quantitative wavelength calculation warm-up using $v = f\lambda$ to reinforce the habit of grounding qualitative claims in numbers. (5) KENYA CONTEXT — Was the Kabete tunnel context motivating for students? Note which students engaged most actively — these students could serve as 'phenomenon ambassadors' who help explain the tunnel mystery to the class in Lesson 12's final synthesis discussion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Using the LabXchange Wave Interference simulation, students observed that waves passing through a gap approximately equal to the wavelength spread out widely (diffract), while waves passing through a much larger gap continue with minimal spreading. Using the PhET Sound simulation with two in-phase speakers, students located audible dead-spots (regions of destructive interference) and loud regions (constructive interference) across the wavefront, recording at least three positions of each type in their evidence tables.
What did I learn?	Diffraction is the bending of waves around obstacles or through gaps, and it is maximised when the gap width is approximately equal to the wavelength. The Principle of Superposition states that when two waves overlap, the resultant displacement at any point is the algebraic sum of the individual displacements — producing constructive interference (amplitude doubles, crests align) or destructive interference (amplitude cancels, crest meets trough). FM radio waves ($\lambda \approx 2.85$ m, f = 105.2 MHz) barely diffract through the Kabete tunnel mouth (~4 m wide) compared with AM waves ($\lambda \approx 300$ m), AND reflected FM waves inside the tunnel create destructive interference dead-zones — together explaining the FM signal blackout.
How does this explain the phenomenon?	The FM radio signal on Classic 105 (105.2 MHz) cuts out in the Kabete tunnel for two compounding reasons: (1) FM wavelengths (~2.85 m) are comparable to the tunnel mouth width (~4 m), so the ratio gap/$\lambda \approx 1.4$, giving very limited diffraction — the signal

	barely bends into the tunnel; (2) inside the tunnel, the incoming FM signal and its reflections off the concrete walls overlap and produce destructive interference at many positions, creating dead-zones where the signal amplitude drops to near zero. AM waves ($\lambda \approx 300$ m) are unaffected because the tunnel mouth is tiny compared with their wavelength (gap/$\lambda \approx 0.013$), so AM waves diffract almost as if the tunnel wall were not there, and their much longer wavelength means reflected AM waves create constructive interference patterns across much wider zones, maintaining a usable signal.

LESSON 10 (80 minutes): Bending Light: Refraction, Snell's Law, and Total Internal Reflection	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how light bends when it passes from one medium to another, derive Snell's Law empirically, and connect refraction and total internal reflection to real-world wave behaviour including fibre-optic internet cables and the Kabete tunnel signal mystery.
Knowledge	By the end of the lesson, the learner should be able to: (a) define refraction as the bending of a wave when it crosses a boundary between two media of different optical densities; (b) state Snell's Law in the form $n_1\sin\theta_1 = n_2\sin\theta_2$; (c) define the refractive index n as the ratio $\sin\theta_i / \sin\theta_r$ for a given medium pair; (d) explain total internal reflection (TIR) and the critical angle; (e) describe how TIR enables fibre-optic signal transmission in undersea cables such as TEAMS.
Skills	By the end of the lesson, the learner should be able to: (a) use the PhET Bending Light simulation to measure angles of incidence and refraction accurately; (b) tabulate $\sin\theta_i / \sin\theta_r$ values and identify the pattern that leads to Snell's Law; (c) construct and annotate a ray diagram showing refraction and TIR at a boundary; (d) calculate unknown angles or refractive indices using Snell's Law.
Attitudes	The learner should: (a) appreciate that systematic measurement and pattern recognition are powerful tools for discovering physical laws; (b) show curiosity about how a phenomenon observed at the school canteen (bent straw) connects to a national infrastructure technology (fibre-optic cables); (c) demonstrate patience and precision when recording angle measurements from a simulation.
Key Inquiry Question	How do waves behave in different media and under different conditions?
Purpose in Storyline	In Lessons 1–9, students gathered evidence about wave properties — mechanical vs. electromagnetic waves, reflection, diffraction, interference, the Doppler effect, and wave speed in different media. They have accumulated strong evidence that the matatu horn pitch-shift is caused by the Doppler effect, but the Kabete tunnel FM signal loss remains only partially explained (diffraction and wavelength were introduced). This lesson adds the critical concept of refraction and TIR: students discover that wave speed change at a boundary causes bending — and that at steep enough angles, waves cannot cross at all (TIR). This directly advances the tunnel mystery (why do FM waves, with shorter wavelengths, fail to diffract or penetrate the tunnel boundary while AM waves do?), and introduces fibre-optics as a career-relevant application of the same physics.
Safety Notes	This lesson uses a computer simulation (PhET Bending Light) — no physical lasers or optical equipment are used. If a physical demonstration with a ray box is added, ensure: (a) the ray box is placed on a stable surface away from water; (b) students do not look directly into the ray box aperture; (c) glass slabs are handled carefully to avoid chipping. Digital safety: ensure students use the simulation only on approved school devices and do not navigate away from the PhET page during class time.

B. LESSON OVERVIEW

Lesson 10 of 12 in the Properties of Waves evidence-gathering sequence. Students open the PhET: Bending Light simulation and systematically vary the angle of incidence of a laser ray at a water–air and glass–air boundary. They record pairs of angles (θ_i, θ_r), compute $\sin\theta_i$ and $\sin\theta_r$, and discover that $\sin\theta_i / \sin\theta_r$ is constant — the refractive index n. Snell's Law is then formalised: $n_1\sin\theta_1 = n_2\sin\theta_2$. Students push the incidence angle beyond the critical angle and observe total internal reflection (TIR). They annotate a cumulative ray diagram in their notebooks. The lesson closes by linking TIR to fibre-optic cables (TEAMS undersea cable carrying internet to Mombasa) and revisiting the Kabete tunnel phenomenon: the concept of a wave being 'trapped' or 'blocked' at a boundary depending on angle and wavelength deepens the class's model of why FM signals disappear in the tunnel. The Driving Question Board is updated to record this new explanatory layer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Refraction: Light Entering Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students are shown a photograph of a straw in a glass of water at the school canteen — the straw appears broken or bent at the water surface. The teacher poses the focus question: 'The straw has not changed shape. So what IS bending?' Students write a 2-sentence individual prediction in their notebooks answering: (a) what is bending, and (b) whether they think the bending would be MORE or LESS pronounced if the straw were in cooking oil (more dense) instead of water. Students then share predictions with a partner (Think-Pair-Share). The teacher cold-calls 4 pairs to share, recording key prediction vocabulary on the board: 'light bends', 'speed changes', 'the boundary causes it', 'denser liquid = more bending'.	1. Display the bent-straw photograph on the projector. Say: 'Before we touch the simulation, I want your honest prediction. Do not worry about being wrong — being wrong is how we learn.' 2. Give students exactly 90 seconds of silent individual writing time. Use a visible countdown timer. 3. Say: 'Now turn to your partner. You have 60 seconds — one person shares, the other listens, then swap.' 4. After pair discussion, say: 'I am going to cold-call four pairs. I want to hear the PREDICTION, not the answer you think I want.' Cold-call pairs (use seating chart, choose spread across the room). For each response, revoice: 'So Akinyi and Kiprop are predicting that … — is that right?' 5. Do NOT confirm or deny any prediction. Say: 'Hold those ideas. The simulation is going to give us evidence.' 6. Write the lesson's focus question on the board: 'What happens to the SPEED and DIRECTION of a wave when it crosses from one medium into another?'	Think-Pair-Share with Prediction Recording. Students externalise prior knowledge before instruction, creating a cognitive anchor. By recording predictions in notebooks, students can later compare their initial mental model with evidence — making conceptual change visible and personally meaningful.	Observable indicator: All students have at least 2 sentences written in notebooks before pair-share begins. Teacher circulates during the 90-second writing period and scans notebooks — look for whether students connect 'bending' to light (not the straw), which reveals readiness. Cold-call responses reveal whether students already hold the idea that medium density matters — this informs how much scaffolding is needed in the Observe phase.
Observe Phase 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Refraction: Light	Students work in pairs at computers (or one device per group of three if devices are limited) with the PhET Bending Light simulation open on the 'Intro' tab, set to 'Ray' mode. The teacher distributes a structured data-collection worksheet. Students: (1) Set the top medium to AIR and the bottom medium to WATER. Rotate the laser to 5 different angles of incidence (20°, 30°, 40°, 50°, 60°) and record the corresponding	1. Before releasing students to computers, model the first data point LIVE on the projected simulation: 'Watch — I set the angle to 20°. I read the refracted angle as … I type that here. Now YOU take over.' This takes no more than 3 minutes. 2. Circulate continuously. Every 4–5 minutes, pause the class with a hand signal and ask one focusing question: (a) 'Is the refracted ray bending TOWARD or AWAY from the	Structured Data Table with Pattern Recognition. By computing $\sin\theta_i/\sin\theta_r$ themselves (rather than being told the formula), students experience the inductive process of scientific discovery. The constant ratio is unexpected and striking — it creates the 'aha' moment that makes Snell's Law memorable rather than merely memorised. Connecting the simulation data to the canteen straw (supporting phenomenon)	Observable indicators: (a) Each pair's data table has 5 rows completed for at least ONE medium pair before the class midpoint pause. (b) During cold-calls on the $\sin\theta_i/\sin\theta_r$ column, students use the word 'constant' or 'the same each time' — if fewer than half do, the teacher re-directs: 'Calculate row 1 and row 3 again carefully and compare.' (c) All students have a TIR ray diagram in their notebooks before Phase 3

Entering Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	angle of refraction shown by the protractor tool. (2) Complete the table: θ_i | θ_r | $\sin\theta_i$ | $\sin\theta_r$ | $\sin\theta_i/\sin\theta_r$. (3) Repeat for AIR–GLASS boundary. (4) Push the incidence angle past ~49° (for water) and record what happens to the refracted ray — students observe TIR. (5) Students sketch two ray diagrams in their notebooks: one showing normal refraction (with angles labelled), one showing TIR with the critical angle marked. Supporting phenomenon link: The teacher pauses the class at the midpoint and asks: 'Think back to the straw in the canteen glass. Which boundary in your simulation matches that situation?'	normal? What does that tell us about speed?' [wait 10 seconds] Cold-call 2 students. (b) 'Look at your $\sin\theta_i/\sin\theta_r$ column. What pattern do you notice?' [wait 12 seconds] Cold-call 3 students. 3. When pairs reach the TIR section, say: 'Before you move the angle, PREDICT in your notebook what you think will happen if the angle gets very large. Write one sentence.' [wait 15 seconds]. Then let them observe. 4. For the supporting phenomenon link, say: 'The straw sits in water with air above. Which medium pair is that? Point to it on your simulation.' [wait 10 seconds] Cold-call 2 pairs. 5. Manage pacing: announce '15 minutes remaining' and '5 minutes remaining' so all pairs reach the TIR observation before time ends.	grounds the abstract numbers in a tangible everyday observation.	begins — teacher checks during the transition.
Explain Phase 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Refraction: Light Entering Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class debrief using a projected data table aggregating values from three pairs. Students observe that $\sin\theta_i/\sin\theta_r \approx 1.33$ for air–water and ≈ 1.50 for air–glass across all angles. The teacher formalises this as the refractive index: $n = \sin\theta_i / \sin\theta_r$, then derives the general form Snell's Law: $n_1\sin\theta_1 = n_2\sin\theta_2$. Students work through three tiered practice problems in pairs: (Easy) Given $\theta_i = 30°$ at an air–water boundary, find θ_r. (Medium) A ray passes from glass (n=1.5) into water (n=1.33) at 40° — find the refracted angle. (Hard) Calculate the critical angle for glass–air TIR using $\sin\theta\_c = 1/n$. After solving, students write a 'Because … Therefore …' sentence connecting their result to the lesson phenomenon: e.g., 'Because light slows down in a denser medium, it bends toward the normal —	1. Open the whole-class debrief by displaying the aggregated data table. Ask: 'What single number appears in every cell of the last column for the air–water rows?' [wait 10 seconds] Cold-call 3 students. Say: 'That number has a name. It is the REFRACTIVE INDEX.' Write: $n = \sin\theta_i / \sin\theta_r$ on the board. 2. Walk through the algebraic rearrangement step-by-step: 'If n_1 is the first medium and n_2 is the second … who can tell me what $n_1\sin\theta_1$ equals?' [wait 12 seconds] Cold-call 1 student at the board to write it. 3. Release pairs to the three practice problems. Circulate; target the hard problem — ask: 'What happens to the refracted angle as θ_i approaches the critical angle? What does the simulation show?' 4. After problems, cold-call one pair per problem level to share working on the board. For each, ask the class:	Data Aggregation → Law Formalisation → Application Ladder. By moving from their own simulation data to a class-level pattern, students see that the law is not invented by a textbook author but REVEALED by evidence. The three-tier problem ladder (easy → medium → hard) builds procedural fluency incrementally. The 'Because … Therefore …' sentence forces causal reasoning — the hallmark of physics sensemaking rather than formula plug-in. The TEAMS cable narrative shows that the law has real economic and social stakes for Kenya.	Observable indicators: (a) At least 80% of students write $n = \sin\theta_i/\sin\theta_r$ correctly in their notebooks with the formula boxed. (b) On the medium-level problem, correct answers ($\theta_r \approx 46°$) are produced by at least two-thirds of pairs — if not, teacher re-teaches the $n_1\sin\theta_1 = n_2\sin\theta_2$ rearrangement with a worked example on the board before moving on. (c) 'Because … Therefore …' sentences include at least one of: boundary, speed, bends, denser — teacher reads 5 notebooks during the transition to Phase 4.

	therefore the straw appears broken at the canteen.' The teacher then introduces the fibre-optic careers link: 'The TEAMS undersea cable that carries internet from Mombasa to the world uses TIR — light bounces inside the glass fibre at an angle beyond the critical angle and never escapes. The same physics you just derived keeps Kenya connected.'	'Do you agree? Thumbs up, thumbs to the side, thumbs down.' Address any thumbs-to-the-side or thumbs-down immediately. 5. For the 'Because … Therefore …' sentence, say: 'I want one sentence that uses the word BOUNDARY and the word SPEED. You have 2 minutes to write it.' [wait 2 minutes]. Cold-call 3 students. 6. Deliver the TEAMS cable narrative with a map of the undersea cable route on the projector. Ask: 'Why does the light not just leak out through the glass wall of the cable?' [wait 10 seconds] Cold-call 2 students — look for 'critical angle' or 'TIR' in the answer.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Refraction: Light Entering Water (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher returns the class's attention to the two-part Driving Question displayed on the wall: 'Why does the matatu horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel?' Students individually write on a sticky note (or an index card if sticky notes are unavailable) one new piece of evidence from today's lesson that helps answer the TUNNEL part of the DQ. They place it under the 'TUNNEL' column of the DQB. The class then does a 3-minute 'gallery walk' of the DQB — silently reading new cards added today and placing a tick (✓) on any card whose idea they agree with. The teacher facilitates a 5-minute whole-class discussion to consolidate: 'We now know that waves bend AND can be totally internally reflected at boundaries. How does this add to our model of why FM signals struggle in the Kabete tunnel?' Students are prompted to connect	1. Redirect attention to the DQB: 'Look at the two columns — Matatu Horn and Kabete Tunnel. Today's lesson gives us new evidence for the TUNNEL column. Take 2 minutes — write your best one-sentence contribution on your sticky note.' [wait 2 minutes, circulate and prompt any student who is stuck: 'Think about what happens when a wave hits a boundary at a steep angle.'] 2. Direct students to place sticky notes and begin the gallery walk: 'You have 3 minutes. Read silently. Tick any card whose idea you agree with.' 3. After gallery walk, cold-call 3 students whose cards received the most ticks: 'Read your card to the class. Then explain why you think it connects to the tunnel.' [wait 10 seconds after each reading for peer response.] 4. Ask for a volunteer to state the updated model in one sentence. Write it verbatim on the board. Say: 'We will refine this in Lesson 11. Does anyone want to	Driving Question Board (DQB) Gallery Walk with Consensus Marking. The DQB is a live artefact of the class's collective understanding. By ticking cards they agree with, students perform rapid peer assessment of ideas. The gallery walk activates reading comprehension in a science context and gives quieter students a voice. The Model Statement written on the board externalises the class's best current explanation — making it available for critique and revision in subsequent lessons, embodying the iterative nature of scientific modelling.	Observable indicators: (a) Every student places at least one sticky note on the DQB. (b) DQB cards reference today's concepts (refraction, boundary, TIR, critical angle) — teacher scans all cards during gallery walk and flags any that are off-topic or vague for a private follow-up conversation. (c) The volunteer Model Statement includes both 'boundary' and at least one of 'wavelength', 'angle', or 'speed' — if it does not, the teacher asks: 'Can someone add one more word to this sentence to make it more precise?'

	today's lesson to previous evidence: diffraction (Lesson 8), wavelength and wave speed (Lesson 6), and the Doppler effect (Lesson 9). The updated class model is voiced aloud by a student volunteer and written as a 'Model Statement' on the board.	challenge or improve this sentence RIGHT NOW?' [wait 10 seconds.] 5. Point to the DQB and say: 'Look how far our model has come since Lesson 1. What was on this board in Lesson 1?' (Rhetorical — builds metacognitive awareness of learning progression.)		
Model Building Phase 📹 **VIDEO:** [Video: Bending Light] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Refraction: Light Entering Water (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Students open their cumulative wave model — a diagram they have been building across all 10 lessons, showing the two-part phenomenon (matatu horn / Kabete tunnel). Today they add a new layer to the TUNNEL half of the model: they draw a cross-section of the Kabete underpass showing an FM wave ray hitting the tunnel wall at a steep angle, bending away, and undergoing TIR — trapping it or deflecting it rather than allowing it to propagate through. They annotate with: (a) the angle of incidence label, (b) the critical angle label, (c) a note on why FM (shorter wavelength, higher frequency) is more affected than AM at the boundary. Students who finish early are challenged to add a second diagram showing the TEAMS fibre-optic cable cross-section with TIR arrows inside the glass core. The lesson closes with students writing a 'Model Reflection' sentence: 'My model is now stronger because I added __________, but I still have a question about __________.'	1. Say: 'Open your cumulative model to the Kabete tunnel half. You are going to add ONE new layer today using what you discovered about refraction and TIR. You have 10 minutes.' Display the model prompt on the projector: 'Draw the tunnel cross-section. Show a wave ray hitting the boundary. Label: angle of incidence, critical angle, refracted ray or TIR. Add a note: WHY is FM more affected than AM?' 2. Circulate — target the FM vs AM annotation: 'FM has a shorter wavelength than AM. How does that affect how much it diffracts around the tunnel opening? We explored diffraction in Lesson 8 — connect those ideas.' [wait 10 seconds for student to think before offering a hint.] 3. With 3 minutes remaining, say: 'Finish your diagram and write your Model Reflection sentence. Be honest about what you still do not understand.' 4. Cold-call 3 students to read their Model Reflection sentences. For each 'still have a question about' clause, write the question on a 'Parking Lot' section of the whiteboard: 'That is an excellent question for Lesson 11.' 5. Summarise: 'Today we moved from a bent straw at the canteen to Snell's Law to fibre-optic cables carrying internet to Mombasa. All of it — the same physics. Waves bend at	Cumulative Model Revision with Annotated Ray Diagram. Progressive model revision — adding one layer per lesson — mirrors the practice of real physicists who refine models as new evidence arrives. Requiring students to label specific quantities (angle of incidence, critical angle) prevents superficial drawing and demands conceptual precision. The 'Model Reflection' sentence develops metacognitive awareness: students distinguish between what they now know and what they still need to investigate — sustaining the inquiry momentum into Lesson 11.	Observable indicators: (a) Cumulative model diagrams show a correctly oriented refracted ray (bending away from normal when going from dense to less dense medium) or TIR arrow at the tunnel wall. (b) The annotation distinguishing FM from AM wavelengths is present — even if imperfect, it shows engagement with cross-lesson evidence synthesis. (c) 'Model Reflection' sentences contain a genuine question (not 'I understand everything') — teacher notes recurring questions to prioritise in the Lesson 11 opening. (d) Exit task photos are captured for at least 70% of student models by end of lesson.

		boundaries. And sometimes they cannot cross at all.' 6. Assign exit task: students photograph their updated cumulative model with a school device (or the teacher photographs selected models to display at the start of Lesson 11).		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) SIMULATION ACCESS — Did all pairs have sufficient device access and simulation load time? If bandwidth was limited and the PhET simulation loaded slowly, what offline alternative (printed ray-diagram worksheet or physical glass-slab ray-box activity) could replace it next time? (2) DATA QUALITY — Were students' $\sin\theta_i/\sin\theta_r$ values consistent enough to clearly reveal the constant ratio? If values varied widely (due to imprecise angle reading on the simulation protractor), consider adding a 'how to read the protractor accurately' micro-lesson at the start of Phase 2. (3) SNELL'S LAW FORMALISATION — Did students make the leap from the constant ratio to the general form $n_1\sin\theta_1 = n_2\sin\theta_2$ without teacher-telling? If not, the inductive gap needs bridging — try asking 'What would happen if the first medium were NOT air?' as a Socratic prompt before writing the general law. (4) DQB QUALITY — Were sticky-note contributions specific (mentioning refraction, TIR, critical angle, boundary) or vague ('waves bend')? Vague contributions suggest students have procedural knowledge (the formula) but have not yet connected it causally to the phenomenon — prioritise the 'Because … Therefore …' sentence in future lessons. (5) EQUITY CHECK — Were female students and students from rural backgrounds (who may be less familiar with fibre-optic infrastructure) equally engaged during the TEAMS cable discussion? Consider assigning a brief pre-reading about the TEAMS cable for Lesson 11 to build shared background knowledge. (6) PACING — Did all pairs reach the TIR observation within Phase 2? If not, the data table has too many rows for the available time — reduce to 4 angle values (20°, 35°, 50°, 60°) in future iterations.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students used the PhET Bending Light simulation to observe a laser ray bending at water–air and glass–air boundaries at five different angles of incidence. They also observed total internal reflection (TIR) when the angle of incidence exceeded the critical angle (~49° for water, ~42° for glass). They connected the simulation to the canteen bent-straw supporting phenomenon.
What did I learn?	Students discovered that the ratio $\sin\theta_i / \sin\theta_r$ is constant for a given pair of media — this constant is the refractive index n. They formalised Snell's Law: $n_1\sin\theta_1 = n_2\sin\theta_2$. They learned that TIR occurs when a wave travels from a denser to a less dense medium at an angle greater than the critical angle ($\sin\theta_c = 1/n$). They connected TIR to fibre-optic cables (TEAMS undersea cable, Mombasa) and linked refraction at boundaries to the Kabete tunnel FM signal loss.
How does this explain the phenomenon?	Waves change speed when they cross a boundary between two media of different optical densities. This speed change causes the wave to bend (refract) — toward the normal when entering a denser medium, away from the normal when entering a less dense medium. When the angle is steep enough (beyond the critical angle), no energy crosses the boundary: total internal reflection occurs. This is why FM radio waves (shorter wavelength, less able to diffract) are more likely to be reflected or attenuated at the Kabete tunnel walls than AM waves — and it is the same physics that keeps internet light signals travelling inside TEAMS fibre-optic cables all the way from Mombasa to the world.

LESSON 11 (80 minutes): Resonance and Stationary Waves — Why the Tunnel Amplifies Some Sounds and Kills Others	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how resonance and stationary waves form, and to connect these phenomena to frequency-selective amplification and cancellation observed in the Kabete tunnel and matatu horn scenario.
Knowledge	By the end of the lesson, the learner should be able to: (a) define resonance as the amplification of vibrations when a driving frequency matches the natural frequency of a system; (b) describe how stationary (standing) waves are formed by the superposition of two waves of equal frequency and amplitude travelling in opposite directions; (c) identify nodes and antinodes in a stationary wave pattern; (d) explain how enclosed spaces (e.g. tunnels, pipes) select for specific resonant frequencies.
Skills	By the end of the lesson, the learner should be able to: (a) set up and observe a standing wave on a stretched string or rubber band; (b) use a PhET simulation to manipulate frequency and observe resonance conditions; (c) construct a labelled diagram of a stationary wave showing nodes, antinodes, and wavelength; (d) apply the relationship between string/tube length and resonant wavelength ($L = n\lambda/2$) to solve simple problems.
Attitudes	The learner appreciates that physical phenomena which initially appear mysterious — such as selective signal loss in tunnels — are explainable through consistent wave principles, and values the use of evidence-based reasoning over guesswork.
Key Inquiry Question	How do waves behave in different media and under different conditions?
Purpose in Storyline	Lessons 1–10 established rectilinear propagation, reflection, refraction, diffraction, interference, and the Doppler effect. Lesson 11 is the penultimate evidence-gathering lesson. Students now investigate resonance and stationary waves — the final major wave behaviour. This directly explains why the Kabete tunnel does not simply block all waves equally: the tunnel cavity resonates at certain frequencies, reinforcing some wavelengths and extinguishing others. Combined with diffraction (Lesson 10), this gives students a near-complete explanation of the FM/AM tunnel mystery, and sets up Lesson 12's final unified model.
Safety Notes	Ensure audio output from speakers/computers is kept at moderate volume (≤ 60 dB) to protect hearing during resonance demonstrations. If using a tuning fork near a resonance tube with water, ensure the tube is secured in a clamp stand to prevent tipping. Wipe up any spilled water immediately to prevent slipping hazards.

B. LESSON OVERVIEW

Students arrive having already investigated interference (Lesson 10) using LabXchange. In Lesson 11, they pivot to the special case of interference that produces stationary (standing) waves, and then to resonance — the condition under which a system dramatically amplifies vibrations at its natural frequency. The lesson opens with a Predict phase in which students hypothesise why a particular note played near the Kabete tunnel entrance causes a sustained hum while other notes do not. Students then Observe stationary wave formation using a PhET simulation (Wave on a String) and a live rubber-band/string demonstration, identifying nodes and antinodes. In the Explain phase, students apply $L = n\lambda/2$ to calculate which FM and AM wavelengths would resonate within the tunnel's approximate length (~120 m), discovering that AM wavelengths (hundreds of metres) cannot form a standing wave in the tunnel at all, while certain FM wavelengths can — partially explaining the selective signal behaviour. The DQB is updated to mark resonance and stationary waves as confirmed explanatory mechanisms. The Model Building phase has students add a resonance cavity diagram to their cumulative wave-behaviour model, annotating which tunnel frequencies are reinforced and which are cancelled, directly tying back to Classic 105 (105.2 MHz, $\lambda \approx 2.85$ m) and a typical AM station (e.g., KBC Radio at 900 kHz, $\lambda \approx 333$ m).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Seismic Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students individually read a short scenario card: 'Kamau, a street musician in Nairobi's CBD, notices that when he plays a low 'A' note on his guitar near the entrance of the Kabete tunnel, the sound seems to swell and echo in a sustained hum. When he plays a higher note, nothing special happens. Why does only one note get amplified? Write your prediction in your science notebook. Include a labelled sketch of what you think the sound wave is doing inside the tunnel.' Students write silently for 4 minutes, then share with a shoulder partner for 2 minutes, then selected students share with the class.	Distribute scenario cards face-down. Say: 'Do NOT flip the card until I say go. This is your thinking time.' After 4 minutes of individual writing, say: 'Turn to your elbow partner. You have 90 seconds each — person closer to the window goes first. Explain your prediction and show your sketch.' After pair-share, cold-call 3 students (aim for at least one who predicted constructive interference, one who mentioned 'echoing', and one who is uncertain). Use exact prompt: 'Tell me what you think is happening to the wave inside the tunnel — not just what you hear, but what the wave is physically doing.' Allow 12 seconds of WAIT TIME after posing the question before calling on anyone. Record 3–4 key prediction phrases on the board under the header 'OUR PREDICTIONS — Lesson 11'. Do NOT correct misconceptions yet. Say: 'Hold onto these ideas. We are going to test every single one of them today.'	Predict-before-observe (anchored prediction): By committing predictions to writing and sketches before any instruction, students activate prior knowledge of reflection and interference from Lessons 9–10, and create a personal cognitive stake in the outcome. The sketch requirement forces students to think spatially about the wave, not just verbally.	Circulate during the 4-minute writing period. Look for: (1) students who draw waves bouncing back and forth inside a tunnel shape — these students are on track toward standing waves; (2) students who write only about 'the tunnel blocking sound' — these students still hold a barrier-only mental model and will need targeted questioning during the Observe phase. Make a mental note of 2–3 students in each category for strategic cold-calling.
Observe Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Seismic Waves (Flexbook) Source: CK-12 🔍 Search ARES	PART A — Live demonstration (5 min): The teacher stretches a long rubber band (or guitar string on a board) between two fixed points visible to the class. A student volunteer plucks it at different positions while others observe. Students sketch what they see in their notebooks, noting any points that appear to stand still vs. points of maximum movement. PART B — PhET simulation (15 min): Students open PhET 'Wave on a String' on ARES offline devices.	PART A: Before plucking, ask the class: 'Watch carefully — I want you to identify any point on this string that does NOT move during vibration. Point at it.' Pluck the string to produce the fundamental mode. After 10 seconds of WAIT TIME, cold-call 2 students to come up and physically point to nodes. Say: 'What do we call a point that never moves in a vibrating system? Turn to a partner and agree on a term — you have 20 seconds.' Confirm 'node' and	Structured Observation Guide with Bridging Question: The observation guide is structured so that the first three tasks build pattern recognition (resonant frequencies are integer multiples of the fundamental), and the fourth task is a deliberate 'bridge' that requires students to transfer the pattern from a simulated string to the real-world tunnel. This prevents the simulation from remaining an abstract exercise and keeps the anchoring	Check student notebooks at the 12-minute mark of the PhET activity. Look for: (1) Correct node/antinode labelling — students should show nodes at the fixed ends and one or more in the middle depending on harmonic; (2) The mathematical relationship between harmonics ($f_2 = 2f_1$, $f_3 = 3f_1$ etc.) written explicitly — if absent, prompt the pair with: 'List your resonant frequencies in a column. Divide each one by the first. What pattern do you see?';

for similar readings	Working in pairs, they follow a structured observation guide: (1) Set to 'Oscillate' mode, fixed ends. Slowly increase frequency from 1 Hz. Record the frequency at which the first clear standing wave pattern appears. Sketch the pattern, label 'node' and 'antinode'. (2) Continue increasing frequency — record the 2nd and 3rd resonant frequencies. What is the mathematical relationship between them? (3) Change the string length (use the clamp slider) and repeat. Does the resonant frequency change? (4) TUNNEL CONNECTION: The observation guide ends with: 'The Kabete tunnel is about 120 m long. If sound waves travel at 340 m/s inside the tunnel, use what you just observed to predict what frequency of sound would resonate most strongly inside the tunnel.' Students record all observations in their notebooks.	introduce 'antinode' with a direct definition: 'An antinode is a point of maximum displacement — the string swings the most there.' PART B: Circulate during PhET work. At the 7-minute mark, pause the class and ask one pair: 'You found your first standing wave at [X] Hz and your second at [Y] Hz. What is the ratio? What does that tell you?' Use 10 seconds WAIT TIME. If students are stuck on the tunnel calculation, prompt: 'You know $v = f\lambda$. You know the first resonant wavelength is twice the tunnel length. Can you work backwards to find f?' Do NOT give the answer — say: 'Try the calculation. I will come back in 2 minutes.' Cold-call a total of 4 pairs during the simulation phase to share one observation each.	phenomenon active throughout the Observe phase.	(3) The tunnel calculation attempted — even an incorrect attempt is acceptable at this stage; correct understanding will be built in the Explain phase.
Explain Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Seismic Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Concept consolidation (8 min): Students read a 250-word structured text (provided on ARES, drawn from the 'Waves – Final Explanation' teacher template content) covering: definition of stationary waves, formation by superposition of two equal waves travelling in opposite directions, nodes (destructive interference always) and antinodes (constructive interference always), and the condition $L = n\lambda/2$ for resonance in a closed-both-ends system. Students annotate the text using a 3-symbol system: '★' for a new idea, '✓' for something they already predicted, '?' for something still confusing. PART B — Application problems (12 min): Students work individually on three	During the reading annotation (8 min), circulate and note which students mark '?' next to 'two waves travelling in opposite directions' — this is the most common confusion point (students ask: 'where does the second wave come from?'). After 8 minutes, pause and address this directly: 'Who put a question mark on the phrase two waves travelling in opposite directions? Raise your hand.' Count hands. Then say: 'In our string experiment — what happened when the wave reached the fixed end?' Pause 12 seconds. Cold-call a student: 'It reflected. So now you have the original wave and its reflection travelling in opposite directions on the same string. THAT is your second wave.'	Annotation + Graded Application (CER — Claim, Evidence, Reasoning): The 3-symbol annotation builds metacognitive awareness of what is genuinely new versus confirmed predictions. The three graded problems use an explicit Claim-Evidence-Reasoning structure in Problem 3, requiring students to synthesise resonance (Lesson 11) with diffraction (Lesson 10) to construct a multi-mechanism explanation of the tunnel phenomenon — moving from single-concept recall to integrated scientific explanation.	Collect written answers to Problem 3 at the end of the Explain phase (students can photograph their notebook page and submit to the ARES platform, or the teacher collects notebooks). Evaluate against this rubric: (4) Student correctly applies $L = n\lambda/2$, concludes AM $\lambda >> 2L$ so resonance impossible, AND connects to diffraction (large λ diffracts around tunnel entrance); (3) Correct resonance argument, weak or absent diffraction link; (2) Correct formula use but incorrect conclusion about AM; (1) Formula not applied, explanation is qualitative only. Students scoring 1–2 are flagged for a 3-minute check-in at the start of Lesson 12.

	graded problems, then compare with a partner: (1) [Recall] A 2 m string vibrates in its third harmonic. Draw the wave pattern and calculate the wavelength. (2) [Apply] The Kabete tunnel is approximately 120 m long. Sound travels at 340 m/s. Calculate the three lowest resonant frequencies for sound waves in the tunnel. Would a 105.2 MHz FM radio wave ($\lambda \approx 2.85$ m) form a standing wave in this tunnel? Show your working. (3) [Extend — Phenomenon Connection] KBC Radio broadcasts on approximately 900 kHz AM ($\lambda \approx 333$ m). Explain, using the resonance condition $L = n\lambda/2$, why AM radio waves cannot resonate inside the Kabete tunnel. How does this combine with what you learned about diffraction in Lesson 10 to explain why the AM signal survives the tunnel but behaves differently from FM?	During the problem phase, cold-call 2 students to write their Problem 2 working on the board. After both have written their answers, ask the class: 'Do both boards show the same resonant frequencies? If not, who can explain the discrepancy?' Use exact language: 'I am not asking who is right — I am asking who can explain the difference.' Allow 15 seconds WAIT TIME. For Problem 3, call on a student who previously struggled with diffraction in Lesson 10 and ask them to explain first — this is a deliberate retrieval practice cold-call. Say: 'You studied diffraction last lesson. How does that connect to what you just calculated about AM wavelengths?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Seismic Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student writes one sticky note (or typed entry on ARES) in this format: 'We now know that [WAVE BEHAVIOUR] explains [SPECIFIC PART OF THE PHENOMENON] because [EVIDENCE FROM TODAY].' Students post their sticky notes on the class DQB, which is organised in columns by wave behaviour: Propagation | Reflection | Refraction | Diffraction | Interference | Doppler | Resonance/Standing Waves. Students then do a 'Gallery Read' — silently reading 3 sticky notes in the Resonance column written by classmates, and placing a tick on any they agree with or a '?' on any they find unclear or incomplete.	Before students write their sticky notes, display the driving question on the board in full: 'Why does the matatu's horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel — and what do these two mysteries reveal about how ALL waves behave?' Point to the Resonance column on the DQB (currently empty or sparse). Say: 'After today, this column should be filling up. Your sticky note must name a specific part of the tunnel or horn story — not just wave behaviour in general.' Give students 3 minutes to write. During the Gallery Read, circulate and identify 2 sticky notes that contain a misconception and 2 that are especially precise. After the	Driving Question Board — Column-Organised Cumulative Tracking: The DQB has been maintained across all 11 lessons, with each column representing one wave behaviour. By physically filling in the Resonance column today, students gain a visible sense of their growing explanatory toolkit. The Gallery Read introduces peer critique as a sensemaking tool — students evaluate claims against the evidence gathered today rather than deferring to teacher authority.	Evaluate DQB sticky notes for: (1) Specificity — does the note reference the tunnel, FM/AM signals, or the matatu horn explicitly? Generic notes ('resonance is when things vibrate together') score low; (2) Evidence citation — does the note mention the PhET finding, the calculation from Problem 2, or the $L = n\lambda/2$ condition? Students whose notes lack both specificity and evidence citation will be paired with a stronger peer during Lesson 12's model-building phase.

		Gallery Read, cold-call the authors of the 2 precise notes: 'Read yours aloud. Why did you frame it that way?' Then ask the class: 'Does anyone want to respectfully challenge or refine either of those two notes?' Use exact language: 'Challenging means improving the science — it is not a criticism of the person.' Allow 10 seconds WAIT TIME before calling on students.		
Model Building Phase 📹 **VIDEO:** Net force Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Seismic Waves (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their Cumulative Wave Behaviour Model — a diagram they have been building since Lesson 1. Today they add a new section: 'Resonance and Stationary Waves in the Kabete Tunnel.' They must draw: (1) A top-down outline of the Kabete tunnel (~120 m long, labelled); (2) An FM wave ($\lambda \approx 2.85$ m, Classic 105) entering the tunnel — show the standing wave pattern that forms if $L = n\lambda/2$ is satisfied, label nodes and antinodes; (3) An AM wave ($\lambda \approx 333$ m, KBC 900 kHz) entering the tunnel — show why no standing wave can form (wavelength >> tunnel length); (4) An annotation connecting today's lesson to Lesson 10 (diffraction): 'AM also survives because its long wavelength diffracts around the tunnel entrance — see Lesson 10 diagram.' Students who finish early add a fifth element: a matatu horn scenario showing Doppler shift combined with resonance in an observer's ear canal (~2.5 cm, resonant frequency ~3400 Hz — near the horn's pitch).	Distribute (or have students open on ARES) the model template showing the tunnel outline and the three annotation boxes. Say: 'Your model must tell the story of what happens to TWO different types of wave — FM and AM — when they enter the same 120-metre tunnel. Use what you calculated in Problem 2.' Circulate after 5 minutes. Stop at any model that shows the FM wave passing straight through without a standing wave pattern and ask: 'Your FM wave — is it being reflected at the far wall of the tunnel?' Pause 10 seconds. 'If it reflects and comes back, and meets the incoming wave — what happens? You solved this in the PhET activity.' Do not provide the correction — let the student self-correct. At the 12-minute mark, ask for two volunteers to hold up their models (or display on ARES projector). Ask the class: 'What is the same between these two models? What is different? Which one more accurately explains why Classic 105 fades but KBC survives?' Cold-call 3 students for responses. Close the model phase by saying: 'In Lesson 12, you will combine EVERY column of the DQB into one final unified model. Make sure	Cumulative Model Revision (Mechanism-to-Phenomenon Mapping): The cumulative model is the primary sensemaking artefact of the unit. Each lesson adds one new explanatory mechanism. By requiring students to draw both the FM and AM wave scenarios in the same tunnel diagram, and to cross-reference Lesson 10 (diffraction), the model-building phase operationalises integration of multiple wave behaviours. The 'early-finisher' extension links resonance back to the Doppler shift (Lesson 7), rewarding students who pursue deeper connections.	Collect or photograph student models at the end of the lesson. Evaluate against three criteria: (1) ACCURACY — does the FM wave diagram show a standing wave pattern with nodes at the tunnel walls? Does the AM diagram correctly show no standing wave? (2) LABELLING — are nodes, antinodes, wavelengths, and frequencies labelled? (3) INTEGRATION — is there an explicit written link to diffraction (Lesson 10)? Students meeting all three criteria are on track for Lesson 12's final synthesis. Students missing criterion 3 (integration) receive a targeted prompt at the start of Lesson 12: 'Look at your Lesson 10 model. How does the wavelength of AM compare to the tunnel entrance width? What does that tell you about diffraction?'

		your model today is complete — it is one piece of a larger puzzle.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) RESONANCE CONFUSION POINT — Did students successfully identify that the second wave in a standing wave system is the reflected wave? If more than one-third of students marked this with '?' during the annotation activity, plan a 5-minute re-teach at the start of Lesson 12 using the rubber band demo: pluck it, then hold one end still and show how the reflected pulse travels back. (2) CALCULATION CONFIDENCE — Were students able to apply $L = n\lambda/2$ independently to the tunnel problem, or did they need step-by-step scaffolding? If fewer than 60% of students completed Problem 2 correctly, add a worked example to the ARES resource library and assign it as a pre-task before Lesson 12. (3) DQB QUALITY — Are students' DQB sticky notes becoming more specific and evidence-linked across the unit? Compare the specificity of today's Resonance column entries with the entries from Lesson 2 (Reflection). Growth in specificity indicates growing scientific reasoning. (4) PHENOMENON CONNECTION — After 11 lessons, can most students articulate a multi-mechanism explanation of both the matatu horn (Doppler) and the tunnel signal (diffraction + resonance/interference)? If not, Lesson 12 must devote extra time to the synthesis activity rather than introducing new content. (5) EQUITY CHECK — Were the same 4–5 students dominating cold-call responses? Review your cold-call log and ensure that Lesson 12 deliberately targets students who have not yet publicly shared their reasoning.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed standing wave patterns forming on a string in the PhET 'Wave on a String' simulation, identifying nodes (points of zero displacement) and antinodes (points of maximum displacement) at discrete resonant frequencies. They also observed a live rubber-band demonstration in which only specific pluck frequencies produced sustained, symmetrical vibration patterns. The tunnel calculation revealed that FM wavelengths (~2.85 m for Classic 105) can satisfy $L = n\lambda/2$ for the ~120 m Kabete tunnel, while AM wavelengths (~333 m for KBC 900 kHz) cannot.
What did I learn?	Stationary (standing) waves form when two waves of equal frequency and amplitude travel in opposite directions and superpose. In an enclosed space like a tunnel, the outgoing wave and its reflection interfere to produce a standing wave only when the cavity length equals a whole number of half-wavelengths ($L = n\lambda/2$). This is called resonance. At resonance, energy builds up at antinodes; at other frequencies, destructive interference dominates. Nodes are points of permanent destructive interference; antinodes are points of permanent constructive interference. FM signals (short wavelength) can satisfy resonance conditions inside the Kabete tunnel, contributing to signal reinforcement OR cancellation depending on exact frequency; AM signals (very long wavelength) cannot resonate inside the tunnel at all.
How does this explain the phenomenon?	The selective behaviour of FM and AM signals in the Kabete tunnel is now explained by two wave behaviours working together: (1) RESONANCE / STATIONARY WAVES — FM wavelengths are short enough to form standing waves inside the tunnel, meaning some FM frequencies are reinforced (antinodes) and others are cancelled (nodes) depending on the tunnel's exact length. AM wavelengths are far too long to form any standing wave in the tunnel and so do not experience resonance-based cancellation or amplification; (2) DIFFRACTION (Lesson 10) — AM waves, having much longer wavelengths, diffract significantly around the tunnel entrance and exits, allowing them to 'bend in' and maintain some signal strength throughout; FM waves diffract less and are more directional. Together, these two mechanisms explain why Classic 105 FM fades inside the Kabete tunnel while the AM signal on the same receiver remains audible.

LESSON 12 (80 minutes): Final Explanation — Pulling It All Together: From the Matatu Horn to the Kabete Tunnel

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all evidence gathered across the unit into a single, coherent scientific explanation for the anchoring phenomenon, and to demonstrate mastery of wave properties through model comparison and driving-question resolution.
Knowledge	Waves are disturbances that transfer energy from one point to another without transferring matter. They exhibit rectilinear propagation, reflection, refraction, diffraction, interference, resonance, and — for moving sources — the Doppler effect. AM waves (longer wavelength, ~300 m) diffract readily around tunnel walls and hills; FM waves (shorter wavelength, ~3 m) diffract very little, causing signal blackout inside the Kabete tunnel. The Doppler effect explains why the matatu horn rises in pitch on approach and falls on departure, even though the driver never changes the horn frequency.
Skills	Constructing and comparing scientific models; synthesising multi-lesson evidence into a written and diagrammatic final explanation; presenting and critiquing claims using evidence; updating the Driving Question Board to show resolved and still-open questions.
Attitudes	Appreciate that seemingly unrelated everyday phenomena (a matatu horn, a fading radio) are unified by the same underlying physics; demonstrate intellectual honesty by acknowledging questions that remain open; value collaborative sensemaking as a scientific practice.
Key Inquiry Question	1. What are the properties of waves? 2. How do waves behave in different media and under different conditions?
Purpose in Storyline	This is the capstone lesson of the sub-strand. Students have spent Lessons 1–11 gathering evidence about wave speed, frequency, wavelength, amplitude, reflection, refraction, diffraction (PhET single/double slit), interference, resonance, and the Doppler effect. In Lesson 12 they weave all of that evidence into a single final explanation for both mysteries in the anchoring phenomenon, compare their initial (Lesson 1) mental models with their final models, and formally close — or flag as still-open — every question on the Driving Question Board.
Safety Notes	No hazardous materials are used. If students present near electrical outlets when using laptops or tablets, ensure cables are secured to avoid trip hazards. Supervise equitable participation during gallery walks to maintain a respectful classroom environment.

B. LESSON OVERVIEW

Students arrive at the final lesson of Sub-Strand 2.1 having already gathered evidence on every major wave property. The lesson opens with a Return-to-Phenomenon gallery walk in which student groups post their best evidence cards from earlier lessons onto a class 'Evidence Wall'. In the Predict Phase, students write a solo 'pre-final' explanation attempt — their best current answer to both mysteries — before any new instruction. The Observe Phase is a structured Evidence Synthesis: students use the PhET Wave Interference simulation (single-slit mode) as a final manipulative check, comparing the diffraction of a 105 MHz FM wave (short $\lambda \approx 2.85$ m) versus a 567 kHz AM wave (long $\lambda \approx 529$ m) through a gap the width of the Kabete tunnel (~10 m). They record the striking difference in spreading. The Explain Phase is the centrepiece: a Jigsaw Expert Presentation in which six home groups each own one wave property and must show — with evidence — how it contributes to explaining the matatu or the tunnel mystery. Students then collaboratively draft the Final Explanation paragraph. In the DQB phase, the class revisits every question posted since Lesson 1, marking each as RESOLVED (green), PARTIALLY RESOLVED (amber), or STILL OPEN (red). The lesson closes with Model Comparison: students physically place their Lesson 1 sketch beside their final annotated model and write a metacognitive 'What changed in my thinking?' exit reflection. A summative oral cold-call round and written exit ticket provide formative closure.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Doppler effect introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives a half-sheet prompt: 'Without looking at your notes, write your BEST explanation right now for (a) why the matatu horn changes pitch as it passes on the Nakuru–Nairobi highway, and (b) why the FM signal fades in the Kabete tunnel but the AM signal survives. Use as many wave-property vocabulary words as you can.' Students write silently and individually for 6 minutes. They then share with a shoulder partner for 2 minutes, noting any vocabulary they forgot or used incorrectly. Pairs circle the ONE strongest idea they want to carry into the class discussion.	Distribute the prompt slips face-down. Say: 'This is NOT a test — it is a snapshot of your thinking TODAY, which you will compare with Lesson 1 at the end of this lesson. Flip over and write.' Start a visible 6-minute timer. Circulate silently for the first 3 minutes — do NOT answer questions; if asked, say: 'Trust what you know.' At the 3-minute mark, quietly tap any student who has written fewer than two sentences and whisper: 'Give me one more idea.' After 6 minutes say: 'Turn to your shoulder partner — 2 minutes — identify the ONE strongest idea between you.' Cold-call 4 pairs (choose mixed-ability): 'Pair 3 — what was your strongest idea?' WAIT 12 seconds after each. Write the 4 ideas on the board under the heading 'WHERE WE ARE NOW'. Do not correct yet — annotate with a question mark next to anything incomplete.	Strategy: SOLO WRITING + PAIR SHARE. This activates prior knowledge from all 11 previous lessons, surfaces residual misconceptions before closure, and gives students a personal benchmark against which to measure their growth at the end of the lesson. Writing before talking prevents the 'fast-talker effect' where confident students dominate.	Observable indicator: Students spontaneously use at least three wave-property terms (e.g., diffraction, wavelength, Doppler effect, frequency) in their pre-final writing. Teacher scans slips during circulation — if fewer than 60% of students use three or more terms, the teacher inserts a 3-minute vocabulary rapid-fire ('shout out a wave property — go!') before proceeding. Collect slips; return at the end of the lesson.
Observe Phase 📹 **VIDEO:** Doppler effect introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open PhET: Wave Interference on laptops or tablets, select 'Interference' mode, then switch to 'Single Slit'. They work in pairs. Using the teacher's guidance card, they set up TWO scenarios side by side: Scenario A — FM wave: frequency set to simulate 105 MHz ($\lambda \approx 2.85$ m), slit width set to 10 m (approximate Kabete tunnel opening). Scenario B — AM wave: frequency set to simulate 567 kHz ($\lambda \approx 529$ m), same 10 m slit width. Students sketch the wave pattern emerging from the slit in each case onto their observation sheet, labelling the spread angle. They answer three	Before opening PhET say: 'We are going to use our simulation one final time — not to learn something new, but to CONFIRM the explanation we are about to build. This is what real scientists do: they test their explanation against evidence before publishing.' Distribute the guidance card. Give 10 minutes of silent pair work. At the 5-minute mark circulate and prompt: 'Have you calculated the λ/slit ratio? What does a ratio much less than 1 mean about spreading?' At 10 minutes, cold-call 3 pairs to share their slit-ratio calculations on the board. Say: 'Class — do you agree with this	Strategy: TARGETED RE-INQUIRY WITH SIMULATION. Rather than a new exploration, students use the simulation as a confirmatory tool — a hallmark of NGSS SEP 3 (Planning and Carrying Out Investigations) and SEP 6 (Constructing Explanations). The slit-width-to-wavelength ratio gives students a quantitative anchor that moves them from qualitative observation ('AM spreads more') to mechanistic explanation ('when λ >> slit width, significant diffraction occurs').	Observable indicator: Each pair's observation sheet shows a correctly labelled diagram with the AM wave spreading widely and the FM wave passing through with minimal spreading, AND a written sentence that correctly links the λ/slit ratio to the degree of diffraction. Teacher collects 6 random sheets and checks within 2 minutes while the next activity begins.

	questions on the sheet: (1) Which wave spreads more — FM or AM? (2) What is the ratio of slit width to wavelength in each case, and what does that ratio predict about diffraction? (3) How does this simulation explain what the student heard on the bus entering the Kabete tunnel?	calculation? WAIT — 12 seconds — anyone want to challenge or add?' After consensus, say: 'Write one sentence in your own words: FM does/does not diffract well through the Kabete tunnel because…' Give 2 minutes for writing. Then display a real satellite image of the Kabete underpass on the projector and say: 'The tunnel opening is roughly 10 m wide. FM wavelength is about 3 m. What does your simulation tell you will happen to Classic 105 inside?' Cold-call 2 students — WAIT 15 seconds each.		
Explain Phase 📹 **VIDEO:** Doppler effect introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Landforms from Wave Erosion and Deposition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	JIGSAW EXPERT ROUND (28–45 min): The class is pre-assigned into 6 home groups of ~5 students, each owning one wave property: Group 1 — Rectilinear Propagation & Reflection; Group 2 — Refraction; Group 3 — Diffraction (FM vs AM in tunnel); Group 4 — Interference & Resonance; Group 5 — The Doppler Effect (matatu horn mystery); Group 6 — Wave Energy, Amplitude & Frequency. Each group has 8 minutes to prepare a 90-second 'Expert Slot': they must (a) state the property in one sentence, (b) show one piece of evidence from their lessons (sketch, data table, or PhET screenshot), and (c) explain EXPLICITLY how their property connects to either the matatu horn or the Kabete tunnel or both. Groups present in sequence (90 seconds each, strict). Audience members complete a listening frame: 'Property: ___ / Evidence given: ___ / Connection to phenomenon: ___'. COLLABORATIVE FINAL	JIGSAW SETUP (28 min mark): 'Each group — you have 8 minutes to prepare your Expert Slot. You must include a piece of EVIDENCE — a sketch, numbers, or a simulation result. No evidence, no slot. Go.' Circulate rapidly; at the 5-minute mark give a '3 minutes left' signal. During presentations, after each slot ask the audience: 'Thumbs up if you can write their connection to the phenomenon in your own words — WAIT 10 seconds.' Cold-call one audience member per slot: 'Rehema, in your own words, how does diffraction explain the tunnel mystery?' COLLABORATIVE EXPLANATION (45 min mark): Display the template. Say: 'We are now going to write the BEST scientific explanation this class can produce. I am the scribe — you are the scientists. Mystery 1 first: who can give me a CLAIM — one sentence that answers why the horn pitch changes?' WAIT 15 seconds. Accept first offer, write it, then say: 'Does anyone want to REFINE this claim — make it more	Strategy: JIGSAW + COLLABORATIVE SCIENTIFIC WRITING. The jigsaw ensures every student is accountable for deep understanding of at least one property (SEP 8: Obtaining, Evaluating, and Communicating Information). The collaborative scribing of the Final Explanation enacts the scientific practice of building shared knowledge — mirroring how Kenyan physicists at JKUAT or the Kenya Meteorological Department write consensus reports. Writing forces precision: students must translate 'waves spread out' into 'FM waves ($\lambda \approx 2.85$ m) diffract minimally through the 10 m Kabete tunnel opening because $\lambda \ll$ slit width, so energy is not redirected into the shadow zone.'	Observable indicator: The co-constructed Final Explanation paragraph for each mystery contains (1) the name of the relevant wave property, (2) a reference to wavelength or frequency with a value, (3) a causal connective ('because', 'therefore', 'this means that'), and (4) an explicit link to the phenomenon. Teacher reads both paragraphs aloud and the class votes (show of hands) on whether each criterion is met before the text is finalised.

	EXPLANATION (45–58 min): After all six slots, the teacher displays a blank 'Final Explanation Template' on the board with two columns: 'Mystery 1 — The Matatu Horn' and 'Mystery 2 — The Kabete Tunnel'. Working as a whole class in a structured Socratic discussion, students co-construct a 5–7 sentence scientific paragraph for each mystery, with the teacher scribing on the board. Students copy the final agreed text into their exercise books.	precise?' Repeat for each sentence. For Mystery 2 say: 'Now explain the tunnel. Start with what property is doing the work.' If students omit wavelength values, probe: 'Can you make this more precise? What is the actual wavelength of FM vs AM?' After both paragraphs are complete, read both aloud and ask: 'Is there anything we CANNOT yet explain? Be honest.' Record any lingering uncertainties on the board — these will go on the DQB as STILL OPEN items.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Doppler effect introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Landforms from Wave Erosion and Deposition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The full Driving Question Board — which has been displayed at the front of the classroom since Lesson 1 and updated every lesson — is now reviewed for the final time. Each student receives three sticky-dot stickers: green (RESOLVED), amber (PARTIALLY RESOLVED), and red (STILL OPEN). Working in silence for 3 minutes, students individually place their dots on the DQB questions they believe match those categories. The teacher then facilitates a 5-minute whole-class tally and discussion: questions that are mostly green are read aloud and celebrated; questions that remain red or amber are acknowledged as genuine open scientific questions — some of which are studied in later strands (e.g., radio wave propagation in space connects to Sub-Strand 4.2 Space Physics). The driving question itself — 'Why does the matatu's horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel?' — is read aloud by a student volunteer, and the class agrees on a one-sentence spoken	Distribute the three sticky dots to each student. Say: 'Three minutes of silence — read every question on the DQB and place your dots. Green = we can fully explain this now. Amber = we have a partial explanation. Red = still a genuine mystery.' WAIT the full 3 minutes without prompting. After dotting, stand at the DQB and point to the question with the most green dots: 'This one we clearly resolved — who can say the answer in one sentence?' Cold-call 2 students — WAIT 12 seconds each. Then point to the most prominent RED question: 'This one is still open — and that is not a failure. Real physicists have open questions too. In fact, some of these connect to what you will study in Strand 4.2 on Space Physics.' Finally, point to the driving question poster at the top of the board. Say: 'Omondi — please read our big question aloud.' After Omondi reads, say: 'Class — together — one sentence. What is the answer?' Facilitate the spoken consensus answer. Write it on the board in a box labelled 'OUR FINAL ANSWER'.	Strategy: DRIVING QUESTION BOARD CLOSURE with TRAFFIC-LIGHT DOT VOTING. This strategy (adapted from NGSS Storyline design) makes the growth of class understanding visible and tangible. The physical act of placing dots on questions that were once mysterious and are now resolved gives students a concrete sense of intellectual progress — a powerful motivator for continued science engagement. Red dots on still-open questions model the iterative, never-fully-finished nature of scientific inquiry.	Observable indicator: At least 80% of the DQB questions related to diffraction and the Doppler effect receive green dots from the majority of students. Any question receiving more than 40% red dots is flagged by the teacher for individual follow-up or re-teaching in a subsequent lesson. The spoken consensus answer to the driving question contains both 'diffraction' and 'Doppler effect' without teacher prompting.

	answer together.			
Model Building Phase 📹 **VIDEO:** [Doppler effect introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Landforms from Wave Erosion and Deposition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their Lesson 1 initial model sketch (stored in their exercise book or a class folder) and place it side by side with their final annotated model. They complete a structured METACOGNITIVE COMPARISON FRAME in their exercise books: (1) 'In Lesson 1, I thought the matatu horn changed pitch because…'; (2) 'I now know it actually happens because…'; (3) 'In Lesson 1, I thought the FM signal faded because…'; (4) 'I now know it actually fades because…'; (5) 'The most surprising thing I learned in this unit was…'; (6) 'One question I still have is…'. After 7 minutes of individual writing, students share item (5) with a partner. The lesson closes with a whole-class oral round: teacher cold-calls 6 students (one per wave-property group) to share their 'most surprising thing'. The exit ticket is the completed pre-final slip from Phase 1, now annotated by the student with corrections in a different colour ink — demonstrating visible self-revision.	Distribute Lesson 1 model sheets (or ask students to open to Lesson 1 in their books). Say: 'Place your Lesson 1 model and your final model side by side. Look at them for 30 seconds in silence.' WAIT the full 30 seconds. Then say: 'Now complete the six-frame comparison. This is about YOUR thinking — there are no wrong answers here, only honest ones. 7 minutes — go.' Circulate and read over shoulders; if a student writes 'I thought it was because the driver changed the horn' for item 1, quietly say: 'That is a perfectly honest Lesson 1 idea — now write exactly why you know that is not true.' At the 7-minute mark: 'Share item 5 — your most surprising thing — with your partner. 90 seconds.' Then cold-call 6 students: 'Wanjiku — most surprising thing?' WAIT 12 seconds. After all 6 share, say: 'Now pick up your Phase 1 slip. In a DIFFERENT colour pen, annotate it — add what was missing, correct what was wrong. This shows your brain at work.' Give 3 minutes. Collect the annotated slips as exit tickets. Close with: 'You began this unit puzzled by a matatu horn and a fading radio. You leave it as physicists who can explain both. That is what science does — it turns mystery into understanding.'	Strategy: LESSON 1 vs FINAL MODEL COMPARISON + METACOGNITIVE WRITING. This strategy (NGSS SEP 2: Developing and Using Models; SEP 6: Constructing Explanations) makes conceptual change visible to both teacher and student. Research in science education shows that explicit comparison of initial and final models is one of the most powerful consolidation tools available, because it forces students to articulate not just WHAT they now know, but HOW and WHY their understanding shifted. The annotated exit slip is a physical artefact of learning.	Observable indicator: In the metacognitive frame, students' 'I now know' sentences for both mysteries correctly name the wave property, include a mechanistic causal chain (e.g., 'The matatu's horn frequency does not change, but because the matatu is moving toward me, more wave crests arrive per second at my ear, so I perceive a higher frequency — the Doppler effect'), and differ meaningfully from their Lesson 1 sentences. Teacher grades 10 randomly selected exit tickets using a 3-point rubric: 3 = correct mechanism with evidence; 2 = correct property named but mechanism incomplete; 1 = misconception persists. Any student scoring 1 receives a follow-up card the next day.

D. TEACHER REFLECTION

After delivering Lesson 12, reflect on the following questions before the next teaching cycle:

1. FINAL EXPLANATION QUALITY: Did the co-constructed Final Explanation paragraphs contain all four required criteria (property name, wavelength/frequency value,

causal connective, explicit link to phenomenon)? If not, which criterion was most commonly missing — and does that reveal a gap in a specific earlier lesson?

2. DQB CLOSURE: How many questions remained RED (still open) at the end of the DQB review? Were these genuinely beyond the scope of the sub-strand, or did they reveal topics that should have been covered more thoroughly? Note any RED questions that could be addressed in the opening lesson of Sub-Strand 2.2 or later strands.

3. MODEL COMPARISON DEPTH: When students compared their Lesson 1 and final models, did the 'I now know' sentences show genuine mechanistic understanding, or surface-level vocabulary substitution (e.g., writing 'Doppler effect' without explaining the mechanism)? If the latter, plan a structured Socratic seminar in the first lesson of the next sub-strand to revisit causal reasoning.

4. JIGSAW ACCOUNTABILITY: Did all six expert groups present with genuine evidence, or did some groups rely on verbal description only? Groups that lacked a diagram or data point in their Expert Slot signal that the corresponding lesson (e.g., Lesson 9 on resonance) may need a stronger hands-on evidence-gathering component in future iterations.

5. KENYAN CONTEXT INTEGRATION: Did students spontaneously use Kenyan context (Kabete tunnel, Nakuru–Nairobi highway, Classic 105 FM) in their final written explanations, or did they revert to abstract/generic language? If the latter, reinforce the contextualisation norm at the start of the next sub-strand.

6. EQUITY CHECK: Review the cold-call log. Were girls, students from pastoralist backgrounds (e.g., Turkana, Maasai), and students with lower prior attainment represented proportionately across the 12+ cold-calls in this lesson? Adjust calling patterns in the next unit to ensure equitable participation from the first lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed in the PhET Wave Interference simulation (single-slit mode) that an FM wave ($\lambda \approx 2.85$ m, simulating 105.2 MHz) passing through a 10 m gap (the Kabete tunnel) produces very little spreading beyond the gap, while an AM wave ($\lambda \approx 529$ m, simulating 567 kHz) passing through the same gap spreads almost uniformly in all directions. Students also observed their own Lesson 1 initial model diagrams — featuring vague or incorrect explanations — placed side by side with their detailed final annotated models, making conceptual change physically visible.
What did I learn?	Students learned that ALL the wave behaviours encountered in this sub-strand — rectilinear propagation, reflection, refraction, diffraction, interference, resonance, and the Doppler effect — are not isolated facts but properties of a single underlying entity: a wave is a disturbance that transfers energy without transferring matter. They learned that the degree of diffraction depends on the ratio of wavelength to gap size (when $\lambda \approx$ gap or $\lambda >>$ gap, significant diffraction occurs; when $\lambda <<$ gap, minimal diffraction occurs), which is precisely why AM waves ($\lambda \sim$ hundreds of metres) bend around the Kabete tunnel walls while FM waves ($\lambda \sim 3$ m) do not. They learned that the Doppler effect — the apparent change in frequency due to relative motion between source and observer — explains why the matatu horn rises in pitch on approach and falls on departure, even though the driver never changes the horn. Together, these two explanations resolve both mysteries of the anchoring phenomenon.
How does this explain the phenomenon?	Students explained in a collaboratively written scientific paragraph that: (1) The matatu horn appears to change pitch because of the Doppler effect — as the matatu approaches, successive sound wave crests are compressed into a shorter distance, increasing the frequency (and therefore pitch) detected at the student's ear; as it moves away, wave crests are stretched out, decreasing the detected frequency. The matatu's own horn frequency never changes — only the perceived frequency does. (2) The FM signal (Classic 105, 105.2 MHz, $\lambda \approx 2.85$ m) fades in the Kabete tunnel because FM waves have a wavelength far smaller than the tunnel opening (~10 m), meaning the ratio λ/gap << 1 and diffraction is minimal — FM waves travel in nearly straight lines and cannot bend around tunnel walls into the shadow zone. AM waves ($\lambda \sim$ hundreds of metres) have a wavelength far larger than the tunnel

	opening, so λ/gap >> 1, strong diffraction occurs, and AM waves bend readily around and into the tunnel, reaching the receiver even in the shadow zone. Both explanations are unified by the same underlying principle: waves transfer energy according to properties — frequency, wavelength, amplitude — that govern how they interact with the physical world.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.